SWIFT SHIP TRACKER

Book. Track. Deliver. Smarter.

User Story

The SwiftShip Tracker is designed to provide a centralized Salesforce-based parcel management solution that enables customers to book parcels, track delivery status, and manage shipments through an intuitive interface and conversational AI support. The system manages Parcel, Delivery, Sender, and Receiver information using Salesforce custom objects and relationships. Flows automate parcel operations and notifications, while AgentForce AI and Prompt Builder enable customers to retrieve parcel information and interact with the system conversationally. Delivery agents can update parcel status in real time, while Experience Cloud, security controls, reports, and dashboards provide secure customer access and operational visibility.

1. Introduction

1.1 Project Overview

The SwiftShip Tracker is a Salesforce CRM solution developed to simplify parcel booking, shipment tracking, and delivery management. It supports the complete parcel lifecycle from booking and dispatch through tracking and delivery confirmation. Customers can access shipment information, while delivery agents can update parcel details and delivery status.

1.2 Purpose

The purpose of SwiftShip Tracker is to:

- Centralize parcel and delivery information.
- Simplify parcel booking and tracking.
- Provide real-time shipment visibility.
- Automate delivery updates and customer notifications.
- Enable conversational parcel tracking using AgentForce AI and Prompt Builder.
- Improve agent efficiency through Salesforce Flows and dashboards.
- Provide secure, role-based access to customers, agents, support staff, and administrators.

2. Ideation Phase

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Customer / Sender	book a parcel easily	booking requires support assistance or multiple applications	parcel operations are not centralized	frustrated
PS-2	Customer / Sender	track my parcel	I do not have a simple real-time tracking experience	shipment information is difficult to access	confused
PS-3	Delivery Agent	update parcel status	status updates require manual handling	there is a need for centralized operational tracking	concerned
PS-4	Customer	receive delivery updates	notifications may not be available at every delivery stage	communication needs to be automated	uncertain
PS-5	Admin	monitor parcel operations	parcel flow and delivery performance are difficult to monitor	operational data needs centralized reporting	concerned

2.2 Empathy Map Canvas

User Persona: Customer / Sender

Think & Feel
- Wants a simple way to book and track parcels without repeatedly contacting support.
- Hopes the parcel reaches on time and safely.
- Feels more confident with real-time updates.
- Gets concerned when there is no tracking information.

Hear
- Delivery confirmations and updates from the parcel service.
- Information about delivery timelines.
- Feedback from friends, family, or colleagues about delivery experiences.
- Announcements or notifications about service changes.

See
- Parcel status and tracking information.
- Delivery details and estimated delivery date.
- User-friendly portal or mobile interface.
- Notifications and emails about shipment progress.



Say & Do
- Books parcels through the portal or app.
- Checks shipment status regularly.
- Asks for delivery updates when needed.
- Provides feedback on delivery experience.
- Shares tracking information with recipients.

Pain
- Delays in delivery.
- Confusion due to lack of clear tracking information.
- Difficulty accessing shipment details.
- Dependence on customer support or multiple apps.
- Uncertainty about delivery timelines.

Gain
- Easy and convenient parcel booking.
- Real-time tracking and delivery updates.
- Timely notifications at every stage.
- Better visibility of delivery progress.
- Peace of mind knowing the parcel is on track.

Figure 1. Empathy Map Canvas - Customer / Sender

2.3 Brainstorming & Idea Prioritization

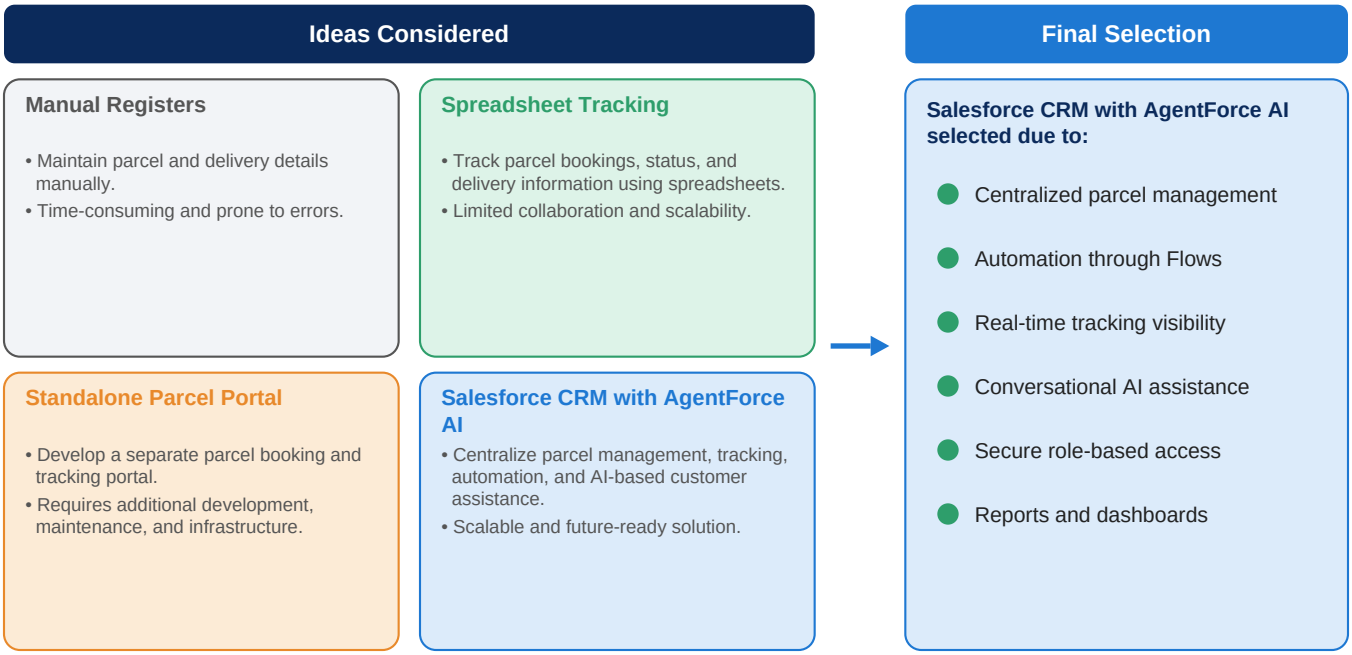


Figure 2. Ideas considered and final selection

3. Requirement Analysis Phase

3.1 Customer Journey Map

Stages: Parcel Booking, Parcel Record Creation, Dispatch & Shipment, In-Transit Tracking, Out for Delivery, Delivery Confirmation, Customer Notification, and Feedback / Support.

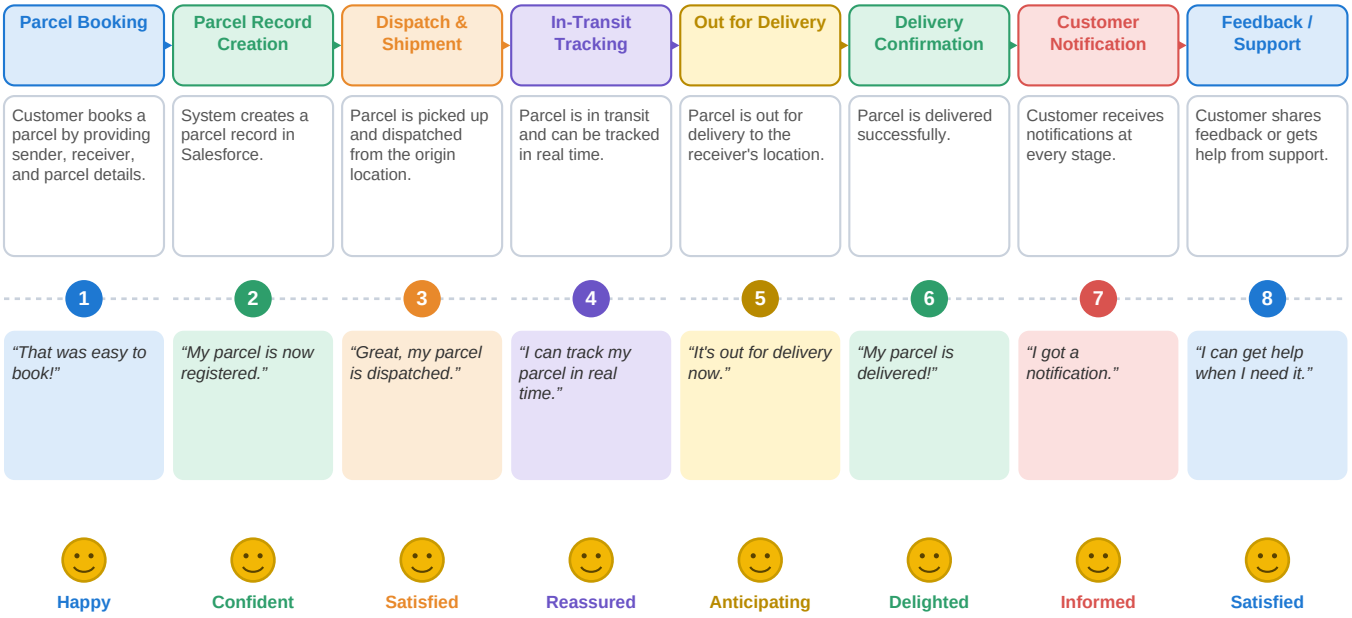


Figure 3. Customer Journey Map - from booking to delivery (top: customer thoughts, bottom: customer emotion)

3.2 Solution Requirements

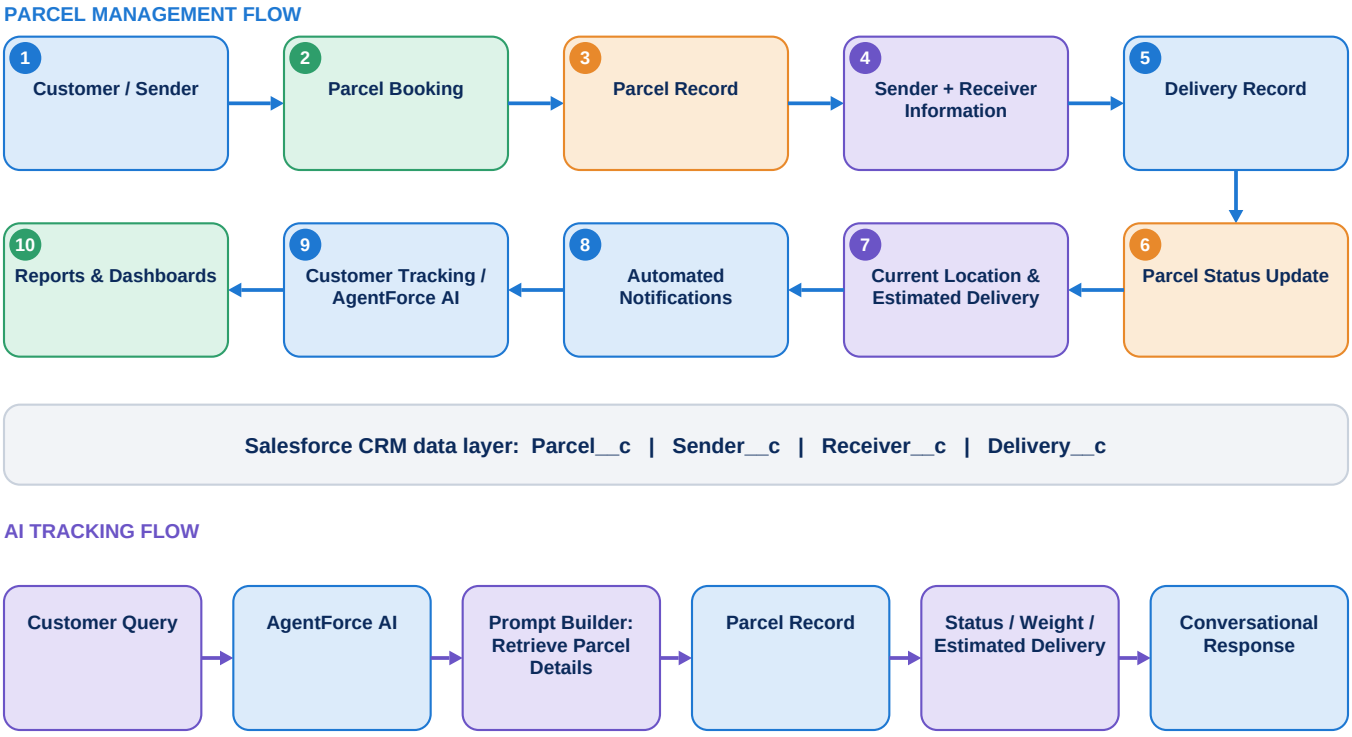
Functional Requirements

FR No	Requirement
FR-1	Parcel booking and record creation
FR-2	Parcel status tracking
FR-3	Weight management
FR-4	Estimated delivery date management
FR-5	Sender information management
FR-6	Receiver information management
FR-7	Delivery and current-location tracking
FR-8	Agent parcel status updates
FR-9	Automated customer notifications
FR-10	AI-powered parcel tracking
FR-11	Reports and dashboards
FR-12	Customer access through Experience Cloud

Non-Functional Requirements

NFR	Description
Usability	Intuitive Salesforce and customer-facing interface
Security	Role-based and field-level access control
Reliability	Accurate parcel status and delivery information
Performance	Timely Flow and AgentForce responses
Availability	Salesforce cloud-based access
Scalability	Supports growing parcel and delivery operations
Data Integrity	Controlled access and relationship-based parcel data
Maintainability	Flows, prompts, topics, and configurations can be updated

3.3 Data Flow Diagram



The Prompt Builder template retrieves Parcel ID, Status, Weight, and Estimated Delivery Date for tracking responses.

Figure 4. End-to-end parcel management flow and AI tracking flow

3.4 Technology Stack

Layer	Technology
Platform	Salesforce Developer Edition
UI	Lightning App Page, Lightning Record Pages, Experience Cloud
Database	Salesforce Custom & Standard Objects
Business Logic	Salesforce Flows, Apex Classes
Automation	Record-Triggered Flows, Auto-Launched Flows, Scheduled Flows
AI	AgentForce AI, Prompt Builder
Notifications	Email Alerts & Email Templates
Processing	Batch Apex
Security	Profiles, Permission Sets, Role Hierarchy, Field-Level Security, Sharing Rules
Reporting	Reports & Dashboards

4. Project Design Phase

4.1 Problem-Solution Fit

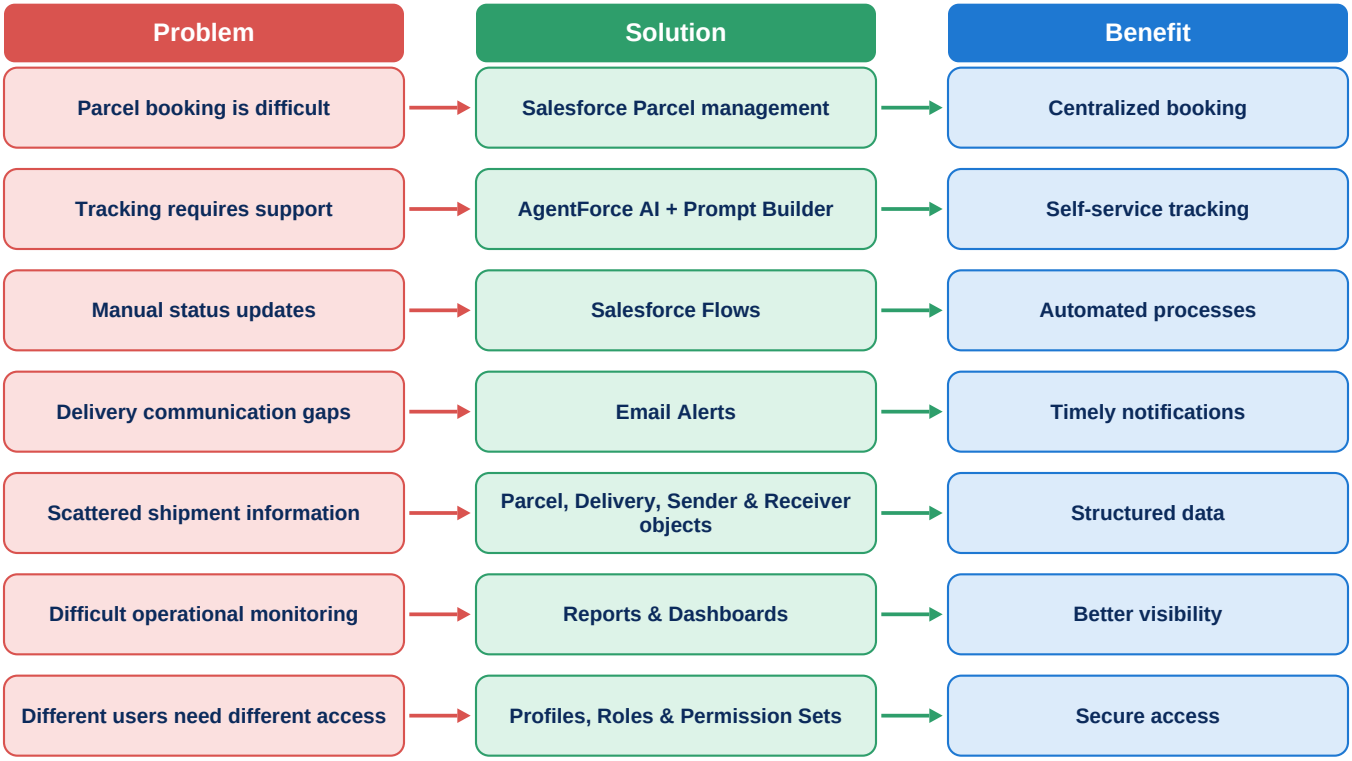


Figure 5. Problem-Solution Fit

4.2 Proposed Solution

Parameter	Description
Problem	Difficult and inefficient parcel booking and tracking
Solution	Salesforce-based Parcel Management System
Innovation	AgentForce AI + Prompt Builder for conversational self-service tracking and automated Salesforce Flows
Impact	Centralized booking, real-time tracking visibility, timely notifications, and efficient delivery management
Business Model	Parcel, customer, delivery-agent, support, and administrator operations
Scalability	Supports growing parcel operations, Experience Cloud, and integration with external logistics systems

4.3 Solution Architecture

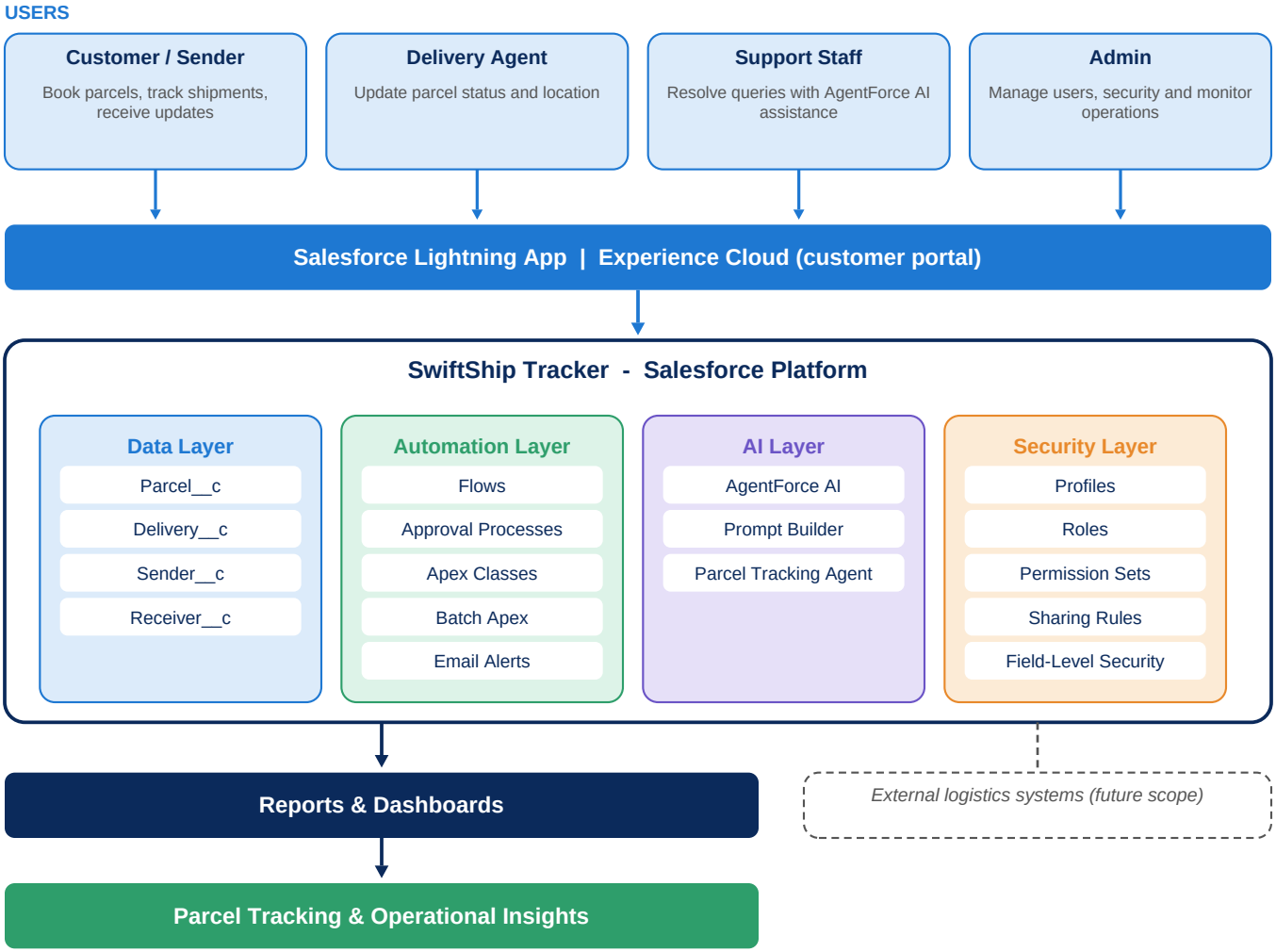


Figure 6. SwiftShip Tracker solution architecture

5. Project Planning & Scheduling

5.1 Project Planning

- Agile methodology
- Epics, Stories, Story Points
- Sprint-based execution
- Velocity calculation used for estimation

Product Backlog, Sprint Schedule, and Estimation

Sprint	Functional Requirement (Epic)	USN	User Story / Task	Points	Priority	Team Member
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create and configure the Salesforce environment so that I can build the SwiftShip Tracker parcel management system.	3	High	Member1
Sprint-2	Data Modeling	USN-2	As an admin, I want to create Parcel, Delivery, Sender, and Receiver objects with proper relationships so that parcel and delivery data is stored and managed systematically.	5	High	Member2

Sprint	Functional Requirement (Epic)	USN	User Story / Task	Points	Priority	Team Member
Sprint-2	Data Modeling	USN-3	As a staff member, I want dedicated tabs and a Lightning App for SwiftShip modules so that I can easily navigate and manage parcel and delivery operations.	5	High	Member3
Sprint-3	Automation	USN-4	As a system, I want to validate parcel details, mandatory fields, and booking inputs so that invalid or incomplete parcel records are prevented.	3	High	Member4
Sprint-3	Automation	USN-5	As a delivery coordinator, I want automated Flows to update parcel status and delivery information so that parcel processing requires minimal manual effort.	3	High	Member4
Sprint-4	Apex	USN-6	As a system, I want Apex triggers and classes to automatically perform critical parcel and delivery actions so that important processes are handled accurately.	5	High	Member1
Sprint-4	Apex	USN-7	As an operations manager, I want Batch Apex jobs to process and update parcel or delivery records automatically so that operational data remains accurate and up to date.	5	High	Member2
Sprint-5	Reports	USN-8	As a manager, I want reports showing parcel status, delivery performance, and operational metrics so that I can monitor SwiftShip operations and make informed decisions.	4	High	Member3
Sprint-6	Dashboards & AI	USN-9	As a manager, I want dashboards showing parcel status, delivery performance, and key metrics, while Agentforce provides AI-powered tracking assistance, so that I can monitor operations efficiently.	4	Medium	Member4

Project Tracker, Velocity & Burndown Chart

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022		
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022		
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022		
Sprint-5	20	6 Days	21 Nov 2022	26 Nov 2022		
Sprint-6	20	6 Days	28 Nov 2022	03 Dec 2022		

6. Project Development Phase

Phase 1: Requirement Analysis & Planning

Objective

“Customers often face delays and confusion while booking, tracking, and managing parcel deliveries because they need to call support or use multiple apps. A conversational AI agent can simplify this by allowing customers to book parcels, get live tracking, and manage deliveries in one place.”

Key Requirements

- Requirement gathering from end users and delivery agents.
- Defining system scope and user personas (Customer, Agent, Admin).
- Designing the data model with relationships between Parcel, Sender, Receiver, and Delivery.

- Identifying use cases such as parcel booking, tracking, and delivery update automation.
- Selecting tools: Salesforce Flows, Prompt Builder, and AgentForce AI.

Defining Project Scope & Objectives

Project Scope

- Build a scalable CRM solution for parcel delivery businesses to manage customer bookings, shipments, and delivery operations efficiently.
- Support the end-to-end parcel lifecycle: parcel booking, dispatch, tracking, and delivery confirmation.
- Include comprehensive data model design, automation (Flows, Apex, Email Alerts), AI-powered Prompt Builder, and reporting dashboards for effective management.
- Provide role-based access for admins, delivery agents, and customers through Salesforce profiles, permission sets, and Experience Cloud.
- Ensure real-time parcel visibility for customers and simplified operational tracking for agents through AI-integrated conversational support.

Objectives Summary

- Streamline parcel operations by automating booking, tracking, and delivery notifications to reduce manual updates.
- Improve customer satisfaction with transparent, real-time shipment status and self-service tracking options.
- Enhance agent efficiency through centralized dashboards and guided Flows for quick status updates.
- Enable business insights with reports and dashboards showing parcel trends, delivery performance, and agent activity.
- Ensure security and reliability by implementing Salesforce role hierarchies, sharing rules, and audit-based data access.

Gathering & Analysing User Needs

Users Involved

- **Customers / Senders** - End users who book parcels, track deliveries, and receive real-time updates through the portal or AI agent.
- **Delivery Agents** - Manage assigned parcels, update delivery status, and ensure timely handover to customers.
- **Customer Support Team** - Handle delivery-related issues, respond to customer queries, and ensure smooth communication.
- **Admin Team** - Configure the Salesforce environment, manage user roles, security settings, and monitor operational performance.
- **System Integrator / Developer** - Responsible for implementing automation, AI agent setup, and backend integrations.

Key Functional Needs

- Seamless parcel booking and tracking interface for customers through Salesforce Experience Site or agent assistance.
- Ability for agents to update parcel status (Booked, In Transit, Out for Delivery, Delivered) in real time.
- Automated notifications and email alerts to customers on every delivery stage update.
- Support for Sender and Receiver details management with address, contact, and email fields for communication.
- AI Agent integration (via Prompt Builder) for real-time parcel status queries and automated assistance.
- Reports and dashboards to monitor parcel flow, delivery performance, and agent activity for admins.
- Security controls with role-based access, profiles, and permission sets for different user levels.

Tools Used

- Salesforce Developer Org - for configuration, development, and testing.
- Prompt Builder (AgentForce AI) - to create intelligent parcel tracking prompts and responses.

- Salesforce Flow Builder - to automate booking, tracking, and delivery updates.
- Google Sheets / Docs - for initial data planning, requirement documentation, and system mapping.

Identifying Key Salesforce Features & Tools Required

Salesforce Features Planned

- **Custom Objects:** Parcel__c, Delivery__c, Sender__c, Receiver__c
- **Standard Objects:** Account, Contact, Case, Task, EmailMessage, User
- **Automations:** Record-Triggered Flows, Auto-Launched Flows, Approval Processes, Scheduled Flows
- **Apex:** Apex Classes for automated parcel operations (e.g., parcel status updates, delivery notifications), Batch Apex for periodic updates or notifications
- **UI:** Lightning App Page (SwiftShip Tracker App), Dynamic Forms for Parcel Details, Lightning Record Pages, and Experience Cloud for Customer Access
- **AgentForce AI & Prompt Builder:** Integrated AI agent trained to fetch parcel data based on Parcel ID and respond to tracking or delivery status queries
- **Email Services:** Custom Email Alerts and Templates for booking confirmations, delivery updates, and completion acknowledgments
- **Security:** Profiles, Permission Sets, Role Hierarchies, Field-Level Security, and Sharing Rules to ensure controlled access for Admin, Agent, and Customer roles

Designing Data Model and Security Model

Data Model Includes

Object	Purpose and relationships
Parcel__c	Stores core parcel details such as Parcel ID, Status, Weight, and Estimated Delivery Date. Linked to Sender__c and Receiver__c objects to fetch sender and receiver details. Connected with Delivery__c object to track real-time parcel location and status.
Delivery__c	Stores delivery-specific information such as current location, estimated delivery date, and linked parcel. Connected with Sender__c and Parcel__c for relationship-based tracking. Used by agents to update and monitor delivery status during the parcel lifecycle.
Sender__c	Captures sender-related information such as address, contact number, and email. Linked to multiple parcels to track sender history and associated deliveries. Used for sending booking confirmations and delivery updates via automated emails.
Receiver__c	Stores receiver details such as address, contact, and email for communication and delivery confirmation. Linked to Sender__c and Parcel__c to associate both ends of the shipment process. Used by agents for delivery reference and automated email notifications upon completion.

Security Model Design

- **Role Hierarchy:** Administrator, Delivery Manager, Agent, Customer
- **Profiles:** Admin Profile - full access to all objects and records. Delivery Agent Profile - Read/Edit access to Parcel, Delivery, Sender, and Receiver objects. Customer Profile - read-only access to their own parcel details via Experience Cloud. Support Staff Profile - access to Cases and Parcel details for issue resolution.
- **Permission Sets:** Used to grant additional privileges (e.g., “Update Delivery Status,” “Access Agent Console”).
- **Record-Level Security:** Sharing Rules applied for Parcel__c and Delivery__c to allow controlled access for agents and managers. Customers can only view records linked to their own Sender or Receiver details.
- **Field-Level Security:** Sensitive fields like Sender Email, Receiver Contact, and Location Coordinates restricted to Admin and Agent profiles only.

Phase 2: Salesforce Development - Backend & Configurations

SwiftShip Tracker

Name: Yamuna P **College:** PB College of Engineering **Date:** 02/10/2026

Org Username: palaniammu1122.2377324c90ea@agentforce.com

This phase focused on developing the core logic of SwiftShip Tracker. Custom objects were created for Parcel, Sender, Receiver and Delivery with relationships. An AI prompt template was built in Prompt Builder and used in an Autolaunched Flow that returns the parcel details.

Milestones

1. Milestone 1 - Salesforce Account
2. Milestone 2 - Objects Creation
3. Milestone 3 - Tabs
4. Milestone 4 - Fields & Relationships
5. Milestone 5 - Prompt Builder
6. Milestone 6 - Flows

Milestone 1 - Salesforce Account

Salesforce is your customer success platform, designed to help you sell, service, market, analyse, and connect with your customers.

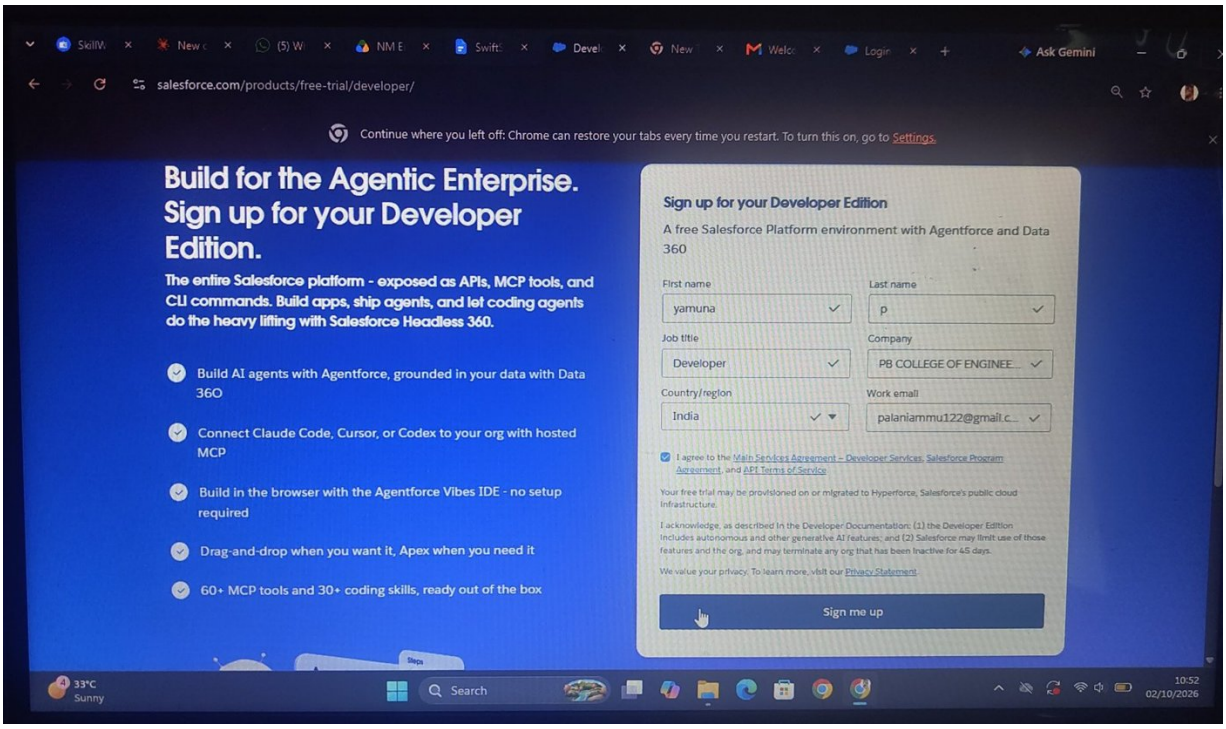
Creating Developer Account

Creating a developer org in Salesforce.

> **How to do it in Salesforce:** Open the sign-up link in a browser, fill the form and submit. Salesforce emails a verification link. After you set a password, Salesforce emails a one-time code to confirm your identity, and then opens your new org.

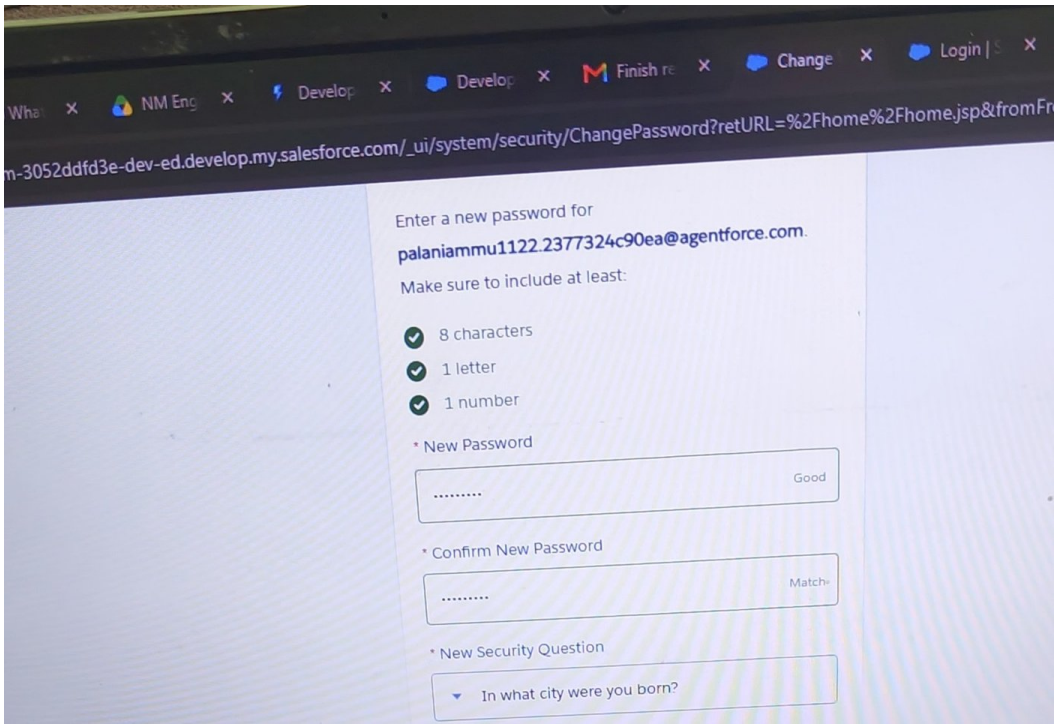
1. Go to https://developer.salesforce.com/signup
2. On the sign-up form, enter the following details:
   - First name: Yamuna; Last name: P
   - Email: palaniammu1122@gmail.com
   - Job Title: Developer
   - Company: PB College Of Engineering
   - Country: India
3. Click on **Sign me up** after filling these.

My org: palaniammu1122.2377324c90ea@agentforce.com



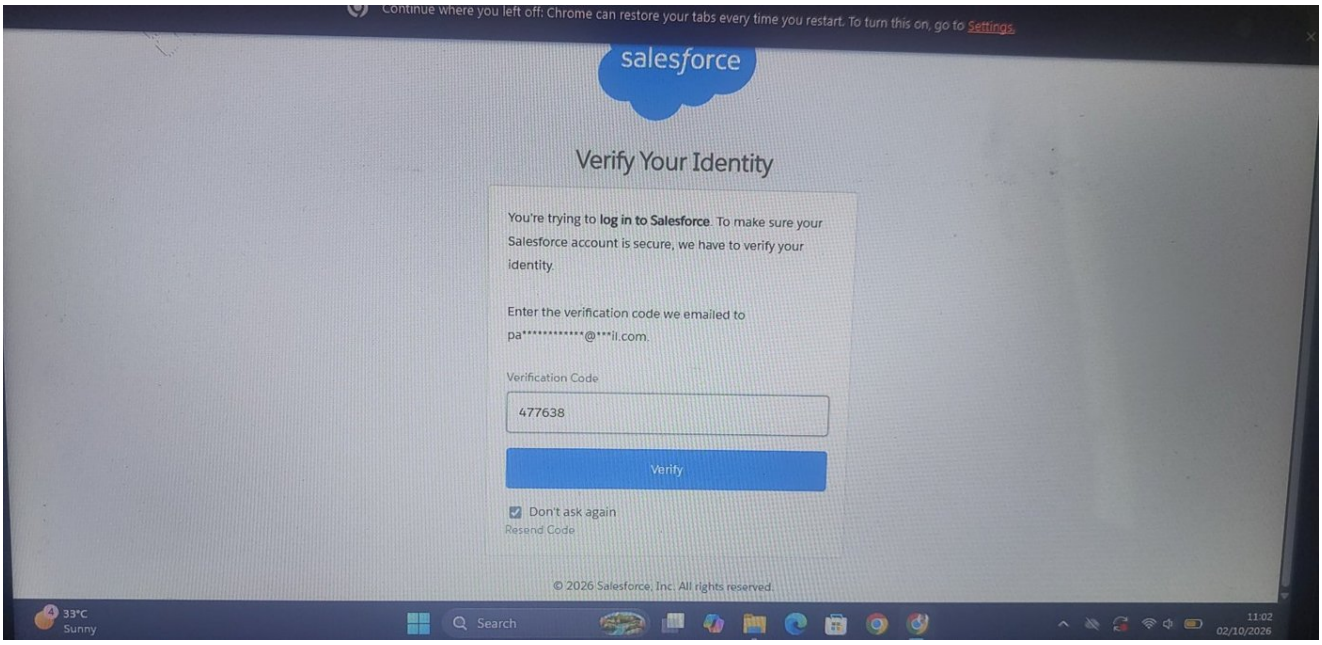
Screenshot 1

1. Go to the inbox of the email that you used while signing up. Click on **Verify Account** to activate your account. The email may take 10-30 mins and sometimes 2 hours.
2. Click on Verify Account
3. Give a password and answer a security question and click on **Change Password**.



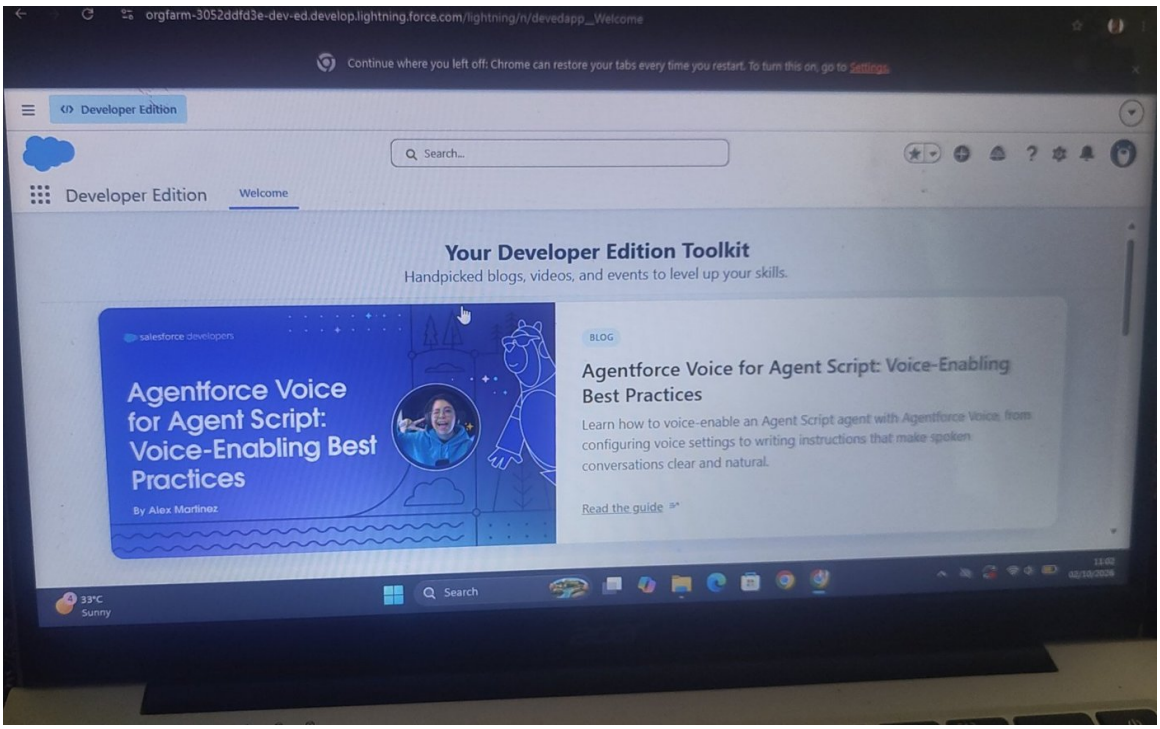
Screenshot 2 - Change Password (username: palaniammu1122.2377324c90ea@agentforce.com)

Salesforce sends a verification code to the registered email. Enter the code in the **Verification Code** box and click **Verify**.



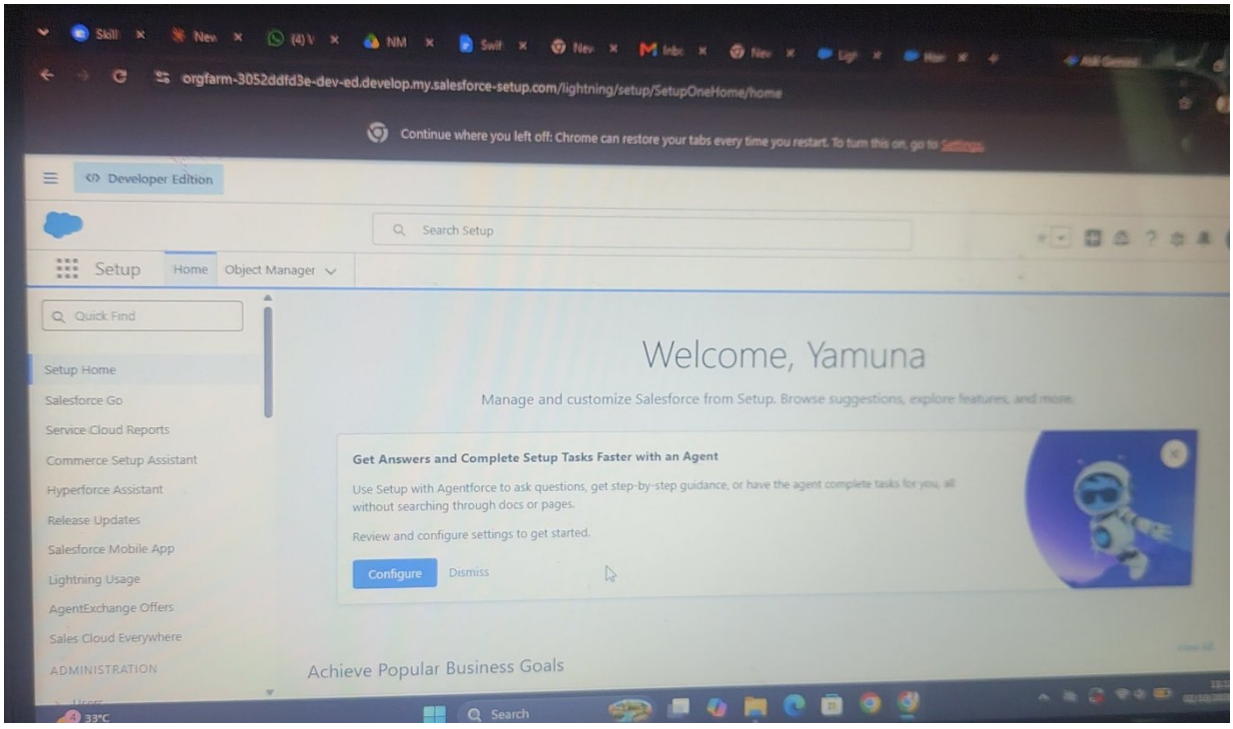
Screenshot 3

Then you will be redirected to your Salesforce setup page. The Developer Edition Welcome page confirms that the org was created successfully.



Screenshot 4

Click the gear icon and select **Setup**. The Setup Home page opens, with Quick Find and Object Manager used for all configuration.



Screenshot 5

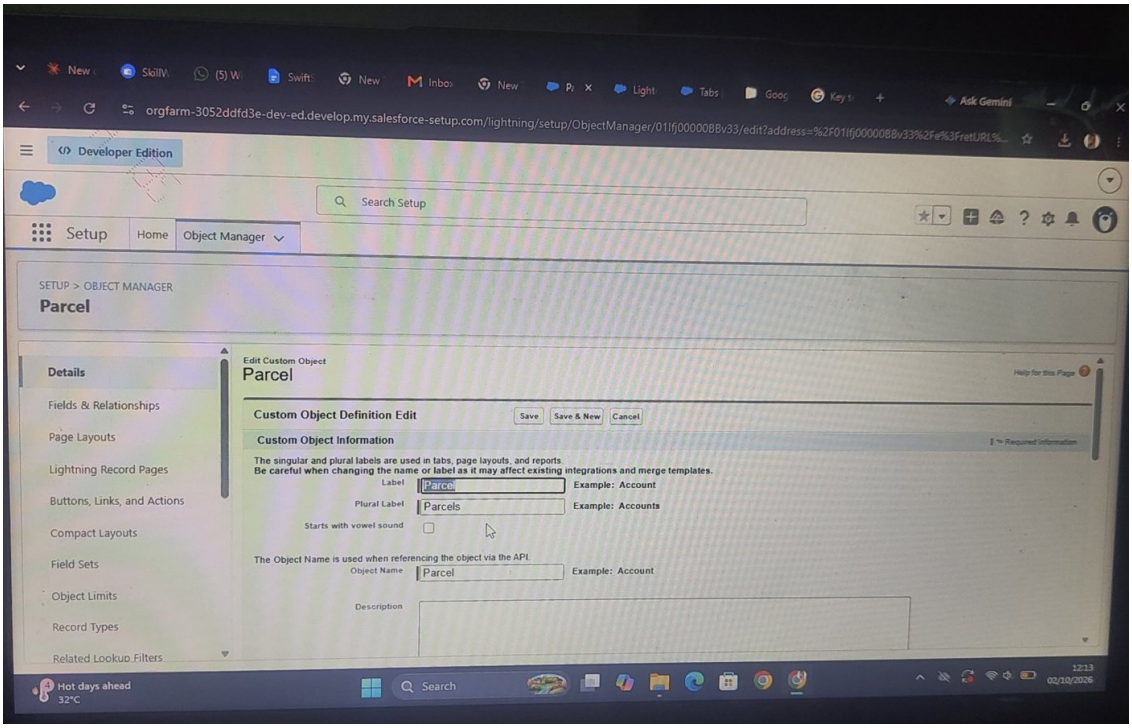
Milestone 2 - Objects Creation

How to do it in Salesforce: An object is like a database table. In Setup, open Object Manager, click Create, then Custom Object, fill in the labels, tick the options and save. Salesforce automatically adds __c to the API name (for example Parcel__c).

Creating Parcel Object

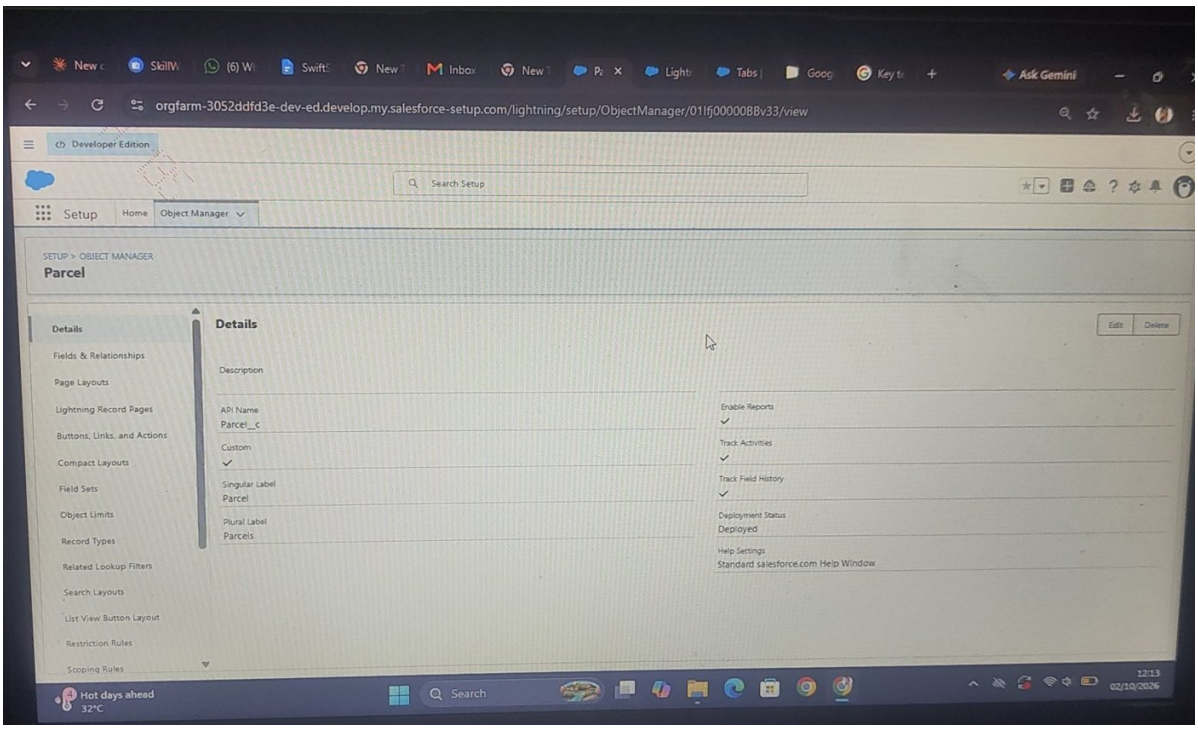
To create an object:

1. From the setup page
2. Click on Object Manager
3. Click on Create >> Click on Custom Object.
4. Enter the label name as Parcel
5. Enter Plural label name as Parcels
6. Enter Record Name as Parcel Name
7. Select Data Type as Text: Parcel Name
8. Select Allow reports.
9. Select Allow search.
10. Allow Track Field History
11. Click on Save and New



Screenshot 6

After saving, the object details page shows API Name **Parcel__c**, Deployment Status **Deployed**, and Enable Reports, Track Activities and Track Field History enabled.

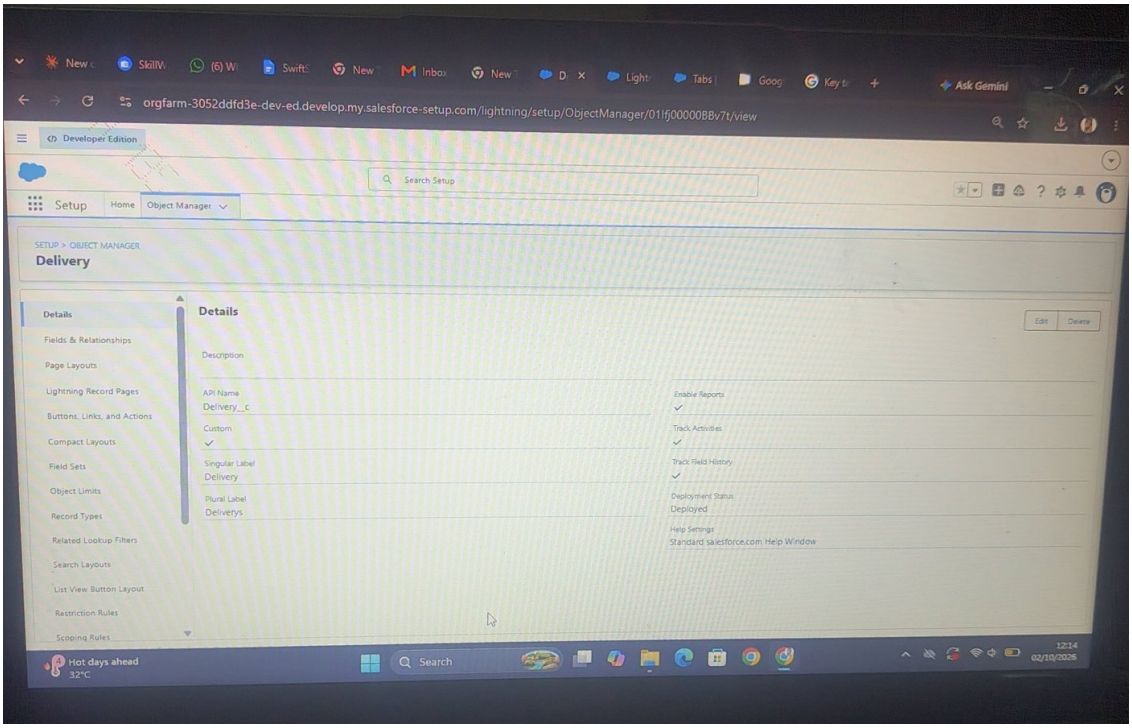


Screenshot 7

Creating Delivery Object

To create an object:

1. From the setup page
2. Click on Object Manager
3. Click on Create >> Click on Custom Object.
4. Enter the label name as Delivery
5. Enter Plural label name as Deliveries
6. Enter Record Name as DeliveryName
7. Select Data Type as Text: DeliveryName
8. Select Allow reports.
9. Select Allow search.
10. Allow Track Field History
11. Click on Save and New

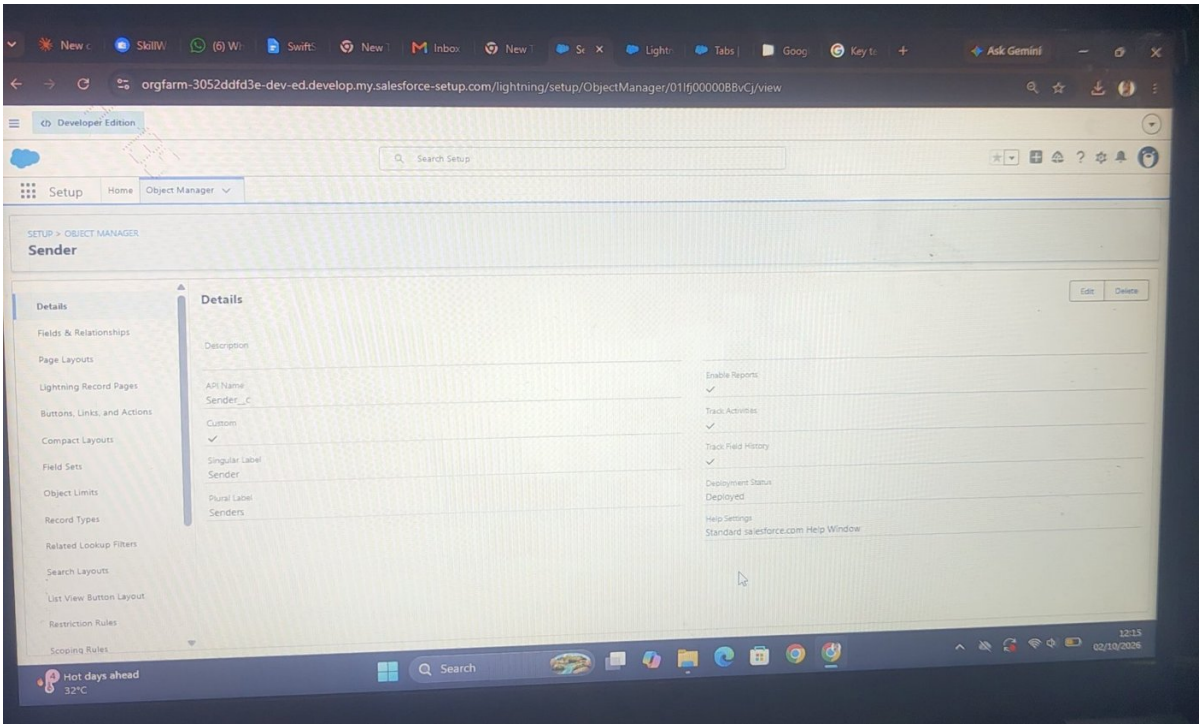


Screenshot 8 - Delivery object (API Name Delivery__c)

Creating Sender Object

To create an object:

1. From the setup page
2. Click on Object Manager
3. Click on Create >> Click on Custom Object.
4. Enter the label name as Sender
5. Enter Plural label name as Sender's
6. Enter Record Name as SenderName
7. Select Data Type as Text: SenderName
8. Select Allow reports.
9. Select Allow search.
10. Allow Track Field History
11. Click on Save and New

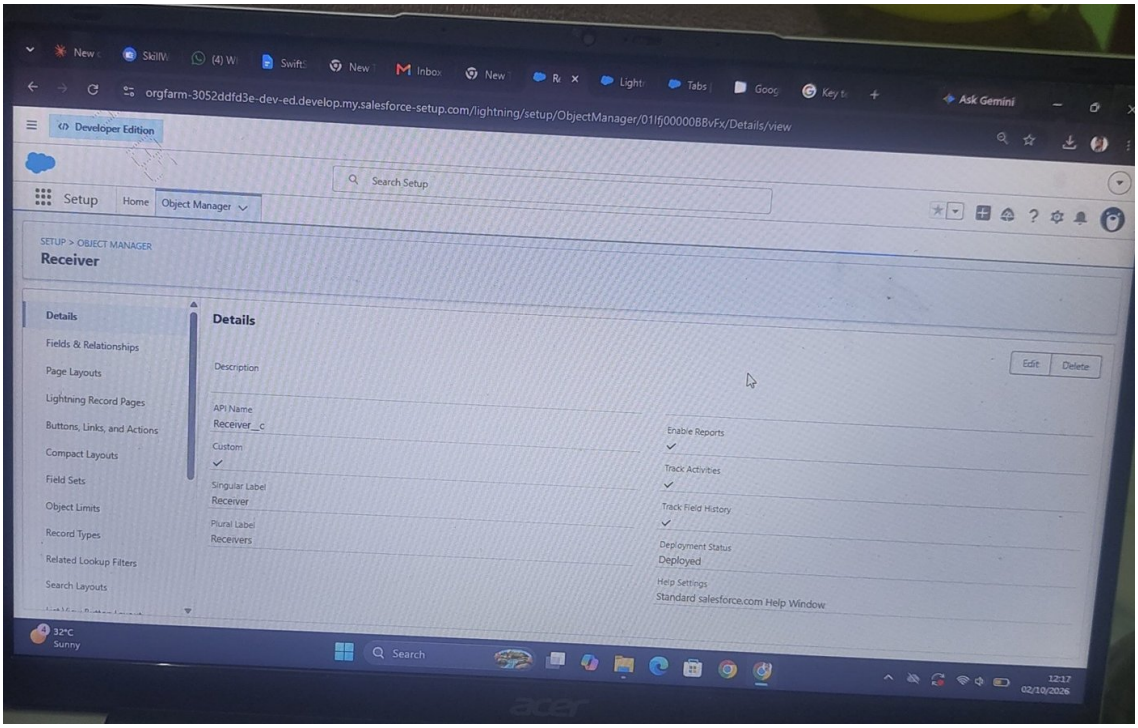


Screenshot 9 - Sender object (API Name Sender__c)

Creating Receiver Object

To create an object:

1. From the setup page
2. Click on Object Manager
3. Click on Create >> Click on Custom Object.
4. Enter the label name as Receiver
5. Enter Plural label name as Receiver's
6. Enter Record Name as ReceiverName
7. Select Data Type as Text: ReceiverName
8. Select Allow reports.
9. Select Allow search.
10. Allow Track Field History
11. Click on Save and New



Screenshot 10 - Receiver object (API Name Receiver__c)

Milestone 3 - Tabs

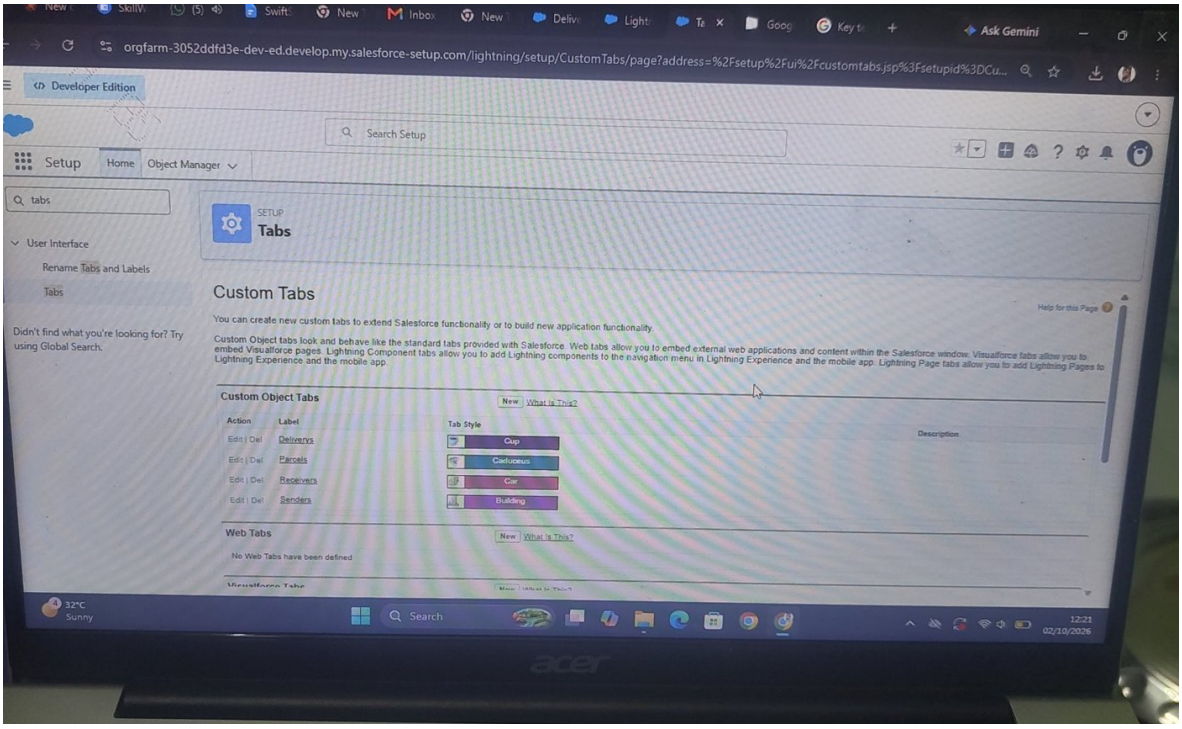
How to do it in Salesforce: A tab gives each object its own menu entry so you can open its records. In Setup, search Tabs, click New under Custom Object Tabs, pick the object and a tab style, and save. Do this once for every object.

Creating a tab for Parcel Object

1. Go to the setup page > type Tabs in Quick Find bar
2. Click on tabs
3. Click on New (under custom object tab).
4. Select Object (Parcel) >> Select the tab style
5. Click on Next >> (Add to profiles page) keep it as default >> Click on Next (Add to Custom App) uncheck the include tab.
6. Make sure that the Append tab to the user's existing personal customizations is checked.
7. Click save

More Tab Creation

1. Go to Setup > type Tabs in the Quick Find box.
2. Click Tabs under the User Interface section.
3. Click New under Custom Object Tabs.
4. Select the object (Delivery, Sender, or Receiver) > choose a Tab Style.
5. Click Next > keep the Add to Profiles settings as default > click Next.
6. On the Add to Custom Apps page, uncheck "Include Tab."
7. Ensure Append tab to the user's existing personal customizations is checked.
8. Click Save and repeat the steps for each object (Delivery, Sender, Receiver).



Screenshot 11 - Custom Object Tabs created: Deliverys (Cup), Parcels (Caduceus), Receivers (Car), Senders (Building)

Milestone 4 - Fields & Relationships

Create fields and Relationships for the custom objects created in the previous milestone.

> **How to do it in Salesforce:** Open Object Manager, select the object, go to Fields & Relationships and click New. Choose the data type (Auto Number, Picklist, Number, Date, Geolocation, Phone, Email or Lookup Relationship), enter the label, click Next, Next, then Save & New. A Lookup field links one object to another.

Creating Fields and Relationships for Parcel Object

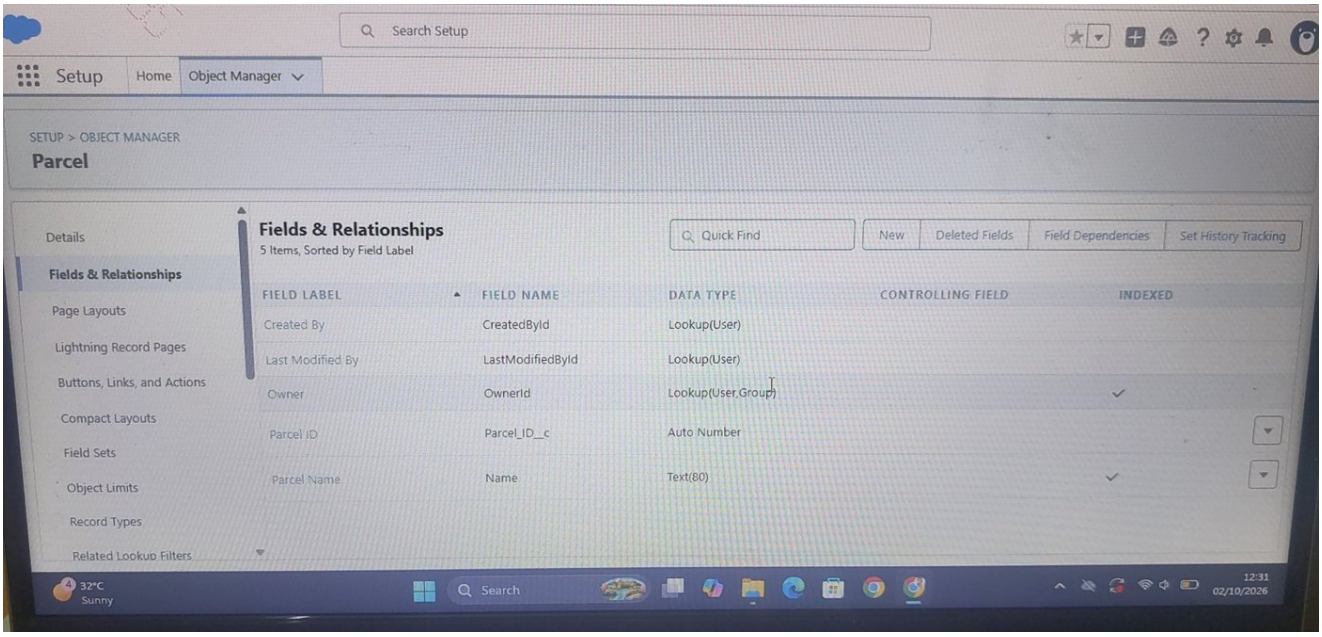
Beside the Quick Find box you have, Object Manager click on it.

1. Search for and select your Parcel object.

Field 1: Parcel ID

- Go to Fields & Relationships > click New
- Data type == auto number > P-{000}
- Click Next > Next > Save & New.

Follow the same steps to create the remaining fields (Status, Weight, Estimated Delivery, Sender) of the Parcel object mentioned in the table below.



Screenshot 12 - Parcel fields: Parcel ID (Auto Number) and Parcel Name (Text 80)

1. Parcel Object - API Name: Parcel__c - Purpose: Store parcel information

Field Name	API Name	Data Type	Description
Parcel ID	Parcel_ID__c	Auto number	Unique ID for each parcel
Status	Status__c	Picklist	[Booked, In Transit, Out for Delivery, Delivered]
Weight	Weight__c	Number	Weight of Parcel
Estimated Delivery	Estimated_Delivery_Date__c	Date	Expected Delivery Date
Sender	Sender__c	Lookup Relationship	Gets Details from Sender

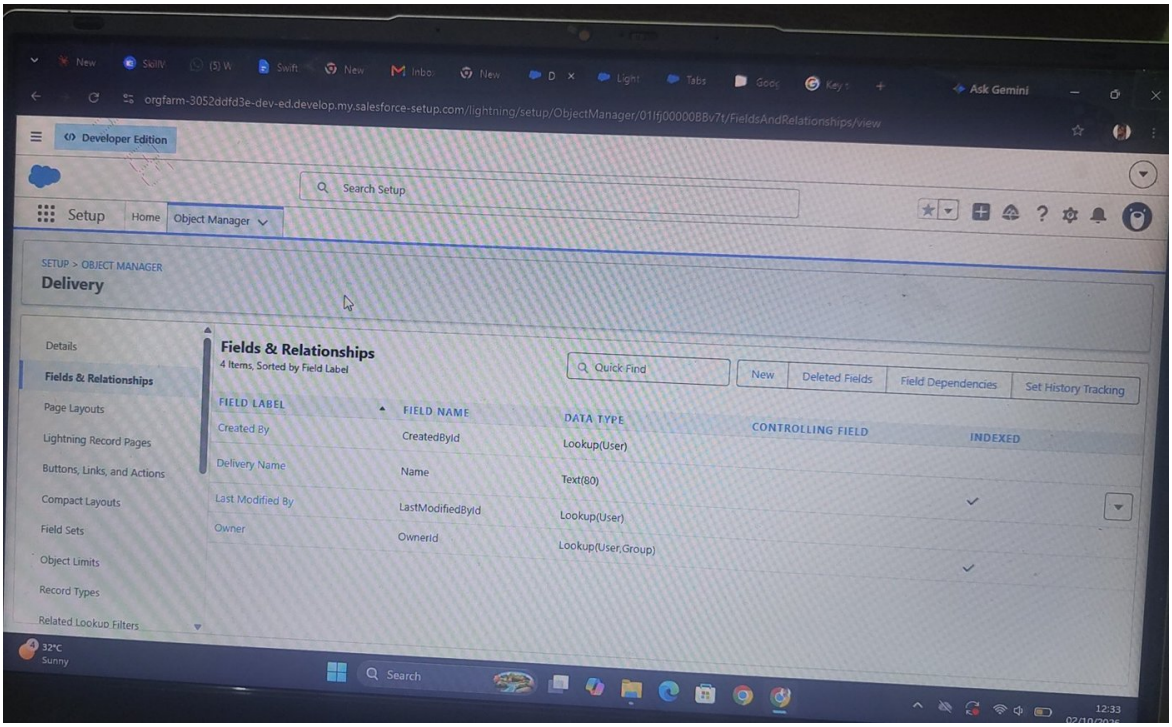
Creating Fields And Relationships For Delivery Object

Beside the Quick Find box you have, Object Manager click on it.

1. Search for and select your Delivery object.

Field 1: Current Location

- Go to Fields & Relationships > click New
- Choose Data Type: Geolocation > click Next. Field Label: Current Location
- Click Next > Next > Save & New.
- Follow the same steps to create the remaining fields (Estimated Delivery, Sender, Parcel) of the Delivery object mentioned in the table below.



Screenshot 13 - Delivery object: Fields & Relationships

Field Name	API Name	Data Type	Description
Current Location	Current_Location__c	Geolocation	Location in terms of longitude and latitude.
Estimated Delivery	Estimated_Delivery_Date__c	Date	Expected Delivery Date
Sender	Sender__c	Lookup Relationship	Gets Details from Sender
Parcel	Parcel__c	Lookup Relationship	Gets Details from Parcel

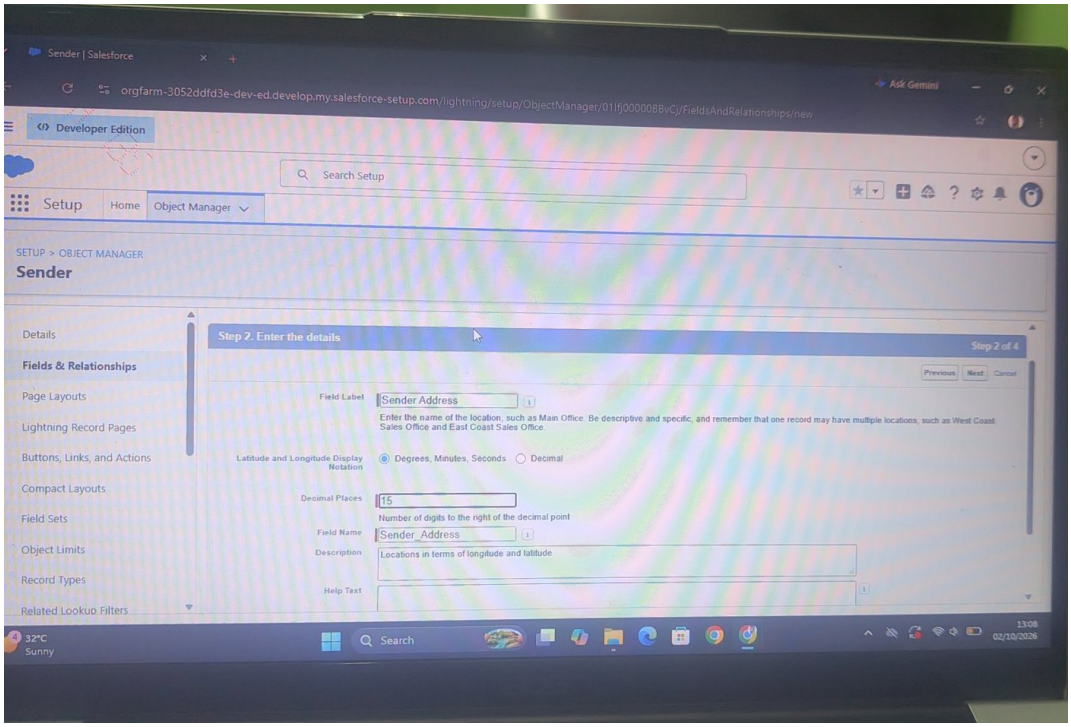
Creating Fields And Relationships For Sender Object

Beside the Quick Find box you have, Object Manager click on it.

1. Search for and select your Sender object.

Field 1: Sender Address

- Go to Fields & Relationships > click New
- Choose Data Type: Geolocation > click Next. Field Label: Sender Address
- Click Next > Next > Save & New.
- Follow the same steps to create the remaining fields (Sender Contact, Sender Email) of the Sender object mentioned in the table below.



Screenshot 14 - New Geolocation field: Sender Address (Decimal Places 15, Field Name Sender_Address)

Field Name	API Name	Data Type	Description
Sender Address	Sender_Address__c	Geolocation	Location in terms of longitude and latitude.
Sender Contact	Sender_Contact__c	Phone	To contact the sender
Sender Email	Sender_Email__c	Email	To send email Notification.

Creating Fields And Relationships For Receiver Object

Beside the Quick Find box you have, Object Manager click on it.

1. Search for and select your Receiver object.

Field 1: Receiver Address

- Go to Fields & Relationships > click New
- Choose Data Type: Geolocation > click Next. Field Label: Receiver's Address
- Click Next > Next > Save & New.
- Follow the same steps to create the remaining fields (Receiver Contact, Receiver Email, Sender, Parcel) of the Receiver object mentioned in the table below.

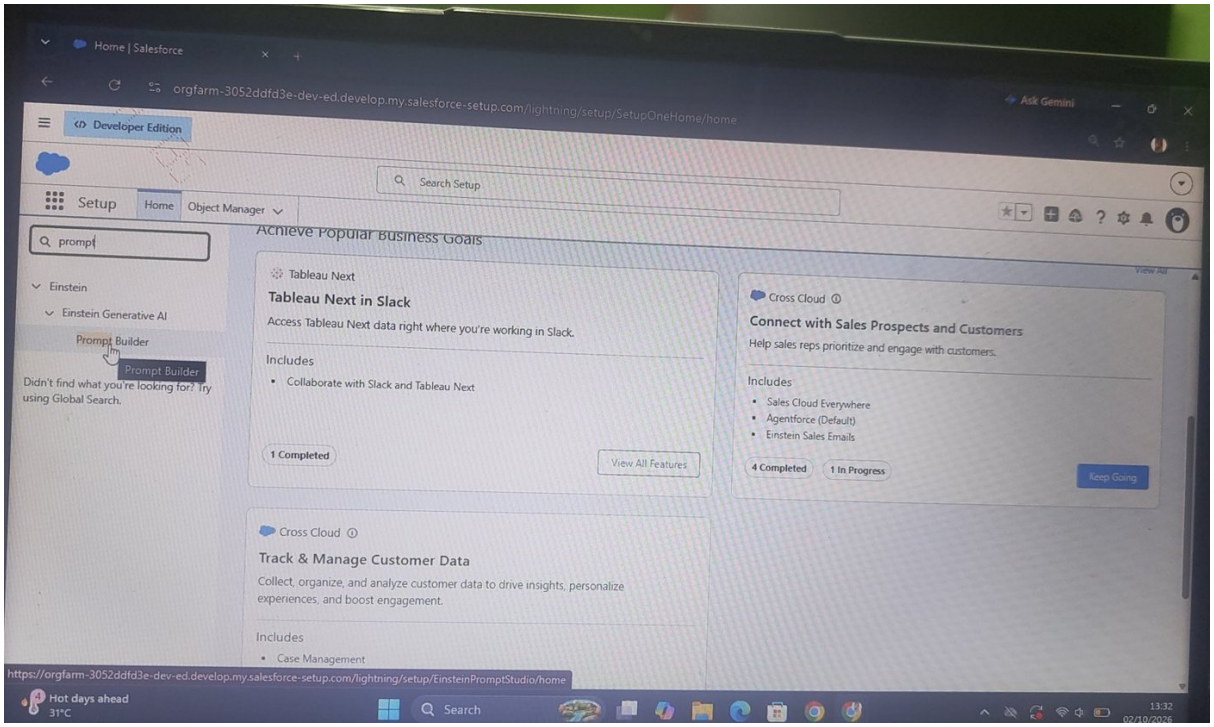
Field Name	API Name	Data Type	Description
Receiver Address	Receiver_Address__c	Geolocation	Location in terms of longitude and latitude.
Receiver Contact	Receiver_Contact__c	Phone	To contact the Receiver
Receiver Email	Receiver_Email__c	Email	To send email Notification.
Sender	Sender__c	Lookup Relationship	Gets Details from Sender
Parcel	Parcel__c	Lookup Relationship	Gets Details from Parcel

Milestone 5 - Prompt Builder

Designing Prompts for Agent Building. In Salesforce, go to Setup.

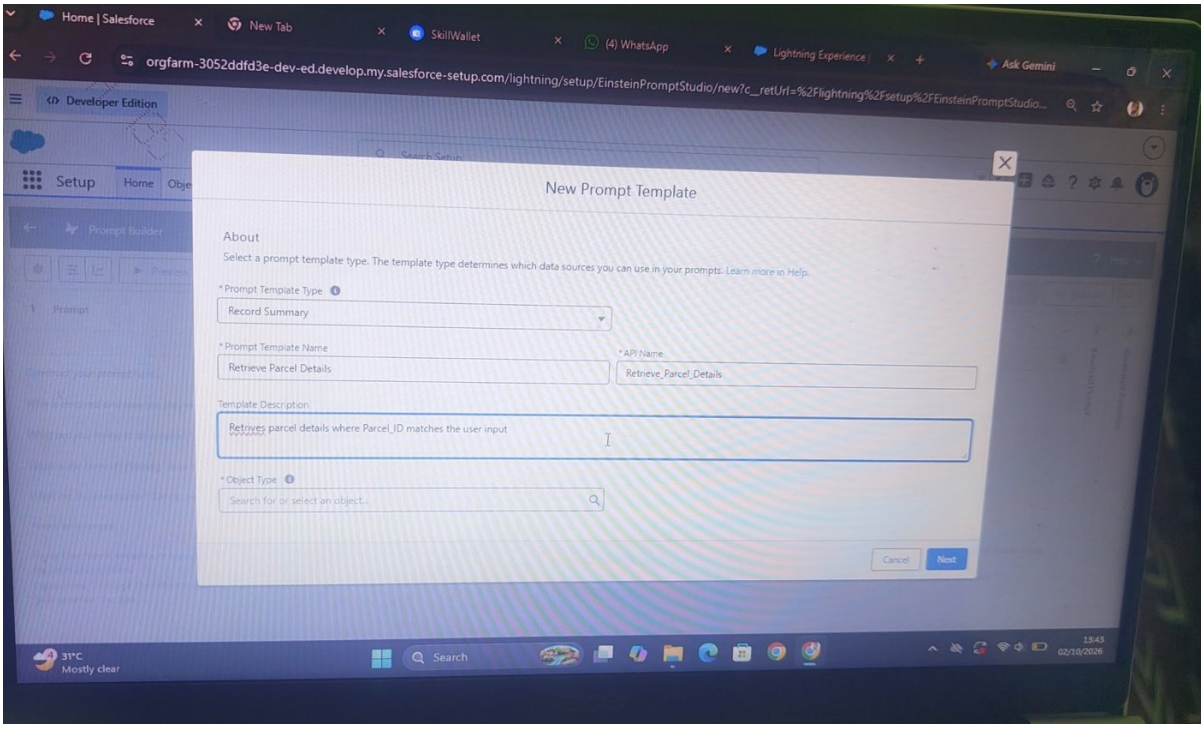
> **How to do it in Salesforce:** Prompt Builder is where you write an AI prompt that uses merge fields from a record. You pick a template type, write the prompt, insert the object fields as resources, preview the AI answer and then activate the template so a flow or agent can call it.

1. In the Quick Find box, type Prompt Templates (or in Agent Console, go to Agent Builder > Prompt Templates).
2. Click New Prompt Template - Record summary template



Screenshot 15 - Quick Find: Prompt Builder

4. Enter the details
    - Label: Retrieve Parcel Details
    - API Name: Retrieve_Parcel_Details
    - Description: Retrieves parcel details where Parcel_ID matches the user input.
    - Object Type: Parcel, then click Next



Screenshot 16 - New Prompt Template (Record Summary)

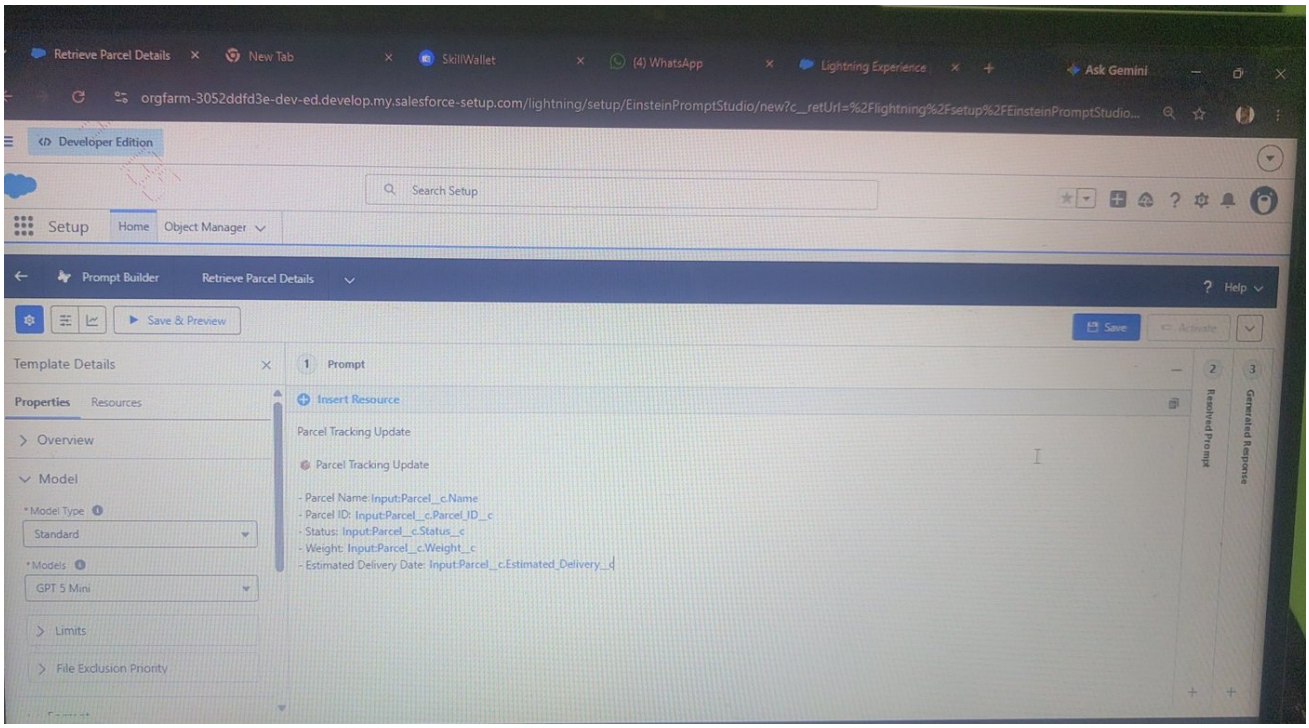
5. Enter the Prompt Template Content

In the Prompt Template Text area, paste the following exactly:

```
Retrieve details from Parcel__c where Parcel_ID__c matches user input. Return response in this
format:

Parcel Tracking Update
- Parcel Name:{!$Input:Parcel__c.Name}
- Parcel ID:{!$Input:Parcel__c.Parcel_ID__c}
- Status:{!$Input:Parcel__c.Status__c}
- Weight:{!$Input:Parcel__c.Weight__c}
- Estimated Delivery Date:{!$Input:Parcel__c.Estimated_Delivery_Date__c}
```

Model Type: Standard, Model: GPT 5 Mini.

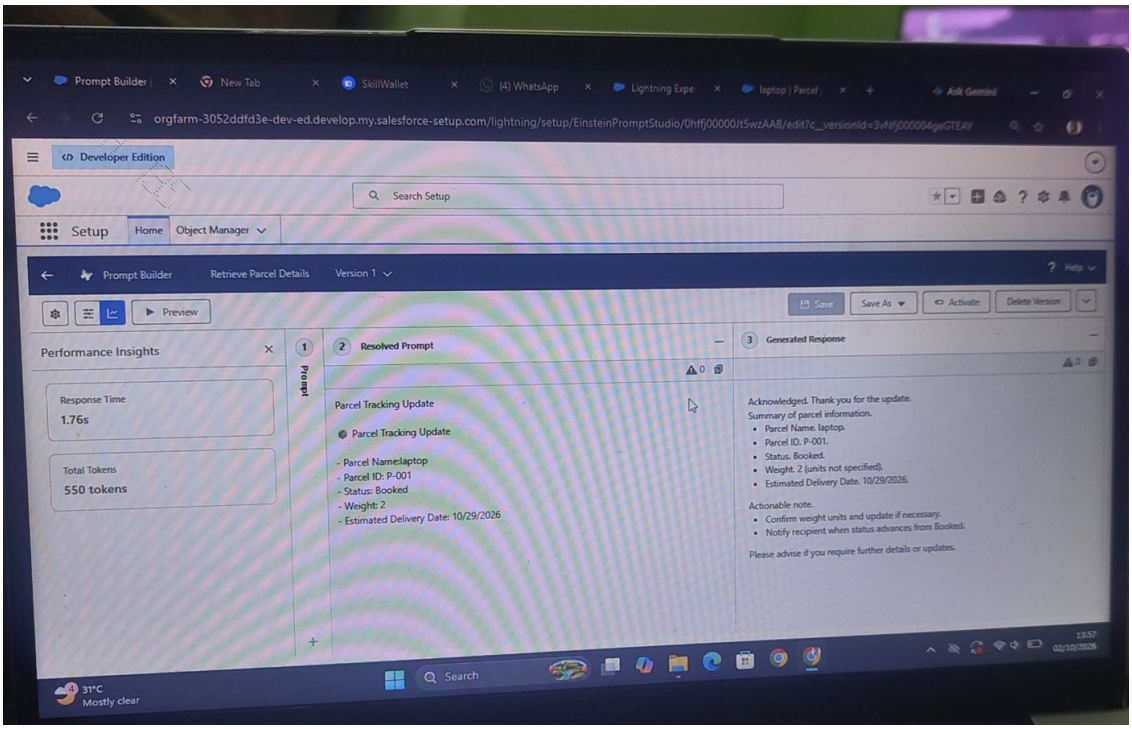


Screenshot 17 - Prompt content with Parcel merge fields and model settings

6. Add Resource Mapping
7. Scroll down to the Resources / Insert Resources section.
   - Click Insert Resource > choose Object.
   - Select Parcel__c.
   - Add the following fields: Name, Parcel_ID__c, Status__c, Weight__c, Estimated_Delivery_Date__c
8. Click Insert Resource (or Add) for each one.
9. Click Save.

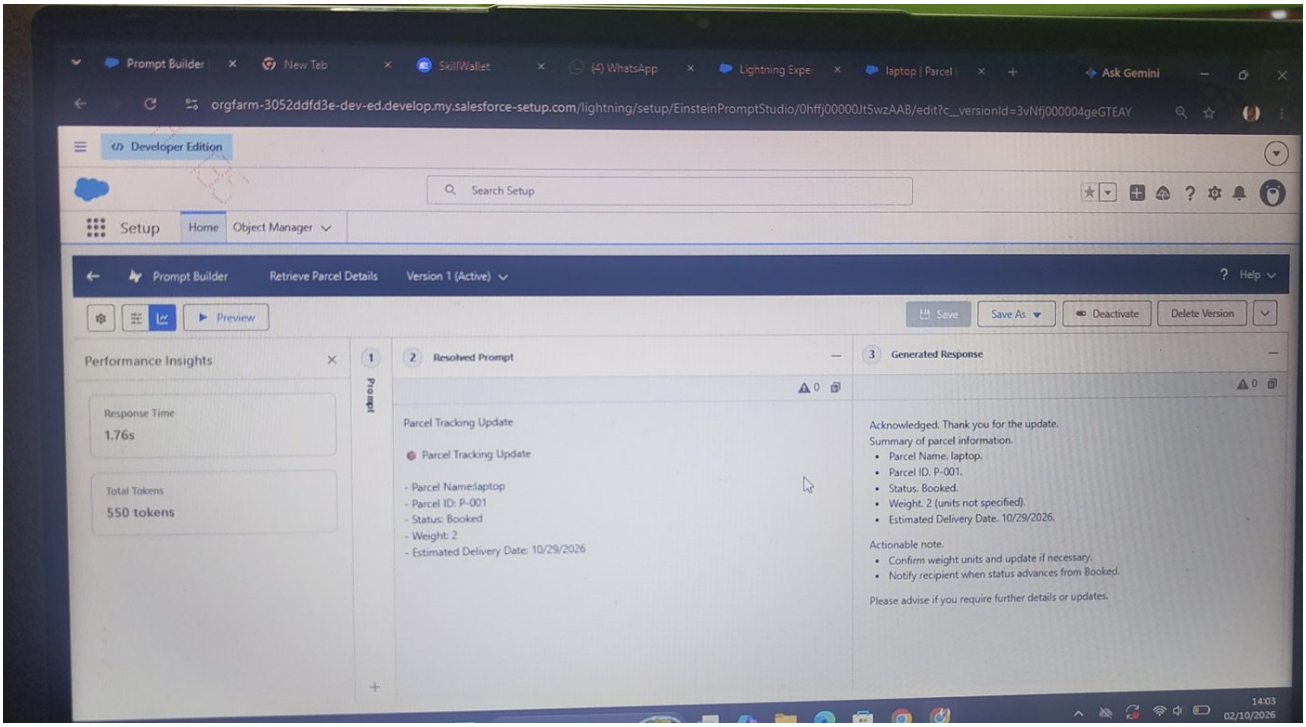
Testing the Prompt

Click **Save & Preview**. The Resolved Prompt shows the real parcel values (laptop, P-001, Booked, weight 2, estimated delivery 10/29/2026) and the Generated Response summarizes the parcel details. Performance Insights show a response time of 1.76 s and 550 tokens.



Screenshot 18 - Save & Preview: Resolved Prompt and Generated Response

10. Then click Activate to make this Prompt Template available for your agent or flow.



Screenshot 19 - Prompt template activated (Version 1 Active)

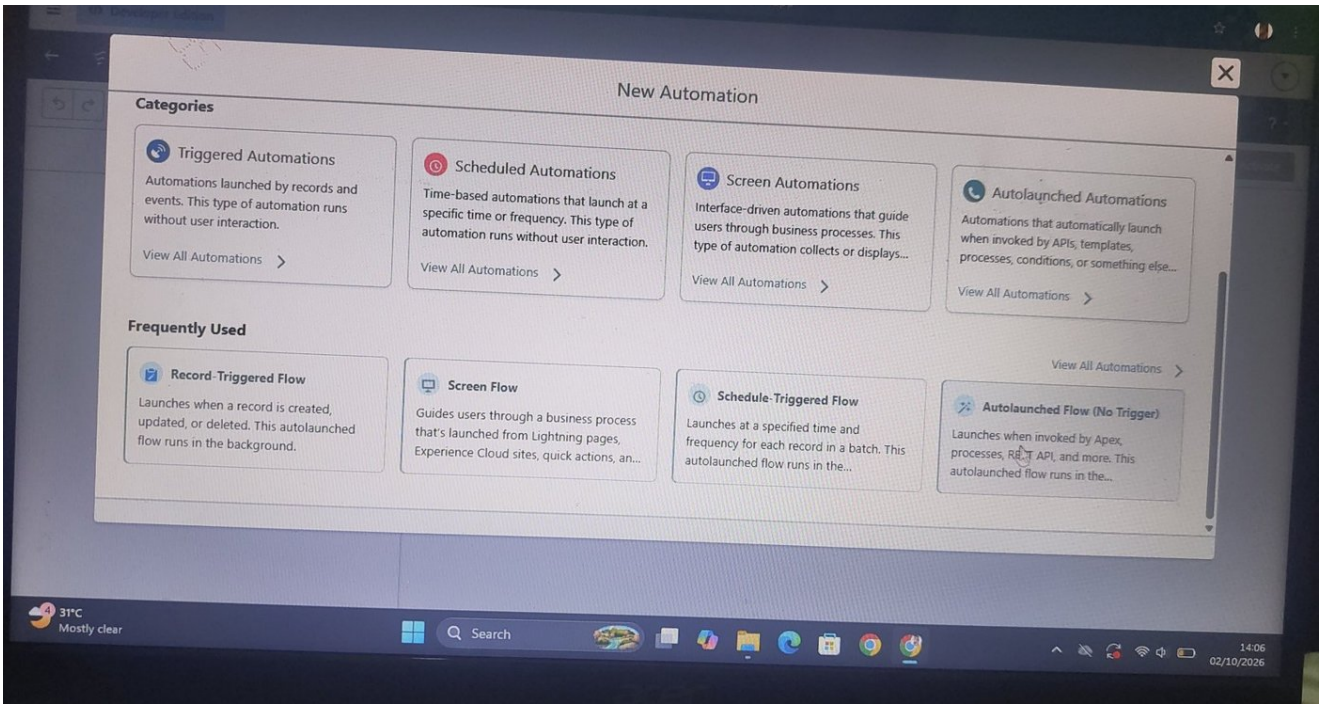
Milestone 6 - Flows

Create flows for capturing the response as shown.

> **How to do it in Salesforce:** A flow is Salesforce's drag-and-drop automation. This Autolaunched Flow takes a Parcel ID as input, finds the parcel record, sends it to the prompt template, and returns the AI response in an output variable.

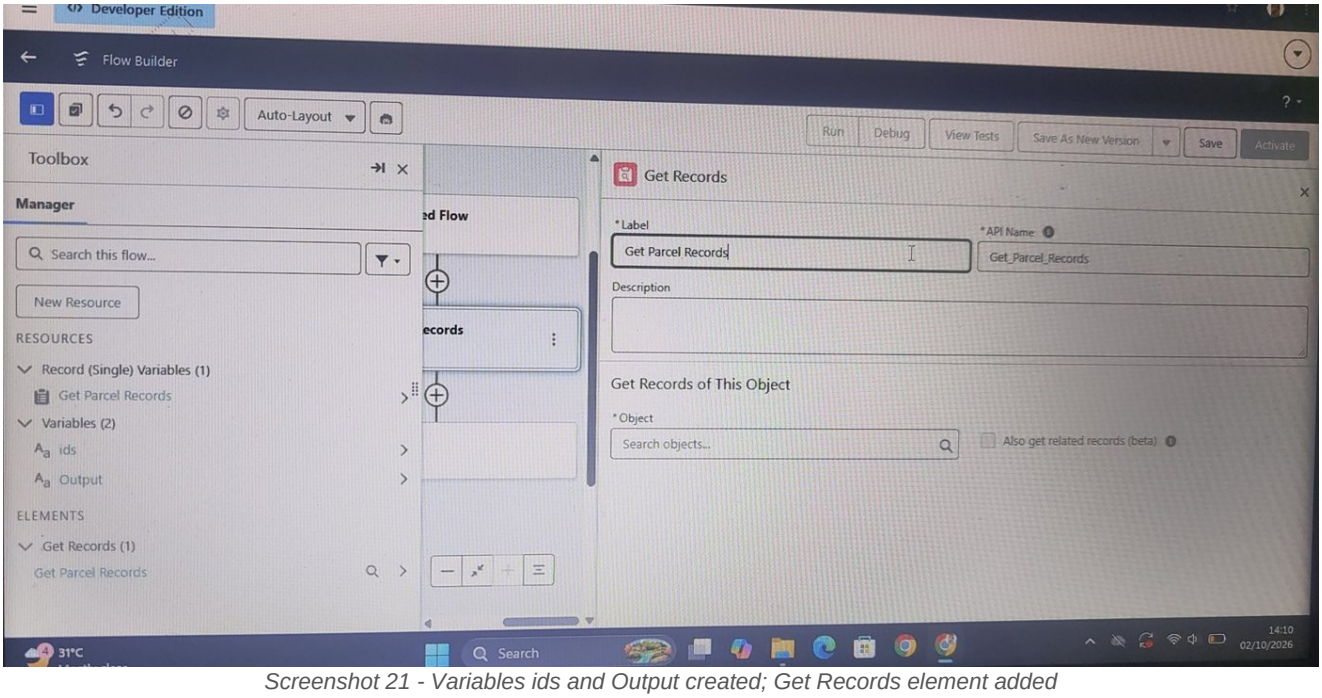
Go to Setup > search Flows > click New Flow.

1. Choose Auto-Launched Flow (No Trigger) > click Create



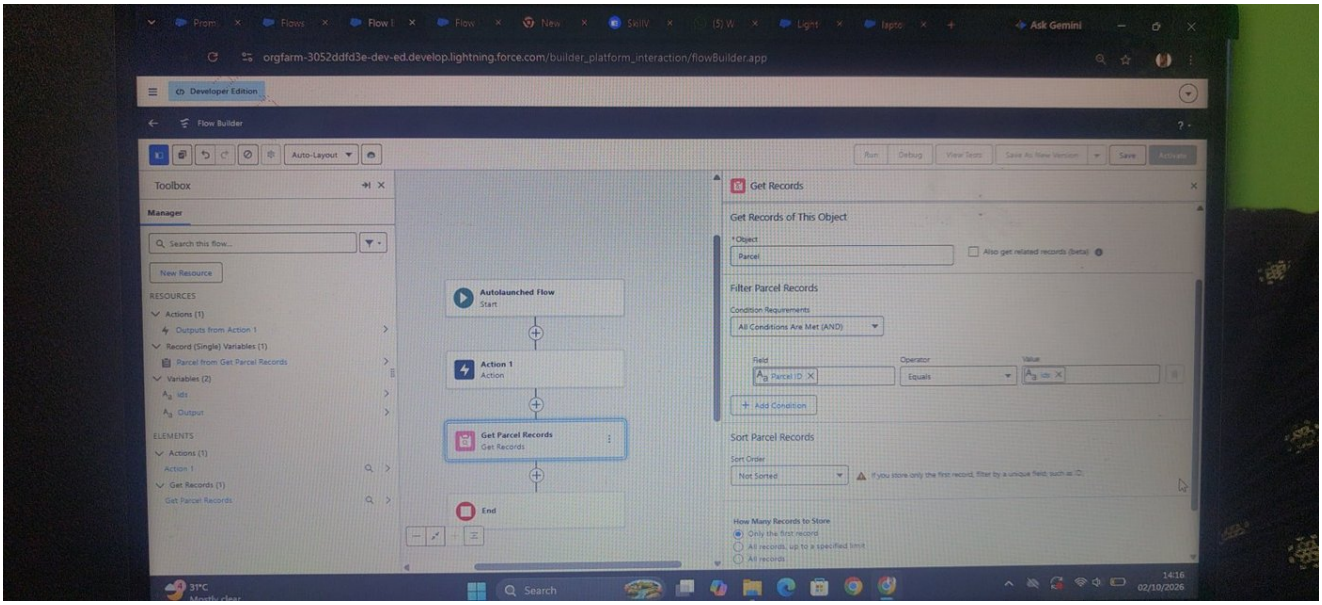
Screenshot 20 - New Automation: Autolaunched Flow (No Trigger)

3. Add Input Variables (to allow Agent to pass values):
   - Click On New Resource > Variable > Available for Input.
   - Ids > Data Type: Text > Available for Input.
   - Output > Data Type: Text > Available for Output.



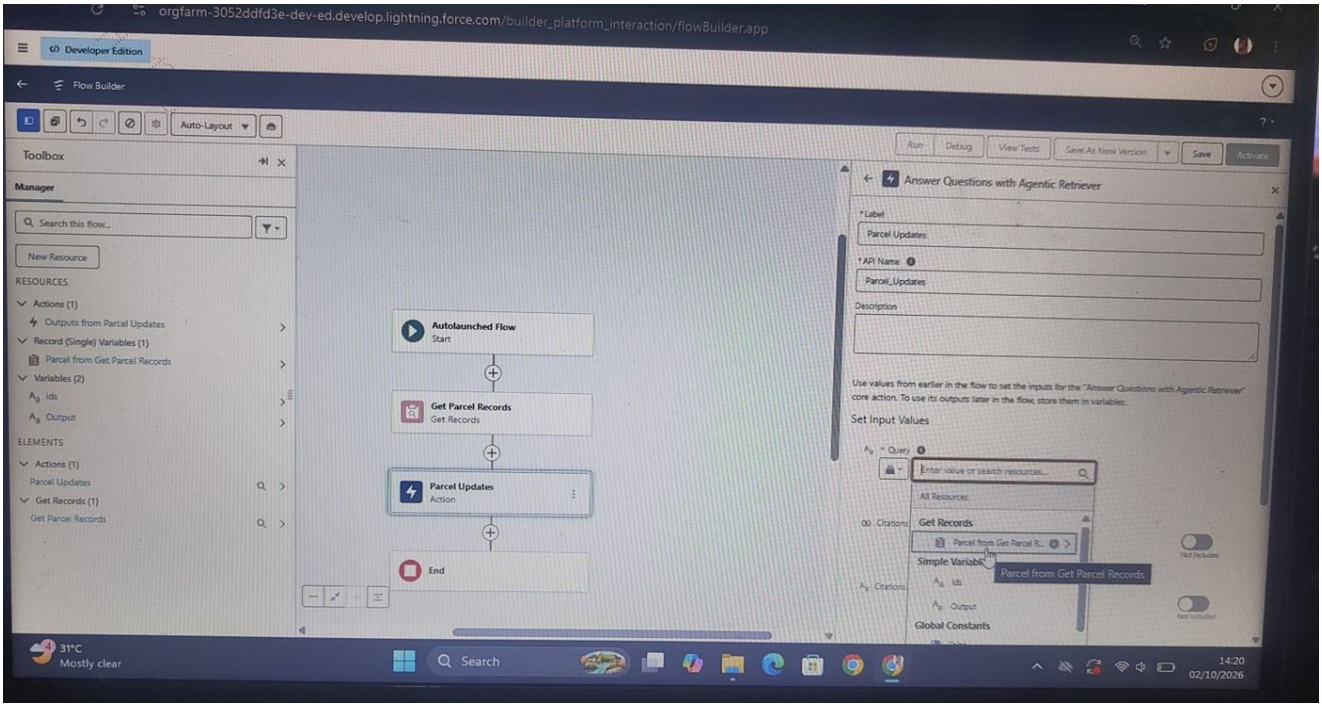
Screenshot 21 - Variables ids and Output created; Get Records element added

4. Click on the Add icon to add Element: Get Records > Label: Get Parcel Records
    - Object: Parcel
    - Field = Parcel_ID__c
    - Operator = Equals
    - Value = ids



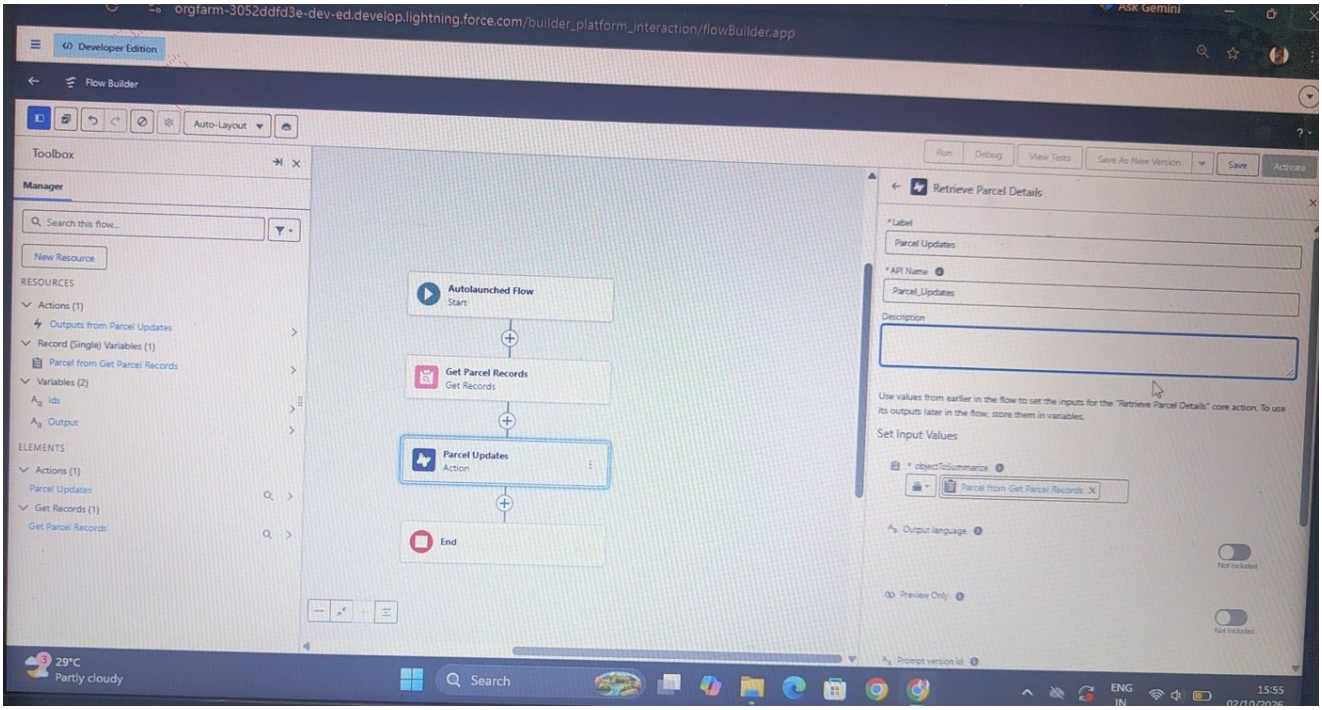
Screenshot 22 - Get Parcel Records filter

5. Element : Action
    - Find Retrieve parcel details
    - Click Retrieve parcel details
    - Label: Parcel Updates.



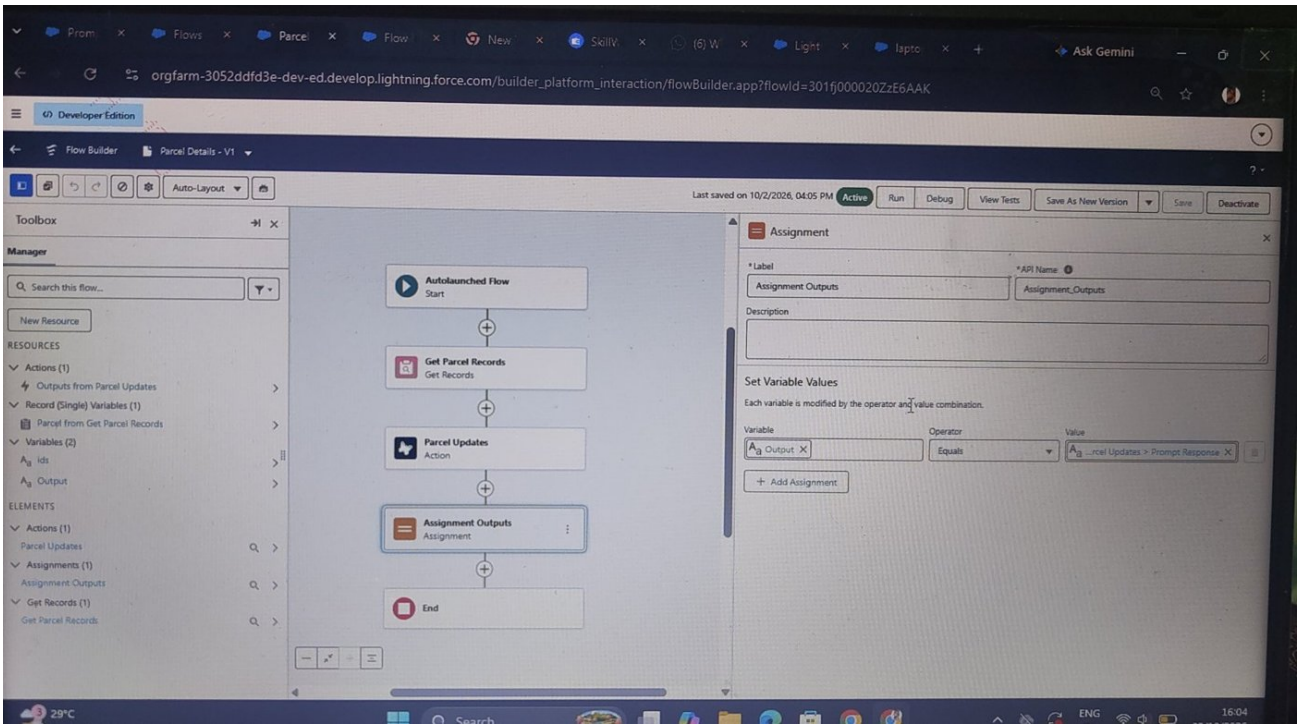
Screenshot 23 - Action added; Parcel from Get Parcel Records is available

- Object To Summarize : Parcel from get Parcel records.



Screenshot 24 - objectToSummarize set

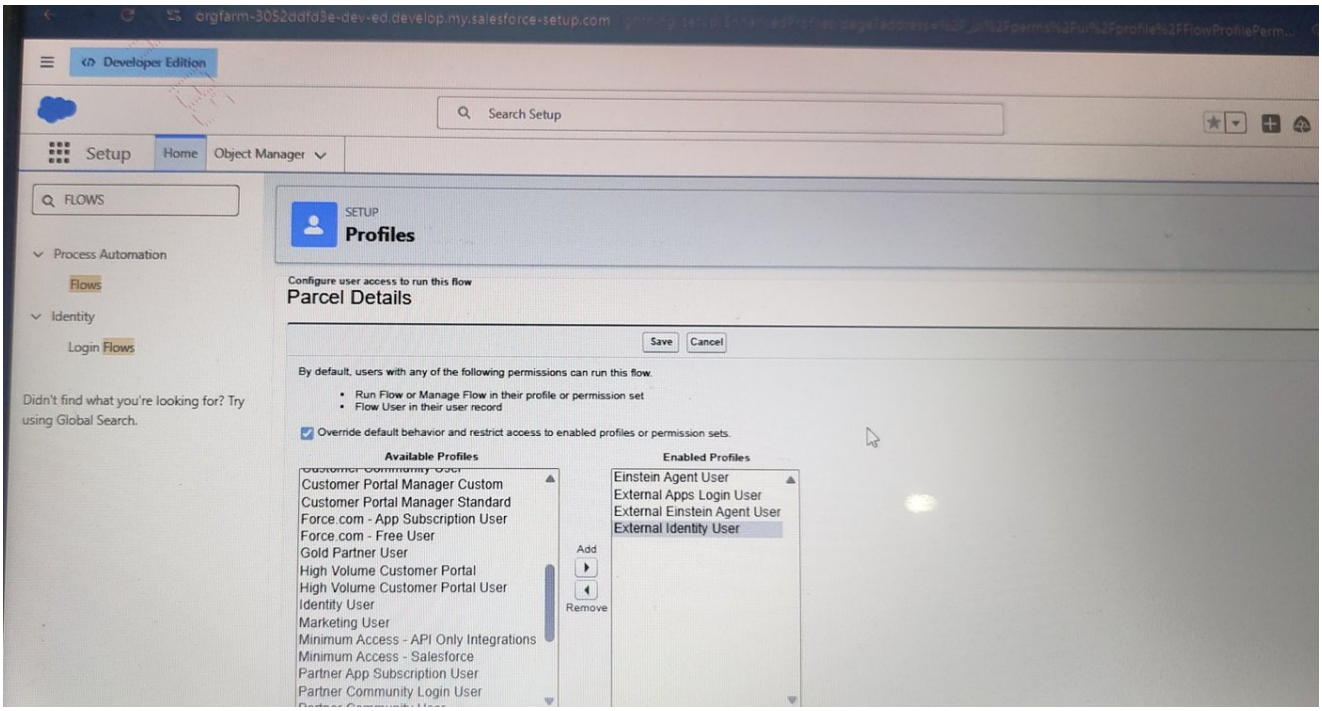
6. Element : Assignment
   - Label: Assignment Outputs.
   - Variable = Output
   - Operator = Equals
   - Value = Outputs from Action > Prompt Response



Screenshot 25 - Assignment Outputs; flow Parcel Details - V1 Active

7. Connect Start > Get Records > Action > Assignment > End.
8. Save Flow > Name: Parcel Details.
9. Activate Flow.
10. Once the flow is saved go back to the flows, Click on the drop down near the flow that is just created, click on edit access.

11. On the top look for flow user access and then assign these available profiles, and click save.



Screenshot 26 - Flow user access: enabled profiles

Phase 3: UI/UX Development & Customization

Milestone 1: Creating Permission Sets

Objective: To create a permission set named **Swift Ship** that gives the Agentforce agent user the access it needs to the SwiftShip custom objects (Parcel, Delivery, Sender and Receiver) and their key fields.

Step 1: Open Permission Sets and click New

- Go to **Setup** and type **Permission Sets** in the Quick Find box.
- Click **Permission Sets** under the **Users** section.
- On the Permission Sets page, click the **New** button.

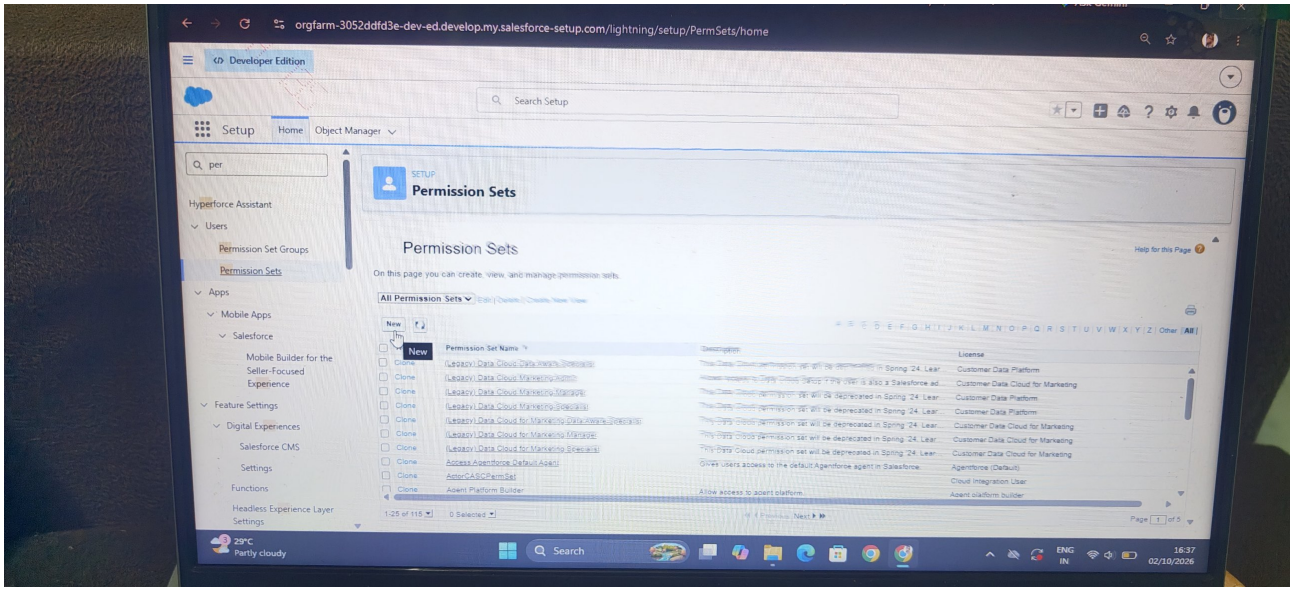


Fig 1: Permission Sets page with the New button

Step 2: Create the Swift Ship permission set

• Enter Label: **Swift Ship**. The API Name auto-fills as **Swift_Ship**.

• Select License: **Einstein Agent**.

• Click **Save**. The Permission Set Overview page opens.

• Under the **Apps** section, click **Object Settings** to set object access.

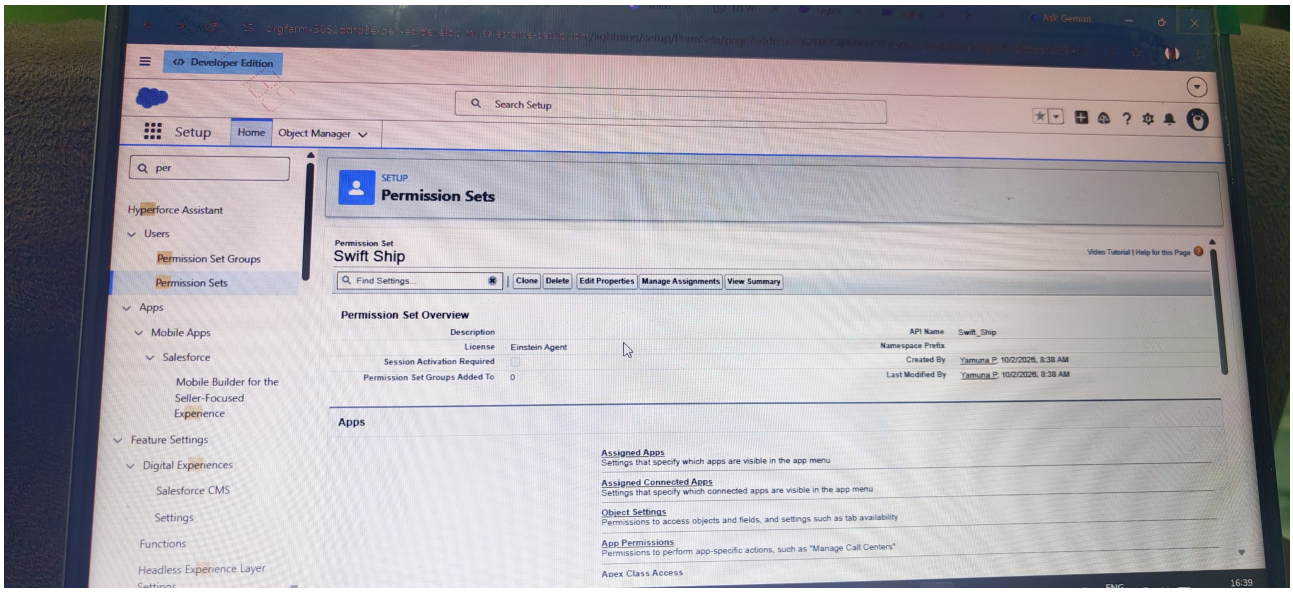


Fig 2: Swift Ship permission set overview (License: Einstein Agent)

Step 3: Configure the Parcel object

• In Object Settings, search for and click **Parcels**, then click **Edit**.

• Under Object Permissions, enable **Read, Create** and **Edit**. Delete is optional.

• Under Field Permissions, tick **Read Access** and **Edit Access** for the key fields: Parcel ID, Status, Weight, Estimated Delivery and Sender.

• Click **Save**.

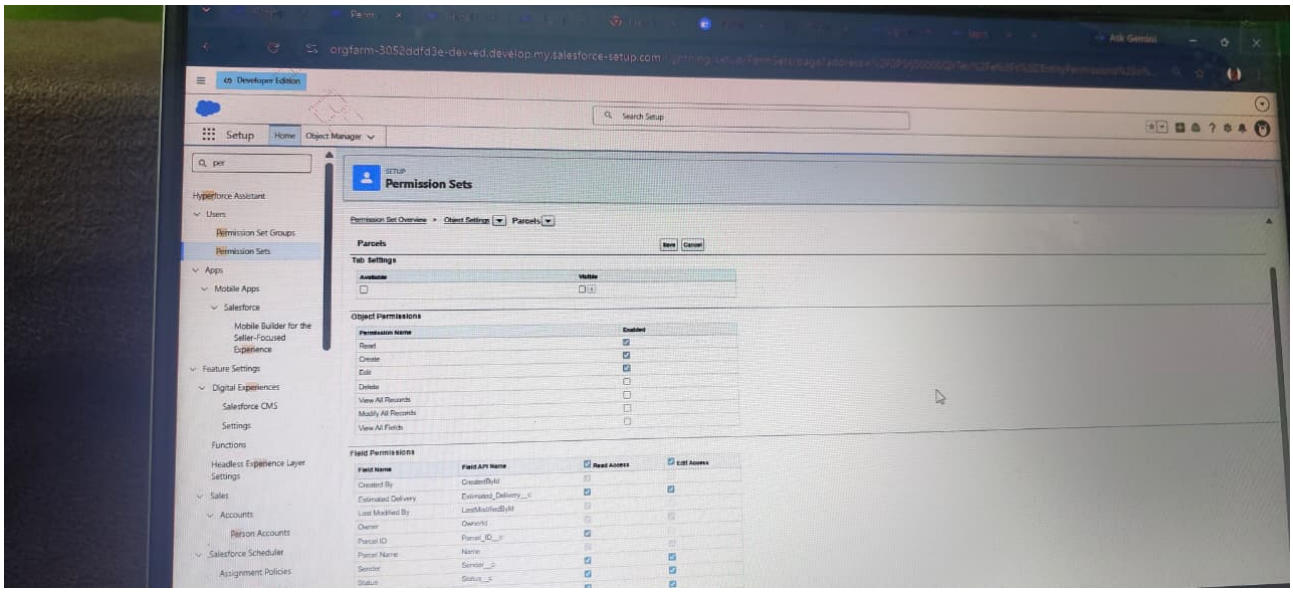


Fig 3: Object and field permissions for Parcel__c

Step 4: Configure the Delivery object

- Go back to Object Settings and click **Deliverys**, then click **Edit**.
- Enable **Read, Create** and **Edit** under Object Permissions.
- Give Read and Edit Access to Current Location, Estimated Delivery, Parcel and Sender.
- Click **Save**.

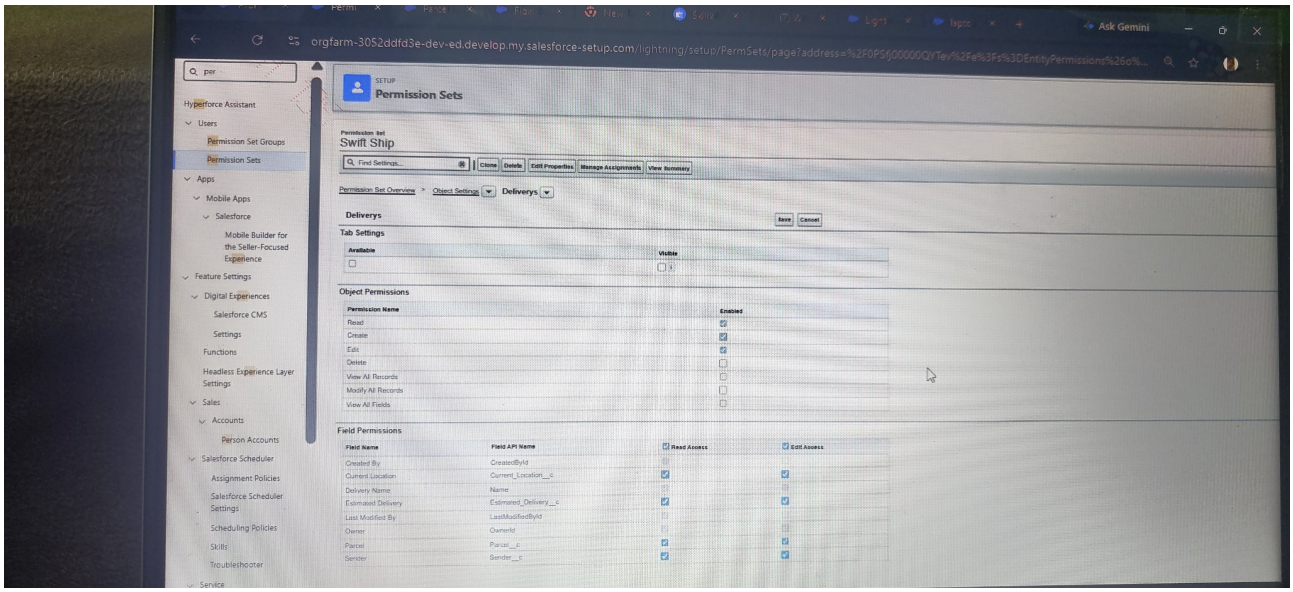


Fig 4: Object and field permissions for Delivery__c

Step 5: Configure the Sender object

- Go back to Object Settings and click **Senders**, then click **Edit**.
- Enable **Read, Create** and **Edit** under Object Permissions.
- Give Read and Edit Access to Sender Address, Sender Contact and Sender Email.
- Click **Save**.

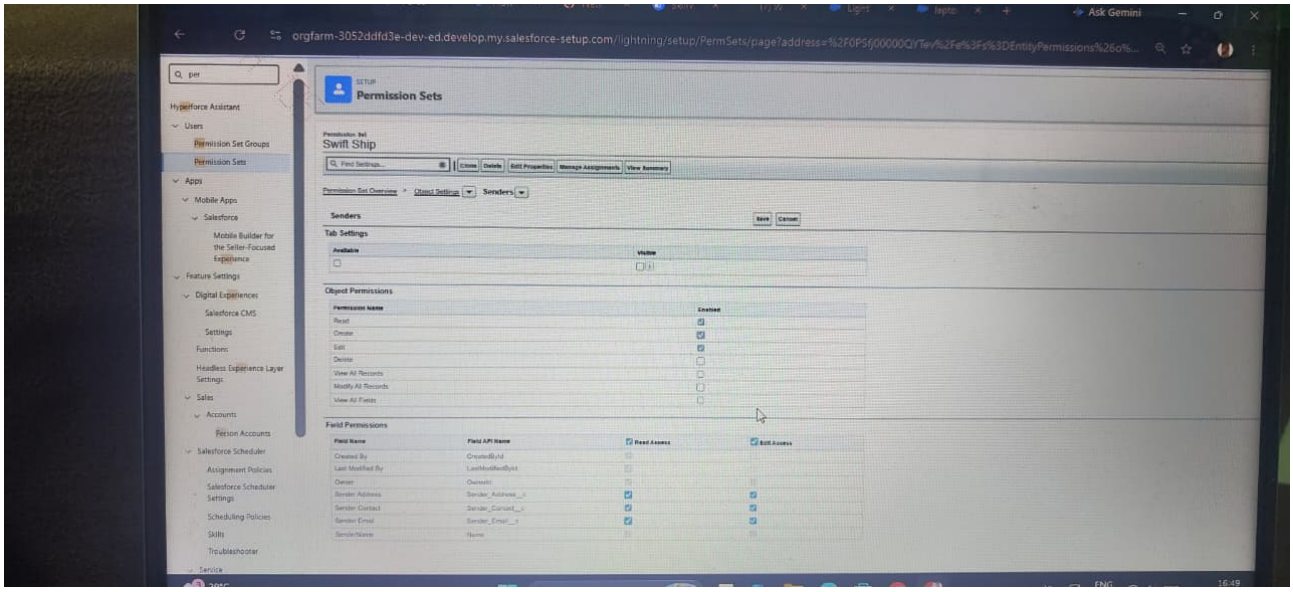


Fig 5: Object and field permissions for Sender__c

Step 6: Configure the Receiver object

• Go back to Object Settings and click **Receivers**, then click **Edit**.

• Enable **Read, Create** and **Edit** under Object Permissions.

• Give Read and Edit Access to Receiver's Address, Receiver Contact, Receiver Email, Parcel and Sender.

• Click **Save**.

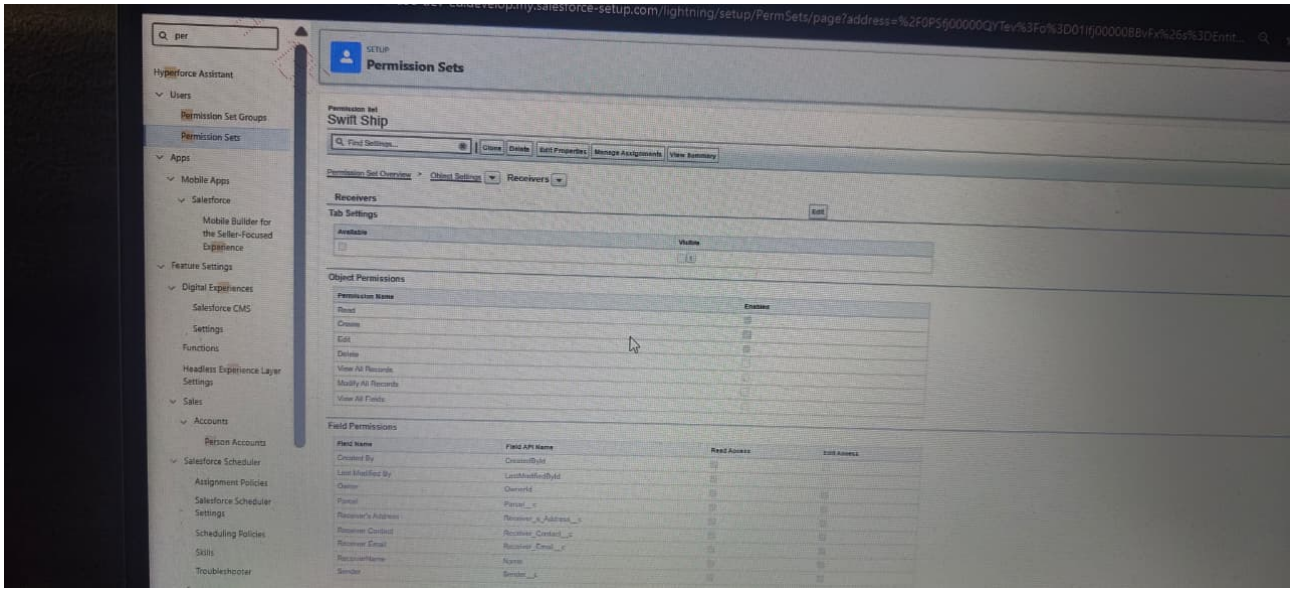


Fig 6: Object and field permissions for Receiver__c

Result

The **Swift Ship** permission set now grants Read, Create and Edit access on Parcel, Delivery, Sender and Receiver, with field-level access to their key fields. In Milestone 2 this permission set is assigned to the Agent user.

Phase 3: UI/UX Development & Customization

Milestone 2: Assigning Permissions to the Agent User

Objective: To assign the **Swift Ship** permission set to the Einstein Service Agent user, so the Agentforce agent can read and update Parcel, Delivery, Sender and Receiver records.

Step 1: Open the Agent user

• Go to **Setup**, type **Users** in Quick Find, and click **Users** under the Users section.

• In the All Users list, find **EinsteinServiceAgent User** (Alias: einstein, Profile: Einstein Agent User).

• Click the Full Name to open the user record.

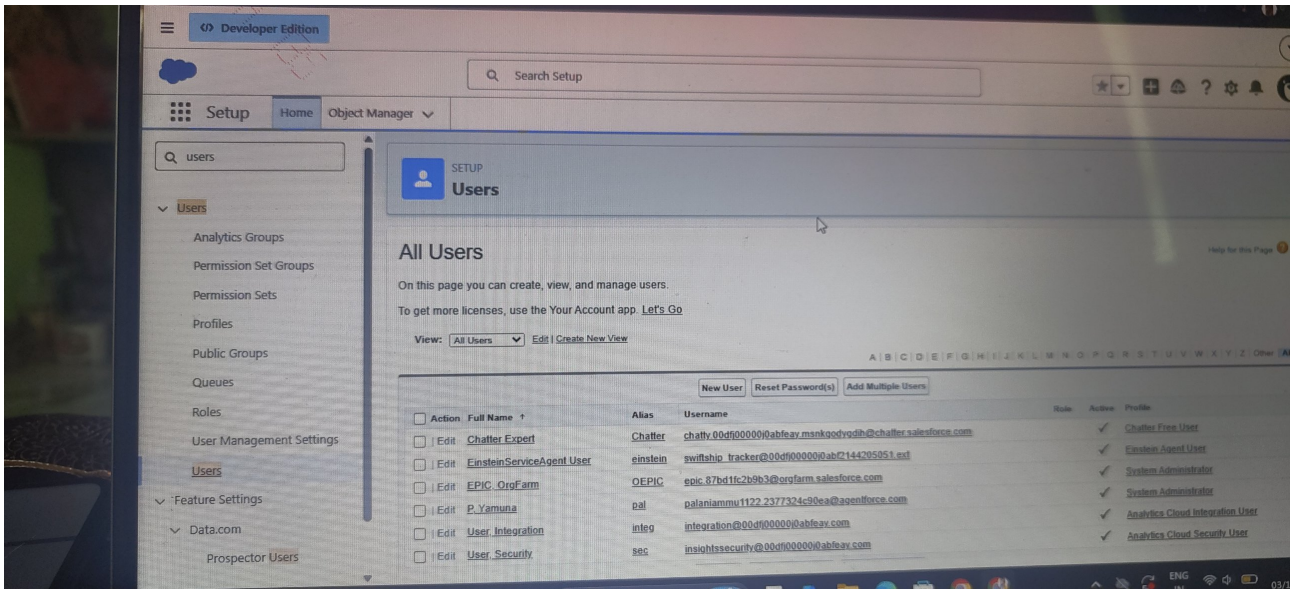


Fig 1: All Users list showing EinsteinServiceAgent User

Step 2: Add the Swift Ship permission set

• On the user record, scroll to **Permission Set Assignments** and click **Edit Assignments**.

• In **Available Permission Sets**, select **Swift Ship**.

• Click **Add** to move it to **Enabled Permission Sets**.

• Click **Save**.

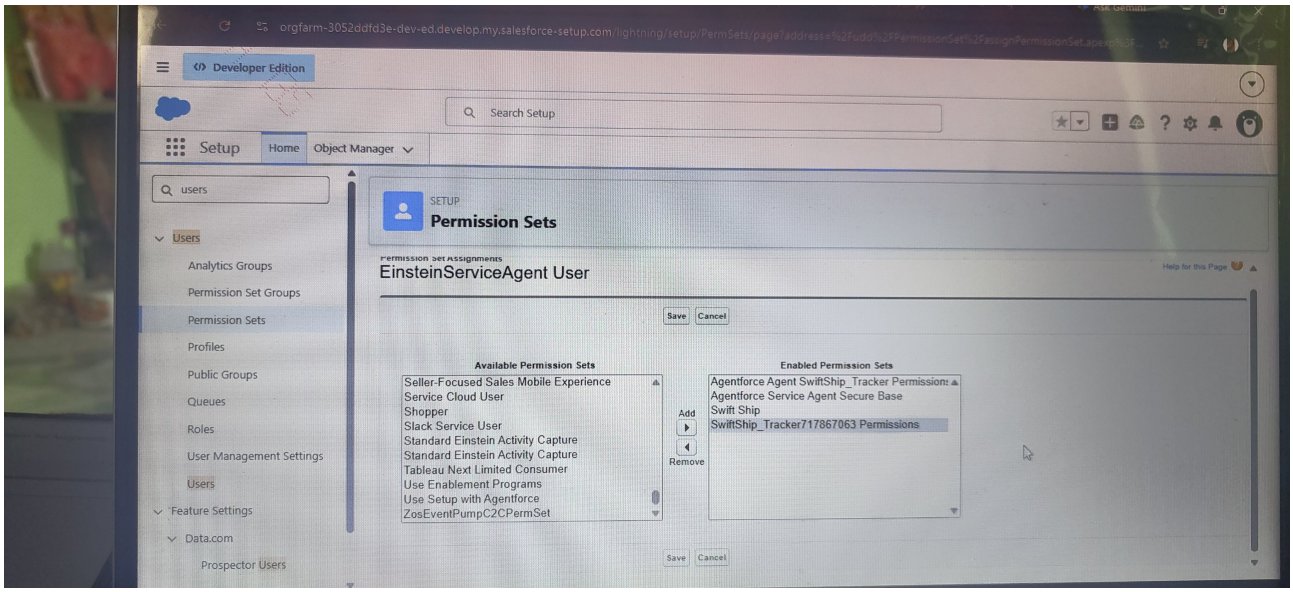


Fig 2: Edit Assignments page with Swift Ship in Enabled Permission Sets

Step 3: Verify the assignment

• Reopen the user record and scroll to **Permission Set Assignments**.

• Confirm that **Swift Ship** is listed along with **Agentforce Service Agent Secure Base**, **Agentforce Agent SwiftShip_Tracker Permissions** and **SwiftShip_Tracker717867063 Permissions**, all assigned on 10/2/2026.

• The **Agentforce Service Agent User** permission set group is also assigned.

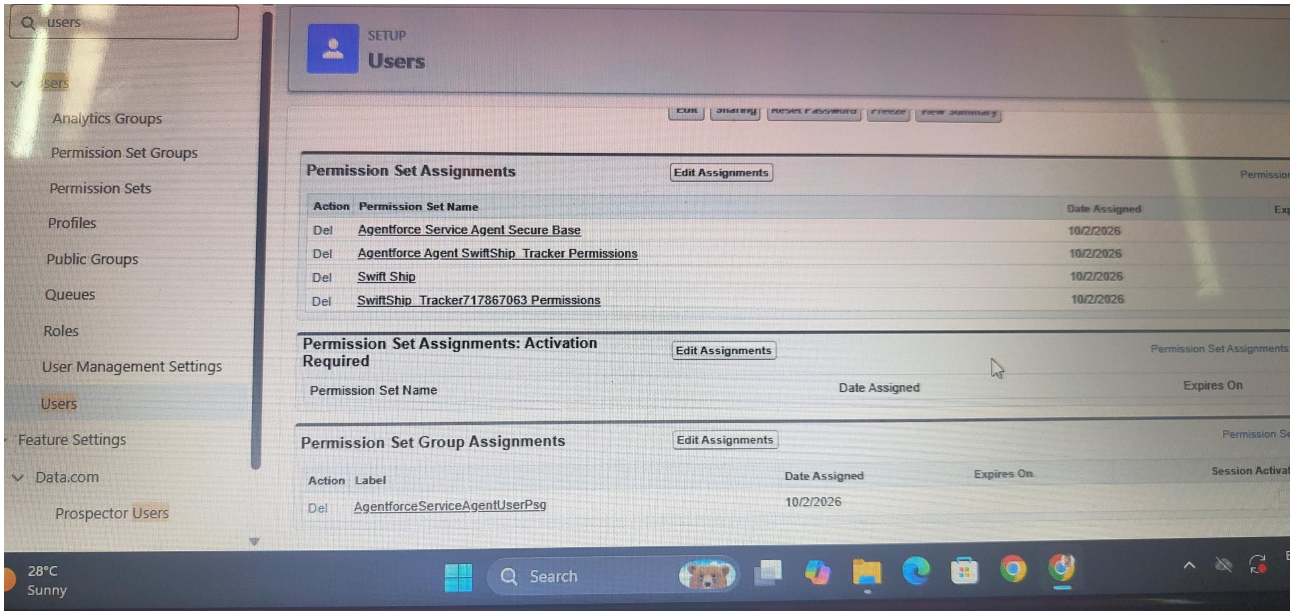


Fig 3: Permission Set Assignments for EinsteinServiceAgent User

Result

The **Swift Ship** permission set is now assigned to the EinsteinServiceAgent User. The agent has Read, Create and Edit access to the SwiftShip objects and their key fields, which is required before the agent is built in Milestone 4.

Phase 3: UI/UX Development & Customization

Milestone 3: Org Setup for Agentforce

Objective: To enable the Salesforce features required to build and run the Agentforce agent: the Default Agent, Einstein generative AI and Agentforce.

Step 1: Enable the Default Agent from Salesforce Go

- In Setup, type **Salesforce Go** in the Quick Find box and click it.
- Search for **Default Agent** and click **Get Started**.
- Turn the Default Agent **On**, then refresh the page.

Step 2: Enable Einstein

- In Setup, type **Einstein Setup** in the Quick Find box and click it. Click **Get Started** if prompted.
- Turn on **Turn on Einstein** to enable generative AI on your Salesforce data.
- Under **Prompt Builder Settings**, turn on **Global Languages** and **Deploy Prompt Templates**.
- Refresh the browser page.

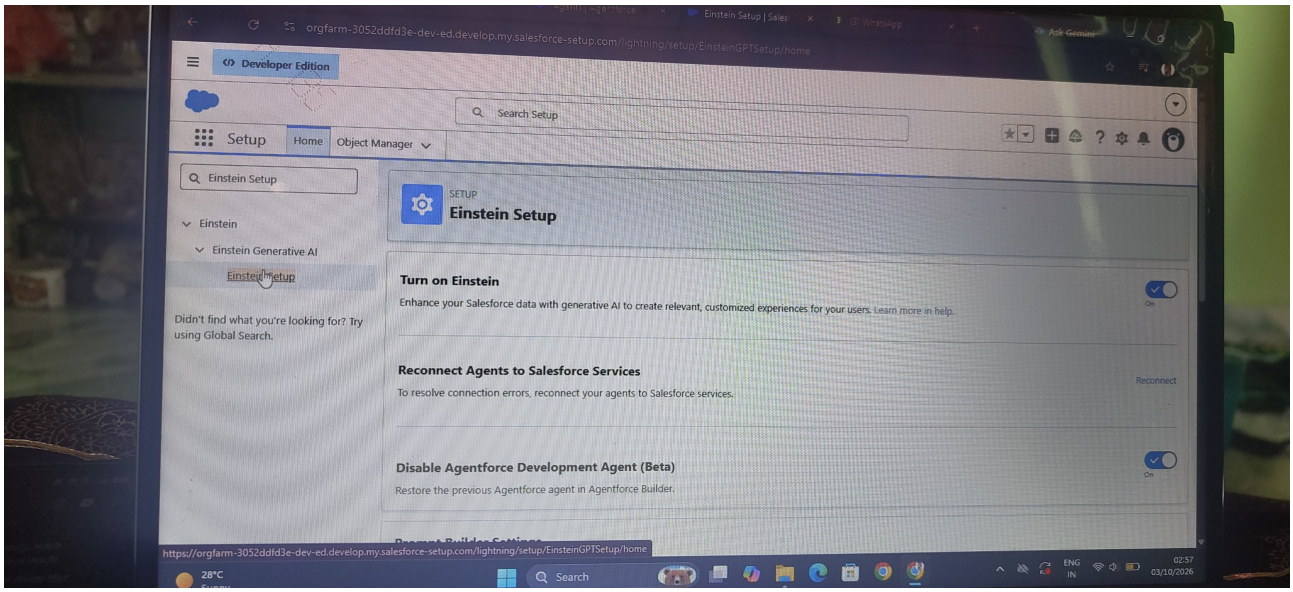


Fig 1: Einstein Setup with Turn on Einstein enabled

Step 3: Enable Agentforce

• In Setup, type **Agentforce Agents** in the Quick Find box and click it (Einstein > Einstein Generative AI > Agentforce Studio > Agentforce Agents).

• Turn the **Agentforce** toggle **On** at the top right.

• Confirm that **Agentforce (Default)** appears in the agents list (Type: Employee).

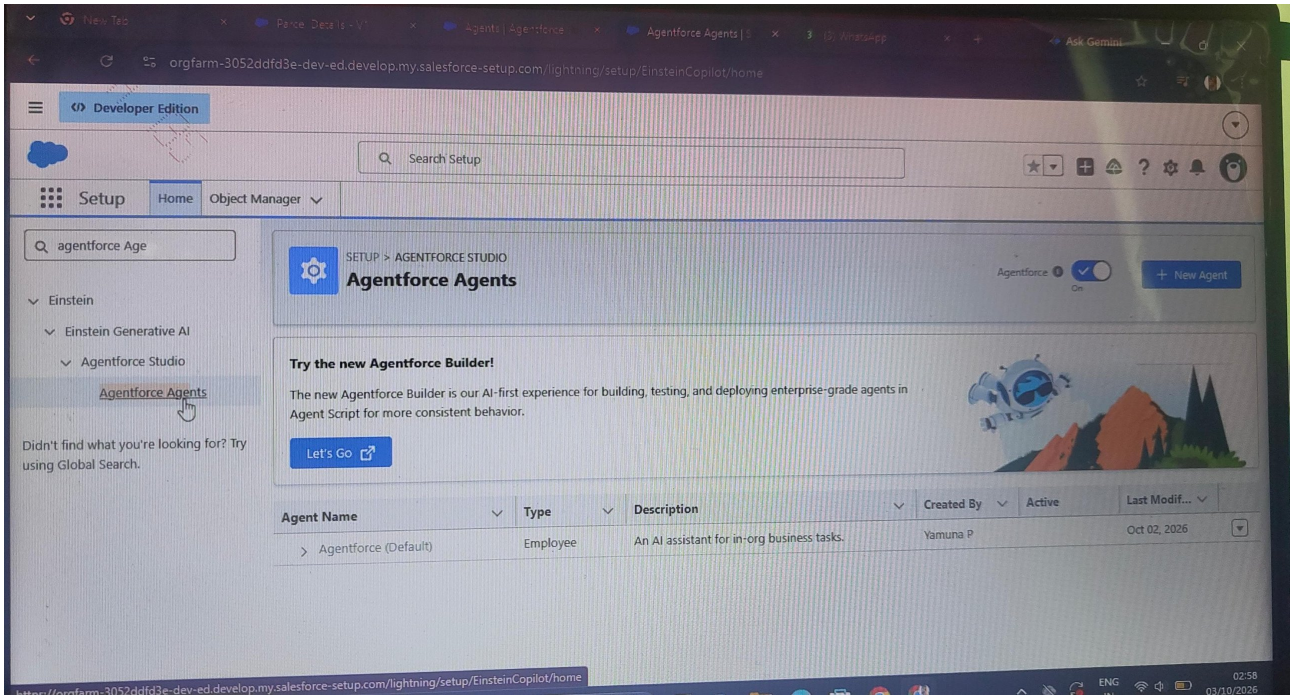


Fig 2: Agentforce Agents page with Agentforce turned On and the Default agent listed

Result

Einstein and Agentforce are now enabled in the org, and the Agentforce (Default) agent is available. It is opened in Agentforce Builder in Milestone 4 to create the SwiftShip Tracker agent.

Phase 3: UI/UX Development & Customization

Milestone 4: Building the Agent

Objective: To build the Agentforce agent that retrieves parcel details by Parcel ID, by creating a subagent and an agent action linked to the Parcel Details flow.

Step 1: Open the Default Agent in Agentforce Builder

• Go to **Setup > Agentforce Agents** and click **Agentforce (Default)**.

• On the Agent Details page (API Name: Copilot_for_Salesforce, Type: Employee), click **Open in Builder**.

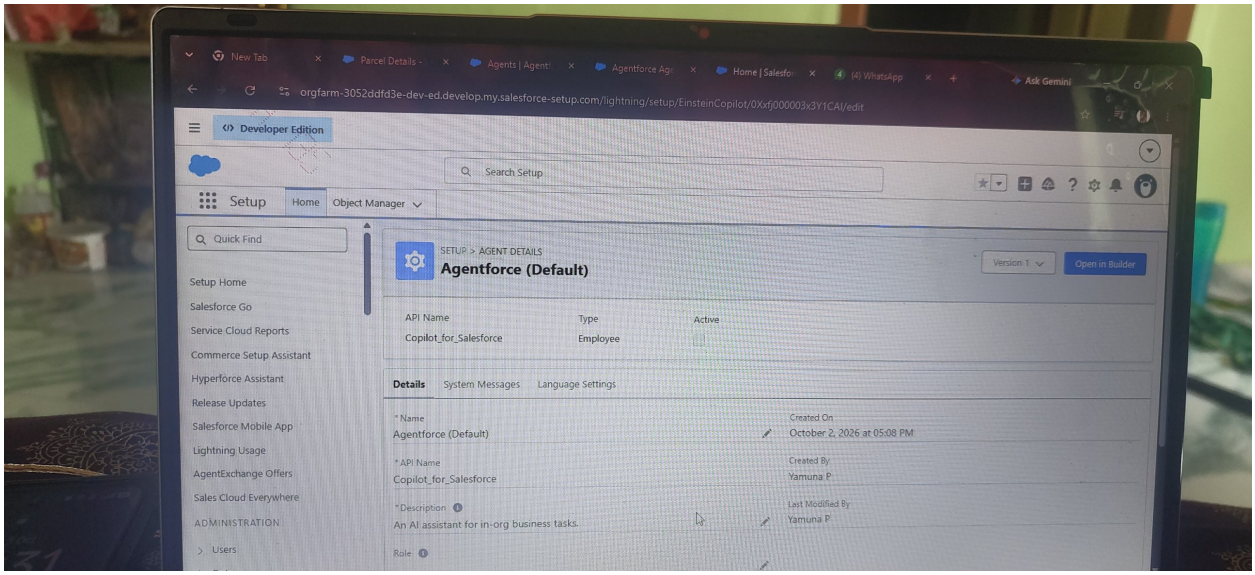


Fig 1: Agent Details page of Agentforce (Default) with Open in Builder

Step 2: Start a new subagent

• In Agentforce Builder, open the **Subagents** panel. It lists General CRM and Single Record Summary.

• Click **New** and select **New Subagent**.

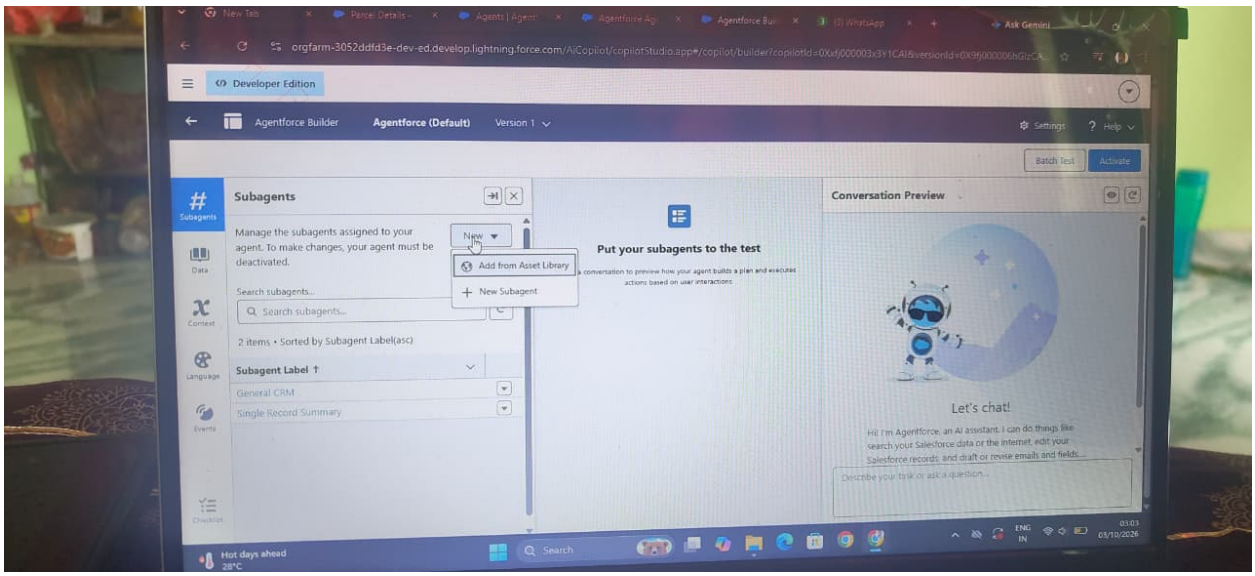


Fig 2: Subagents panel with the New > New Subagent option

Step 3: Describe the subagent

• In the **Create a Subagent** window, enter what the subagent should do: *This Agent should give the parcel details to the user, by asking for the parcel id.*

• Click **Next** and complete the wizard. The subagent is named **Parcel Details Inquiry**.

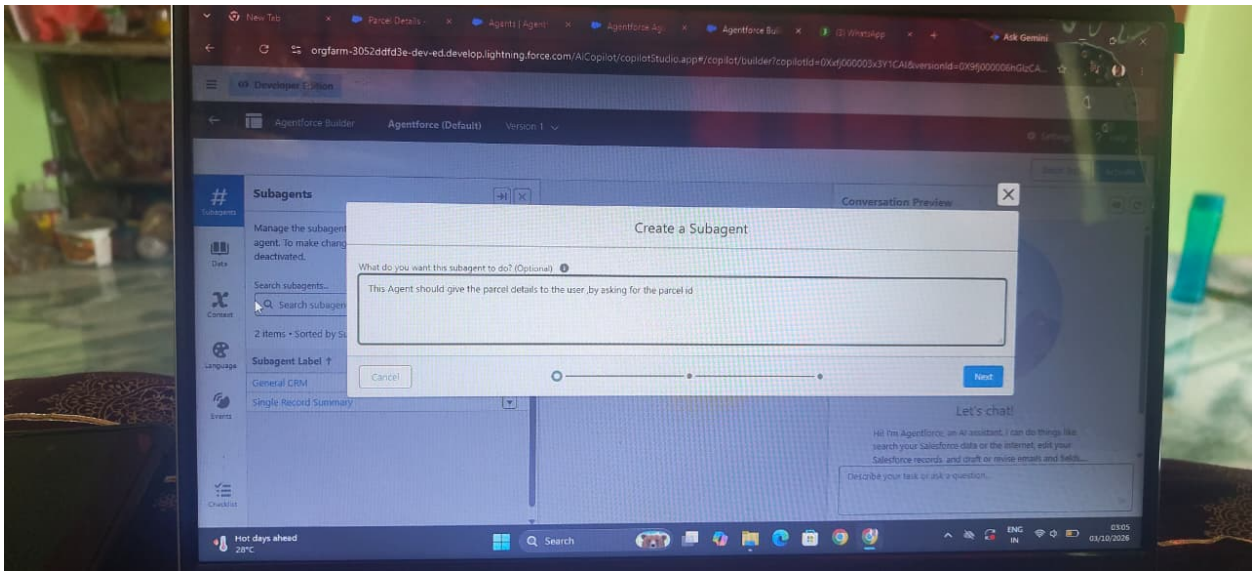


Fig 3: Create a Subagent dialog

Step 4: Open the Agentforce Asset Library

• Go to **Setup > Agentforce Asset Library** and open the **Actions** tab.

• Agent actions are listed here (134 items). The **New Agent Action** button is at the top right.

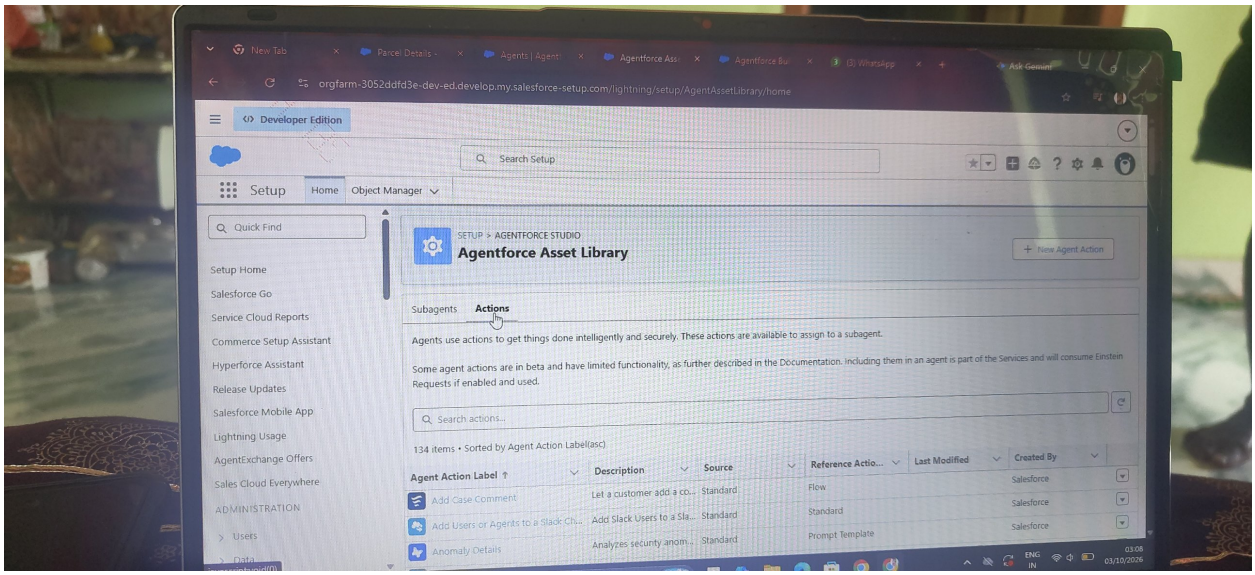


Fig 4: Agentforce Asset Library, Actions tab

Step 5: Configure the Parcel Details agent action

• Create a new agent action with Reference Action Type **Flow** and choose the **Parcel Details** flow.

• Agent Action Label: **Parcel Details**. Agent Action API Name: **Parcel_Details**.

• Agent Action Description: *Use this action to get or checked the parcel details and tracking status for a shipment.*

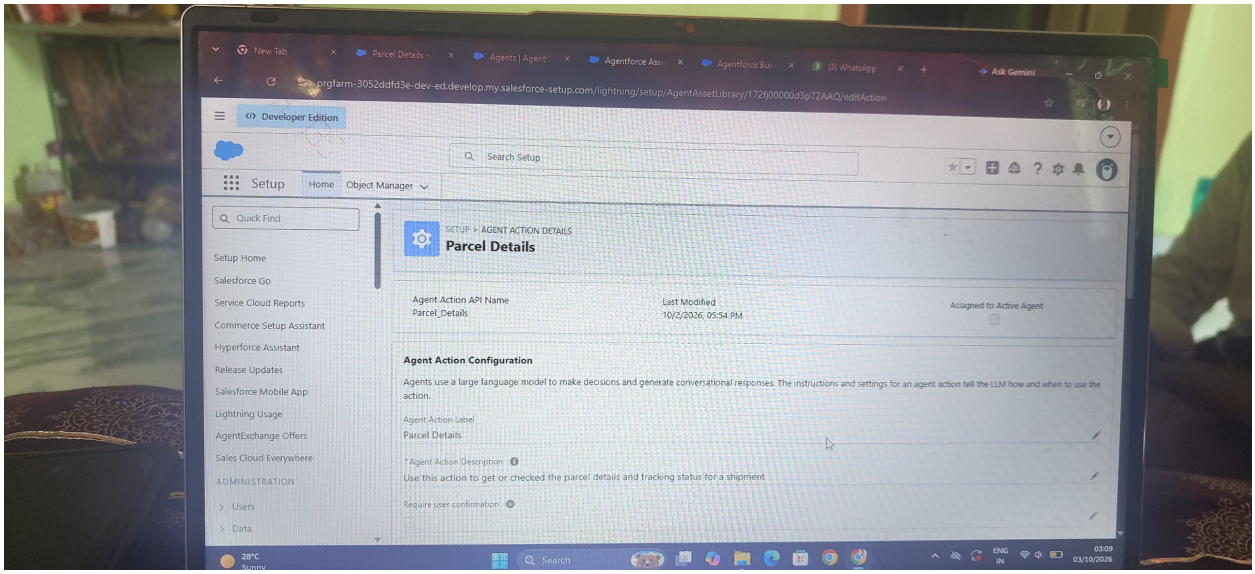


Fig 5: Parcel Details agent action configuration

Step 6: Verify the action in the Asset Library

• Return to the Actions list and scroll to **Parcel Details**.

• Its Source is **Custom**, which confirms the action was created from the flow.

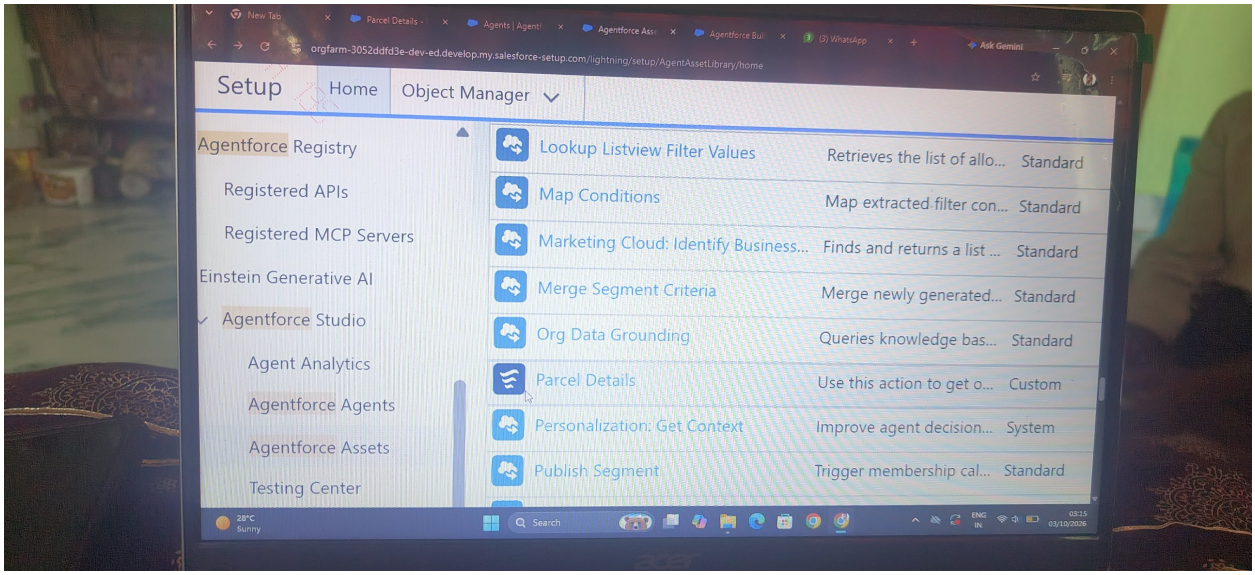


Fig 6: Parcel Details listed as a Custom action

Step 7: Confirm the subagent

• Back in Agentforce Builder, the Subagents list now shows **Parcel Details Inquiry** (3 items).

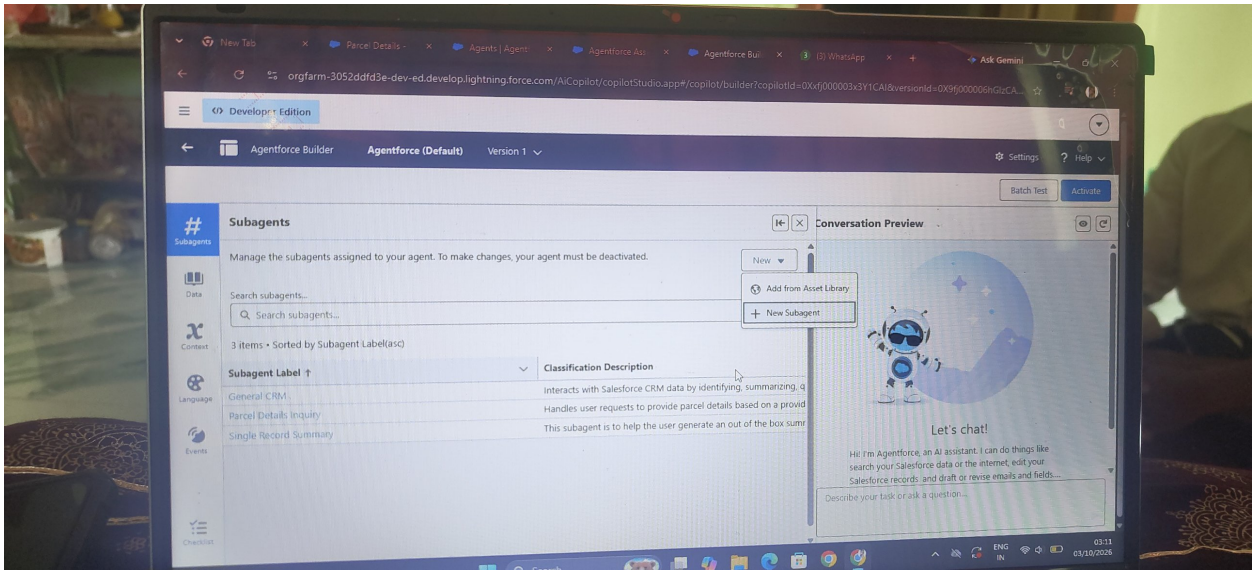


Fig 7: Subagents list with Parcel Details Inquiry

Step 8: Assign the action to the subagent

• In Agentforce Builder, under Subagents, click the **Parcel Details Inquiry** subagent.

• Click **This Subagent's Actions**, then click **New** and select **Add from Asset Library**.

• Select the flow action created earlier, **Parcel Details**, and click **Finish**.

• The action now appears under the subagent's actions.

Step 9: Rename the agent

• Click **Settings** at the top right of Agentforce Builder.

• In Edit Agent Details, change the Name to **SwiftShip Tracker**.

• Click **Save**.

Step 10: Activate the agent

• Click **Activate** at the top right.

• If a **Configuration Issues Detected** warning appears, click **Ignore & Activate**, then click **Activate**.

• Go to **Setup > Agentforce Agents** and confirm that **SwiftShip Tracker** is listed and Active.

Result

The Parcel Details Inquiry subagent and the Parcel Details agent action are created. The action is assigned to the subagent, and the agent is renamed **SwiftShip Tracker** and activated, so it can answer parcel tracking queries using the Parcel ID.

Phase 4: Data Migration, Testing & Security

Objective: To validate that the SwiftShip Tracker agent routes parcel queries to the correct subagent, runs the Flow, retrieves parcel data, and works without access or permission issues. The goal is to confirm that the AI-driven agent gives accurate responses to parcel tracking queries within Salesforce.

1. Verify the Agent

Go to Setup → Agentforce Agents and confirm that the SwiftShip Tracker agent (Type: Employee) appears in the list.

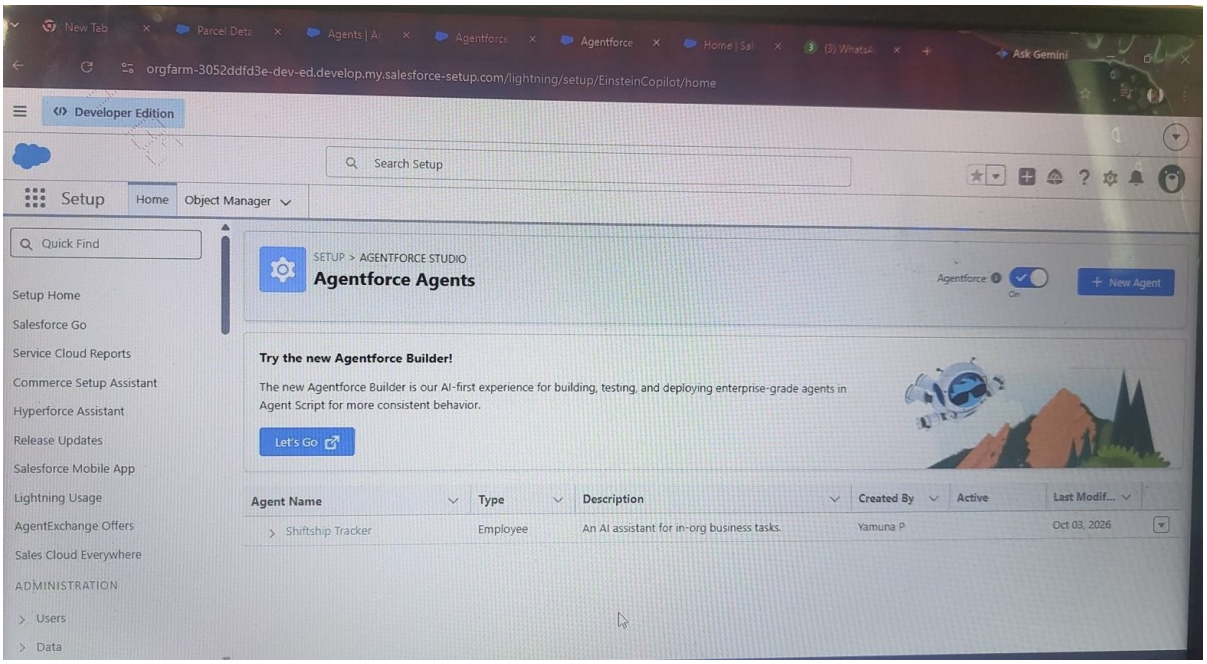


Figure 1: Agentforce Agents list showing the agent

2. Validate the Subagent

In Agentforce Builder (Version 1, Active), open the Parcel Details Inquiry subagent and confirm its configuration:

- **Classification Description:** Handles user requests to provide parcel details based on a provided parcel ID.
- **Scope:** Retrieve and provide parcel details to the user by asking for and using the parcel ID. Unrelated inquiries are not addressed.

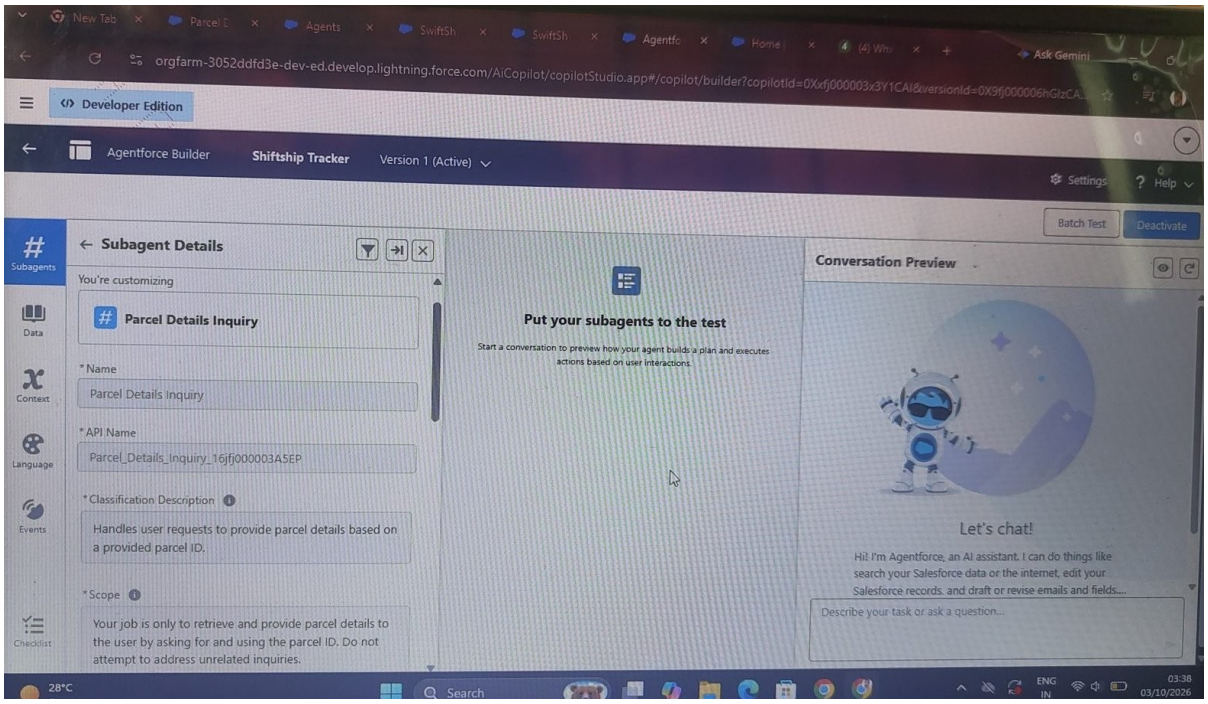


Figure 2: Subagent configuration

Under This Subagent's Actions, confirm that the Parcel Details action is linked.

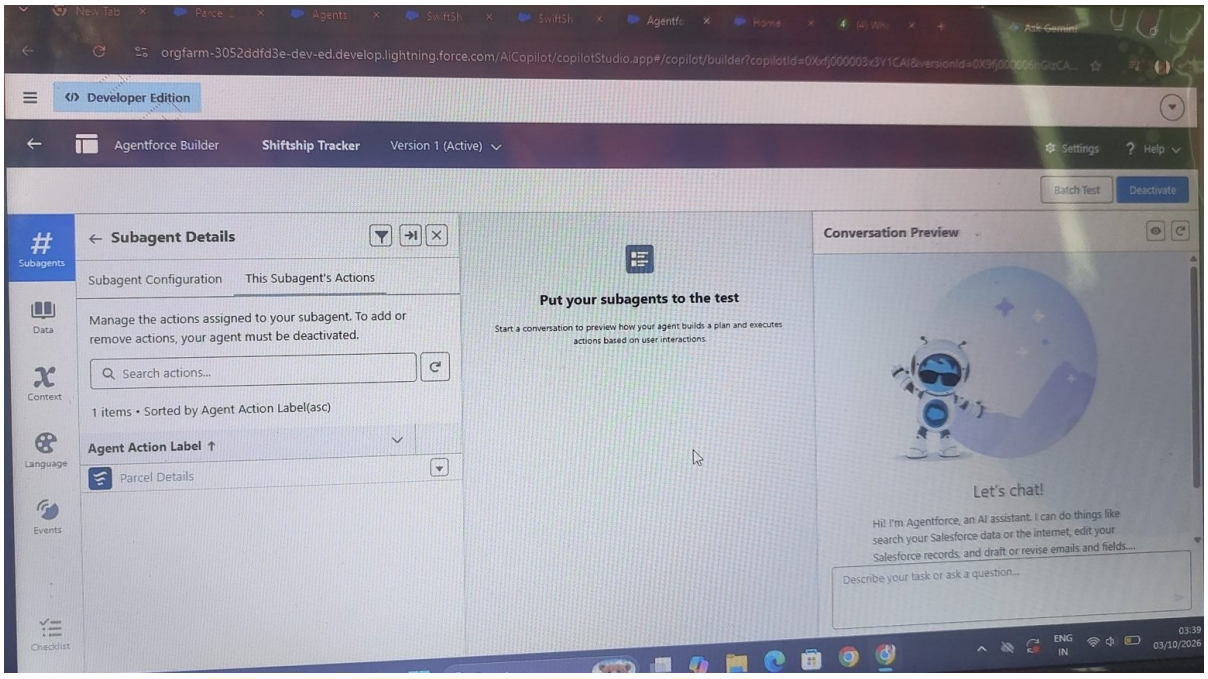


Figure 3: Parcel Details action linked to the subagent

3. Check the Flow

Go to Setup → Flows and open Parcel Details – V1. Confirm that the flow is Active and follows this sequence: Start → Get Parcel Records → Parcel Updates (prompt template action) → Assignment Outputs → End.

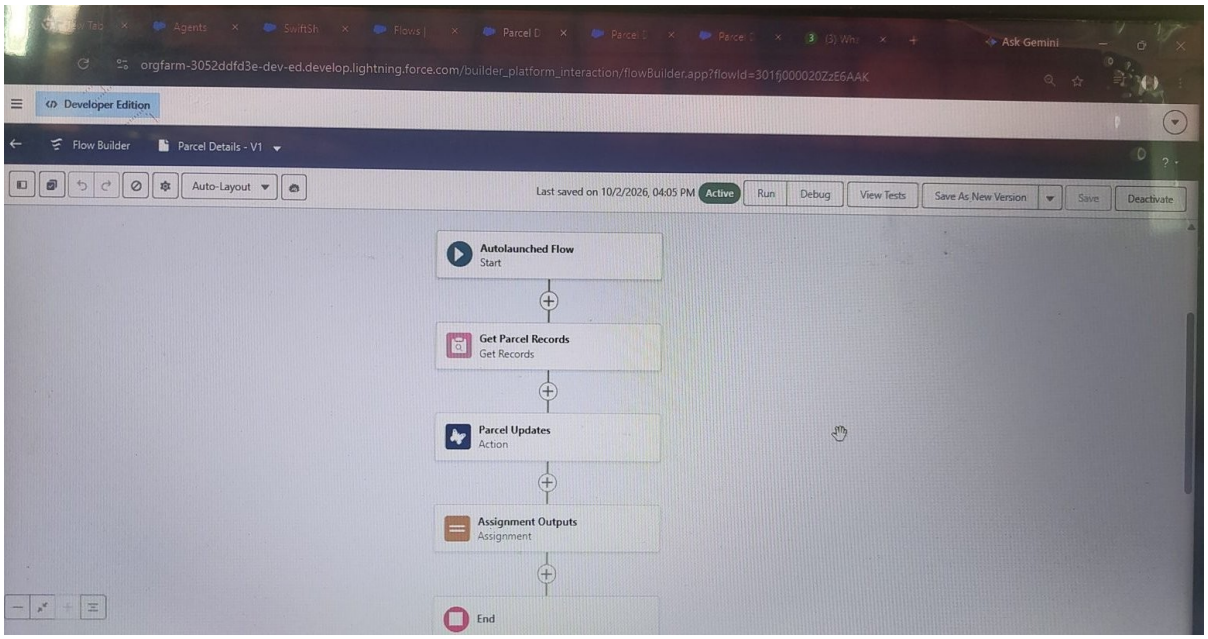


Figure 4: Parcel Details auto-launched flow (Active)

4. Security: Assign the Permission Set

Open the Swift Ship permission set and go to Manage Assignments. Click Add Assignment, select the EinsteinServiceAgent User, and click Next. The Assignment Summary shows the status as Success.

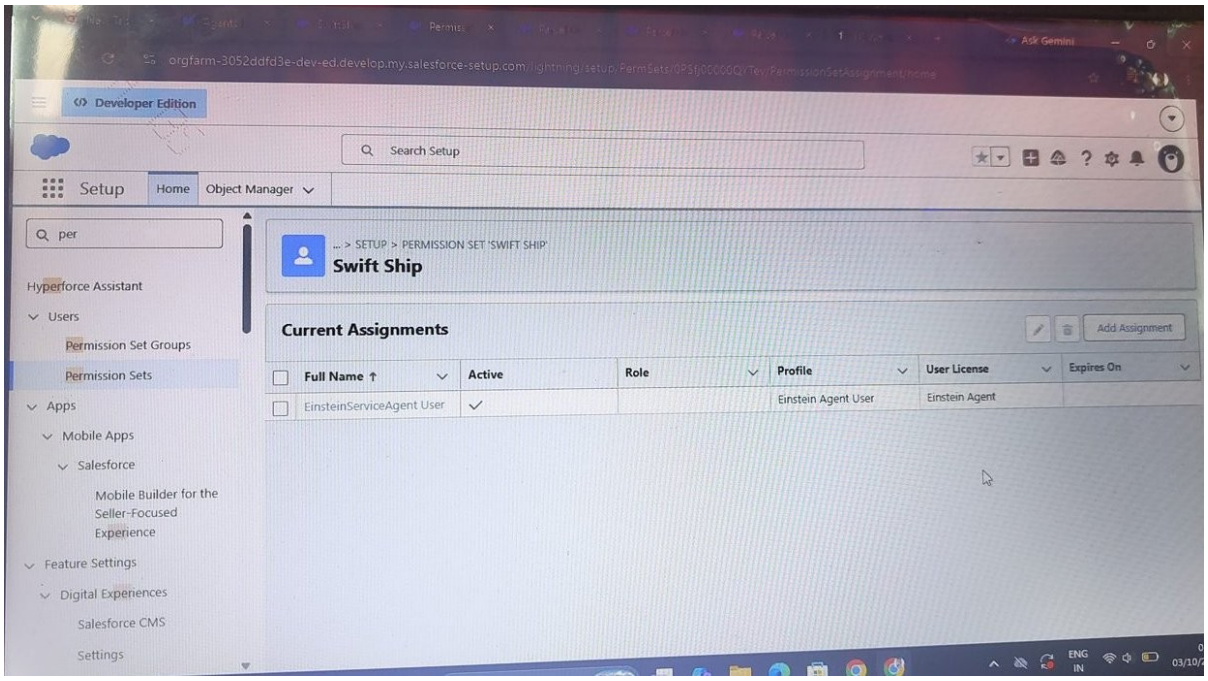


Figure 5: Swift Ship permission set – current assignments

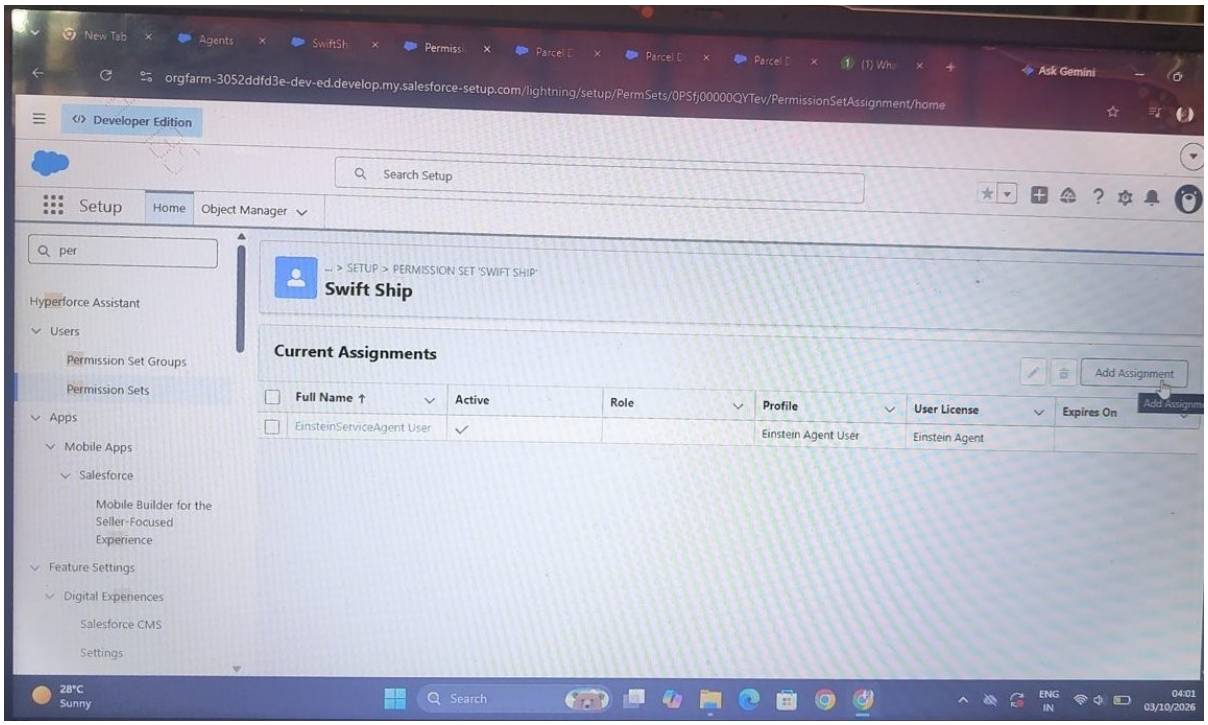


Figure 6: Add Assignment option

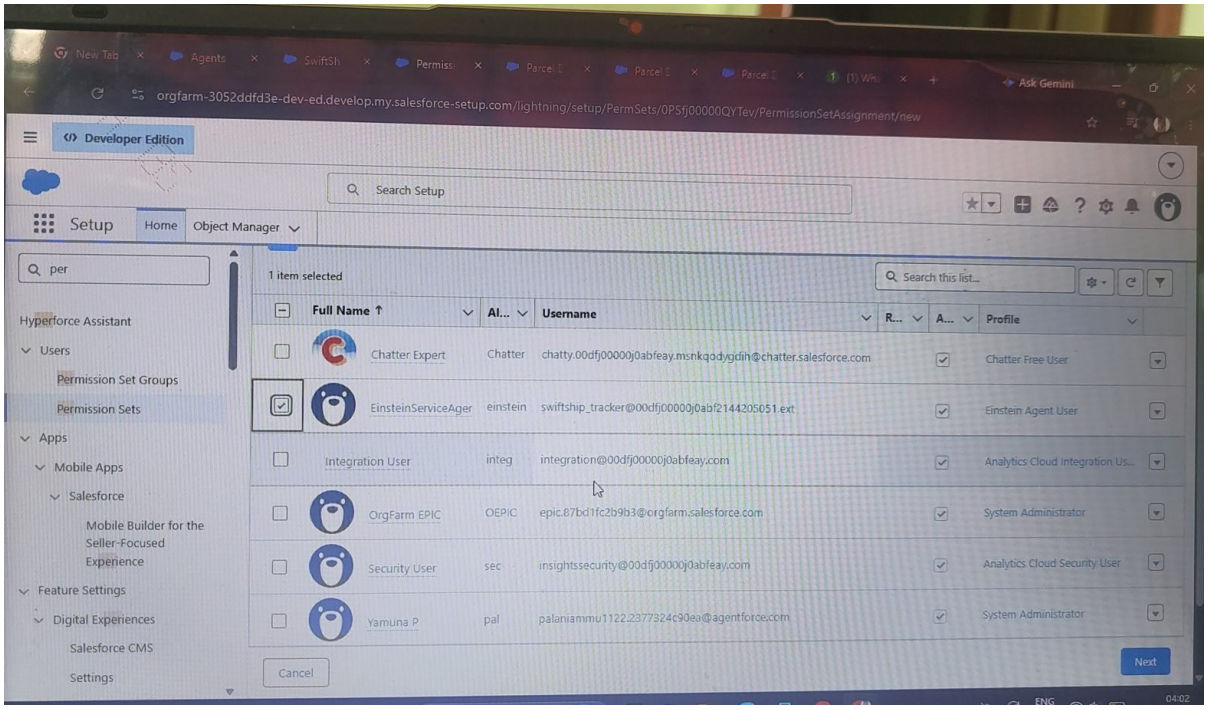


Figure 7: Selecting the EinsteinServiceAgent User

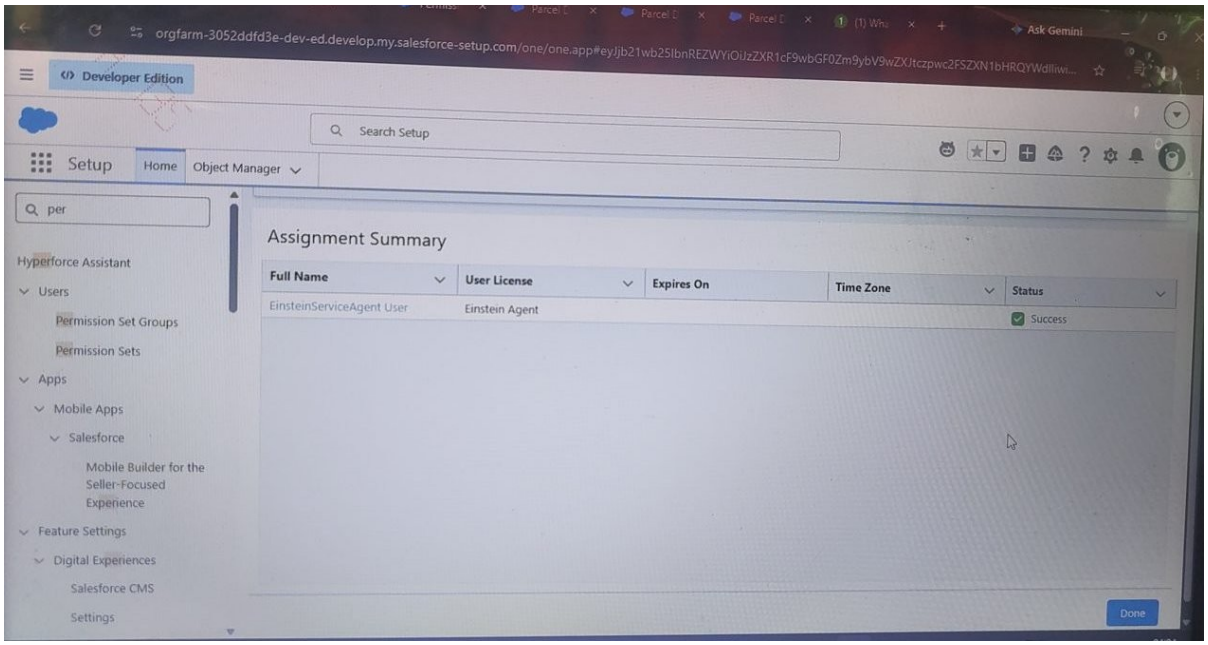


Figure 8: Assignment Summary – Success

5. Configure and Activate the Agent

In the Agentforce Builder, the Parcel Details action is configured as follows:

- **Input:** ids (String) – “Enter the tracking ID or Parcel ID”, with “Require Input to execute action” enabled.
- **Output:** Output (String) – returns the current status of the parcel.

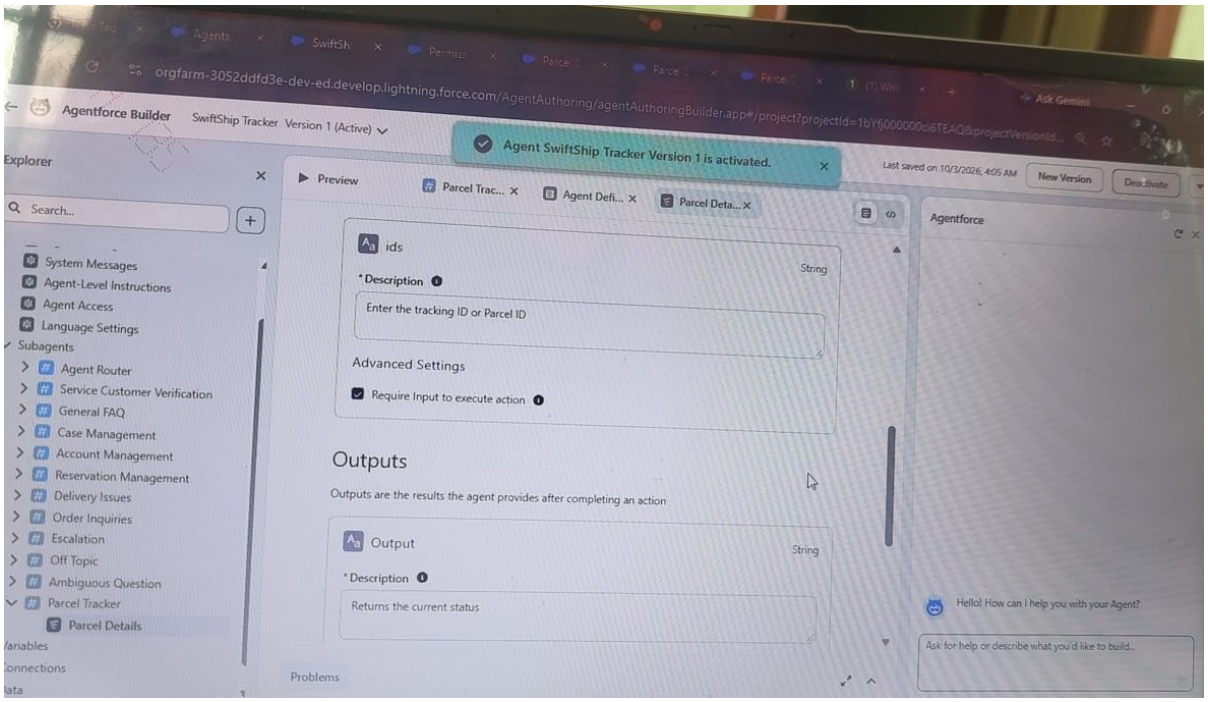


Figure 9: Parcel Details action – inputs and outputs

The version is then committed, and the agent is activated. A confirmation message shows that the agent SwiftShip Tracker Version 1 is activated.

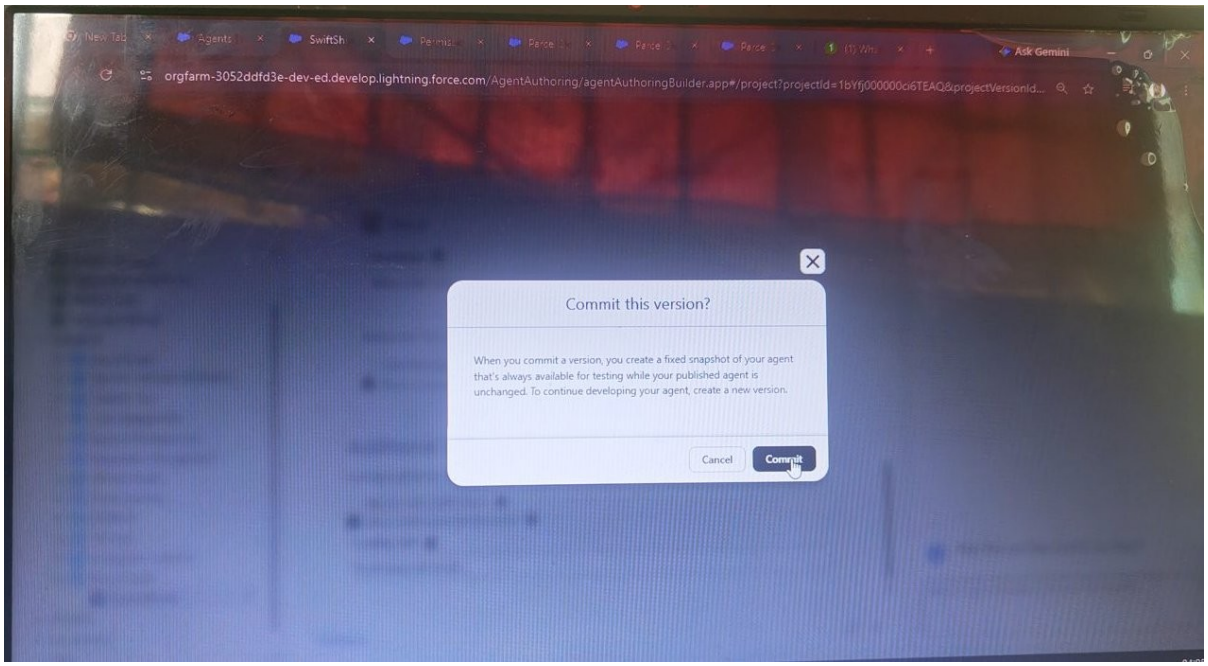


Figure 10: Commit this version

6. Test Data in Salesforce

A test Parcel record was created in the Developer Org with the following values:

Field	Value
Parcel Name	laptop
Parcel ID	P-001
Status	Booked
Weight	2
Estimated Delivery	10/29/2026

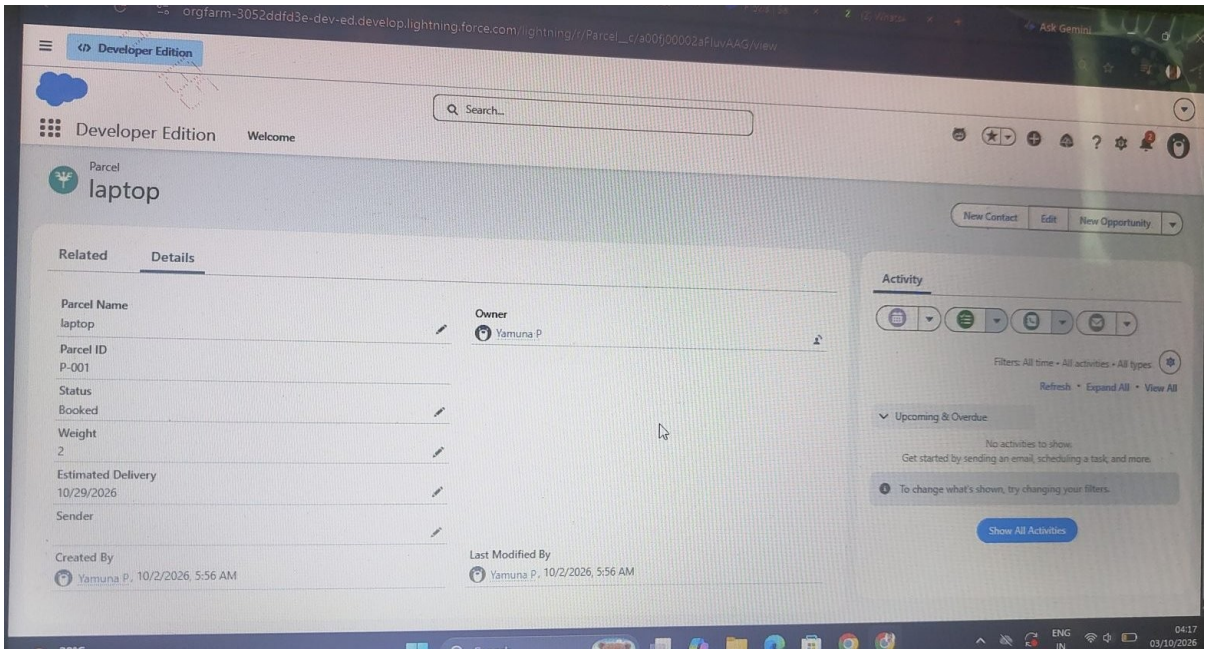


Figure 11: Test Parcel record (P-001)

7. Test the Agent (Live Test)

In the Preview panel (Live Test), the following prompt was entered:

"Please provide parcel information about parcel id=P-001"

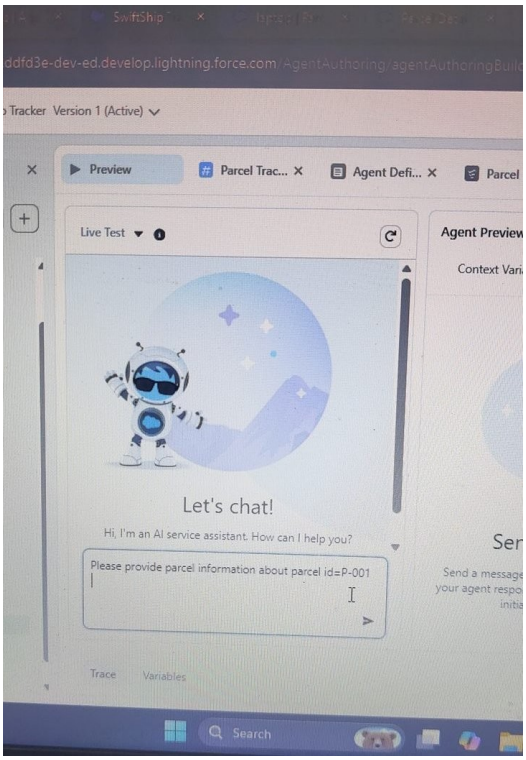


Figure 12: Prompt entered in the Live Test

The agent responded with the parcel details:

- **Parcel Name:** laptop
- **Status:** Booked
- **Weight:** 2 kg
- **Estimated Delivery Date:** 10/29/2026

The Agent Preview trace shows that the request was transitioned to the Parcel Tracker subagent, the Parcel Details action was invoked, and the output evaluation was GROUNDED.

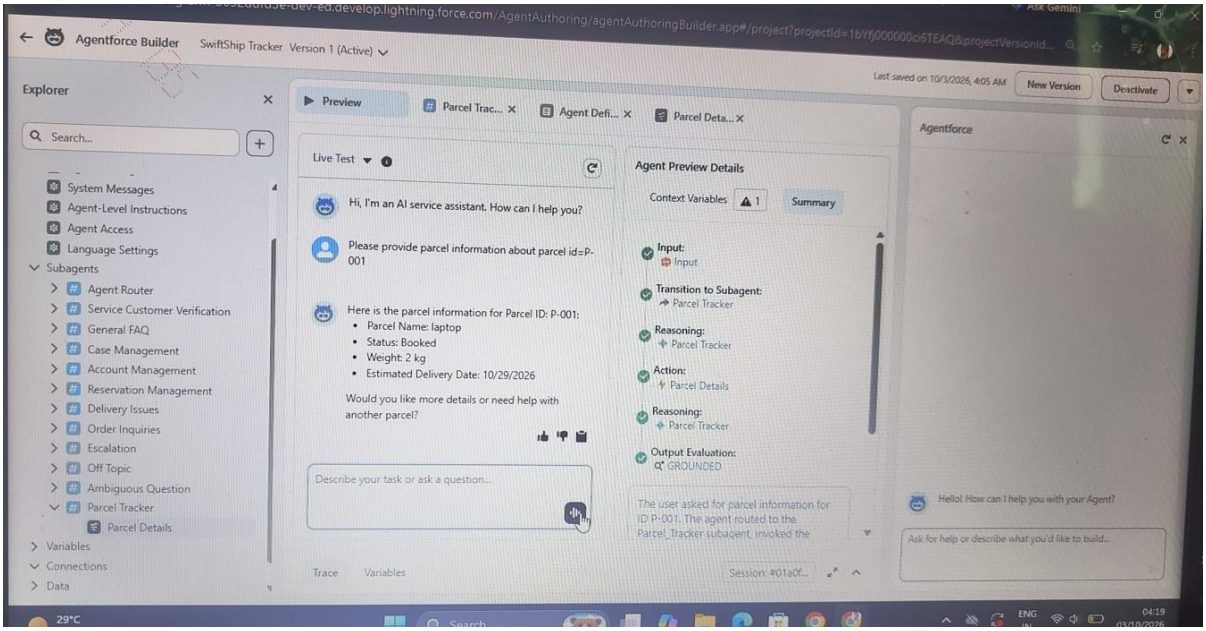


Figure 13: Agent response and reasoning trace

Result

The agent's response matches the Parcel record exactly. This confirms that topic routing, Flow execution, permissions, and data retrieval all work correctly.

Data Migration

Sample records were entered manually in the Developer Org, so no bulk data migration was required.

Phase 5: Deployment, Documentation & Maintenance

This phase validates that the SwiftShip Tracker agent, created using Salesforce Agentforce Builder, works as intended within the Salesforce Developer Org environment, and describes how the solution is documented and maintained.

1. Deployment

The agent was built and tested in a Salesforce Developer Org. Version 1 of the agent was committed, creating a fixed snapshot of the agent, and then activated. Agentforce Builder confirmed the activation with the message “Agent SwiftShip Tracker Version 1 is activated.”

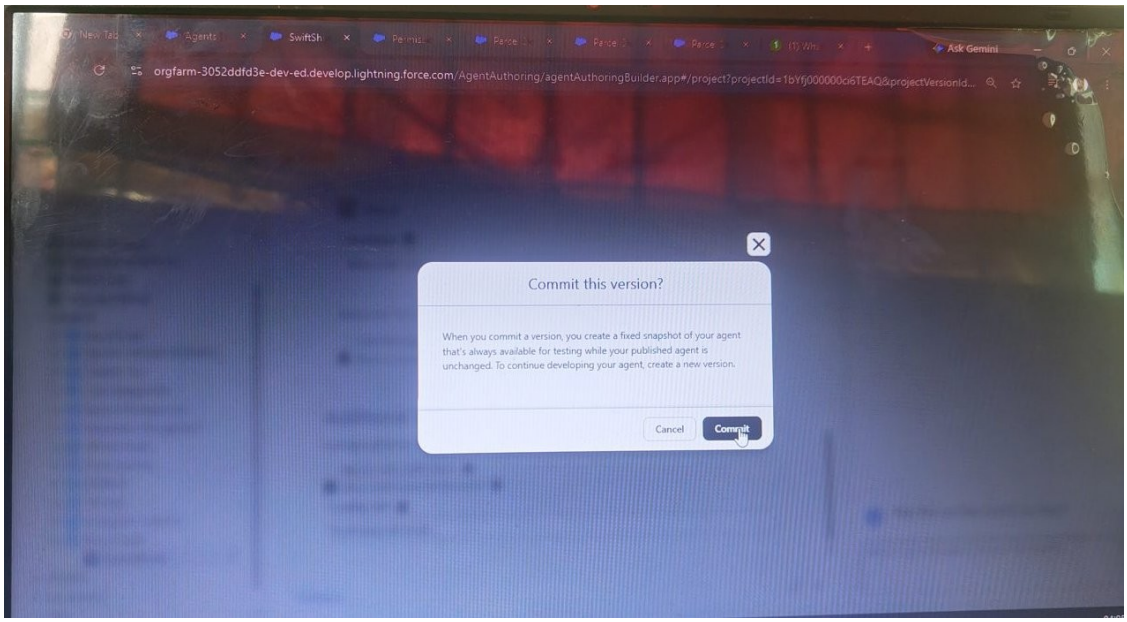


Figure 1: Committing Version 1 of the agent

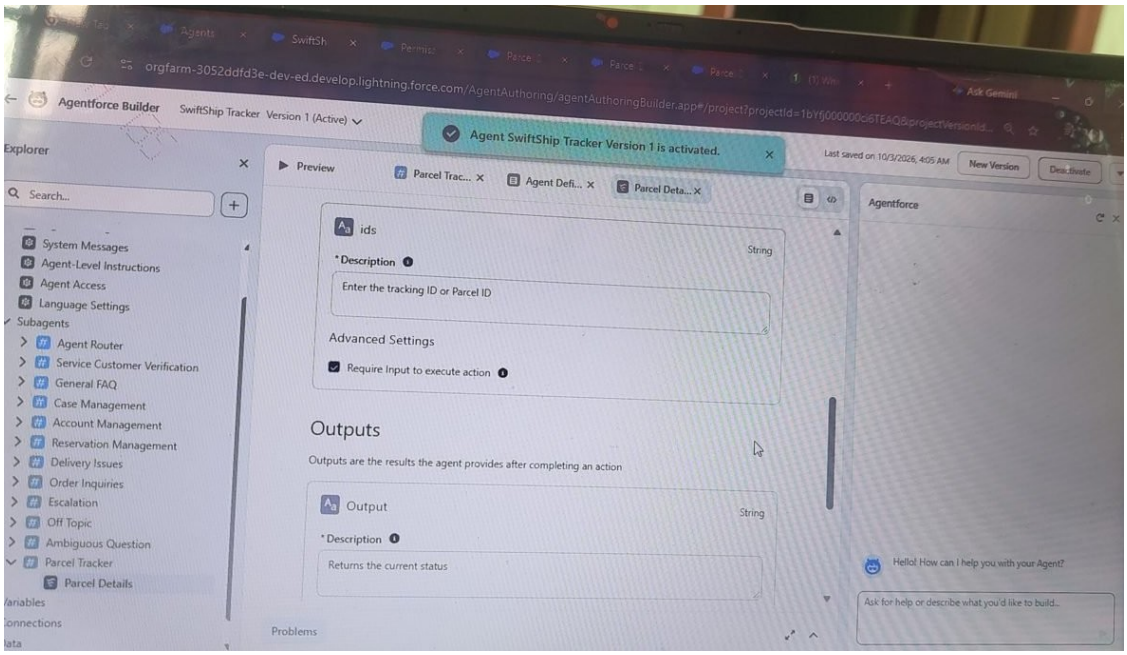


Figure 2: Activation confirmation in Agentforce Builder

The deployed solution includes:

- **Custom objects:** Parcel, Delivery, Sender, and Receiver
- **Prompt template:** “Retrieve Parcel Details” (Prompt Builder)
- **Flow:** Parcel Details (auto-launched Flow)
- **Subagent:** Parcel Details Inquiry, with the Parcel Details action
- **Permission set:** Swift Ship, assigned to the EinsteinServiceAgent User

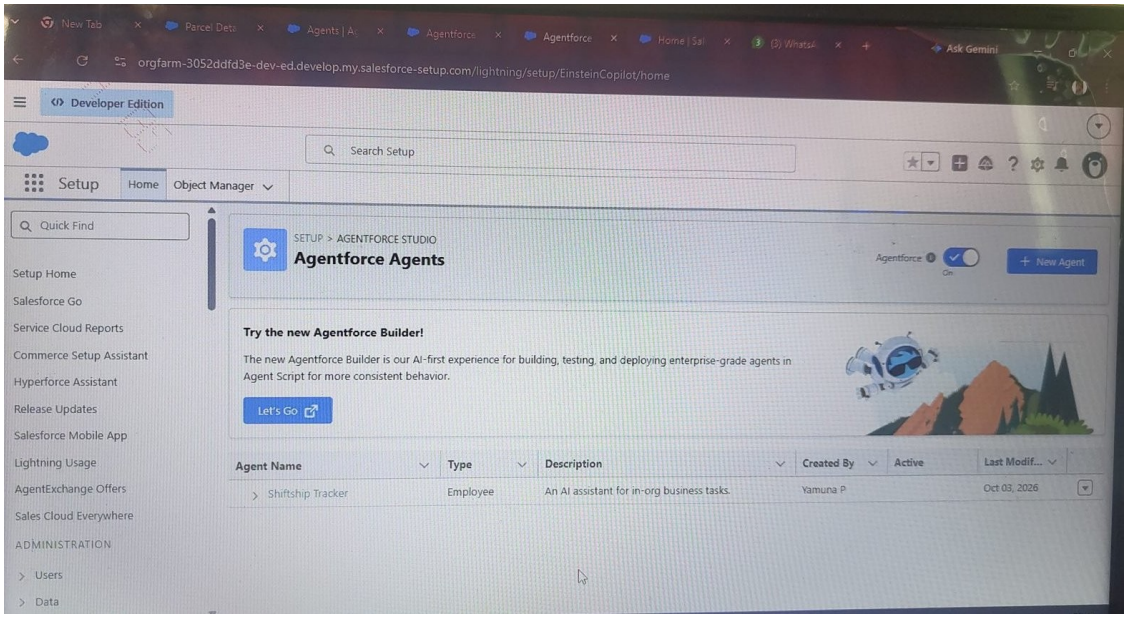


Figure 3: SwiftShip Tracker agent listed in Agentforce Agents

2. Testing in the Developer Org

Testing was conducted within the Developer Org to simulate how the agent would perform in a live setup. The main goal was to confirm correct topic recognition, Flow execution, and data retrieval from the Salesforce custom objects, and that permissions and interactions function correctly.

A test Parcel record (Parcel ID P-001, Status: Booked, Weight: 2, Estimated Delivery: 10/29/2026) was created, and the agent was asked for its details.

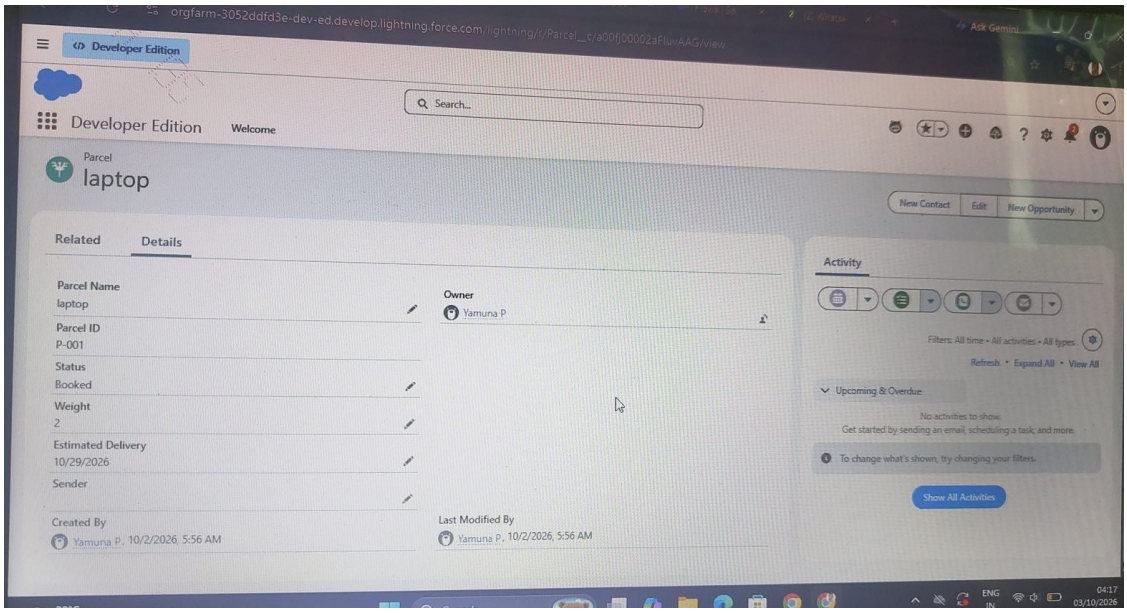


Figure 4: Test Parcel record P-001 in Salesforce

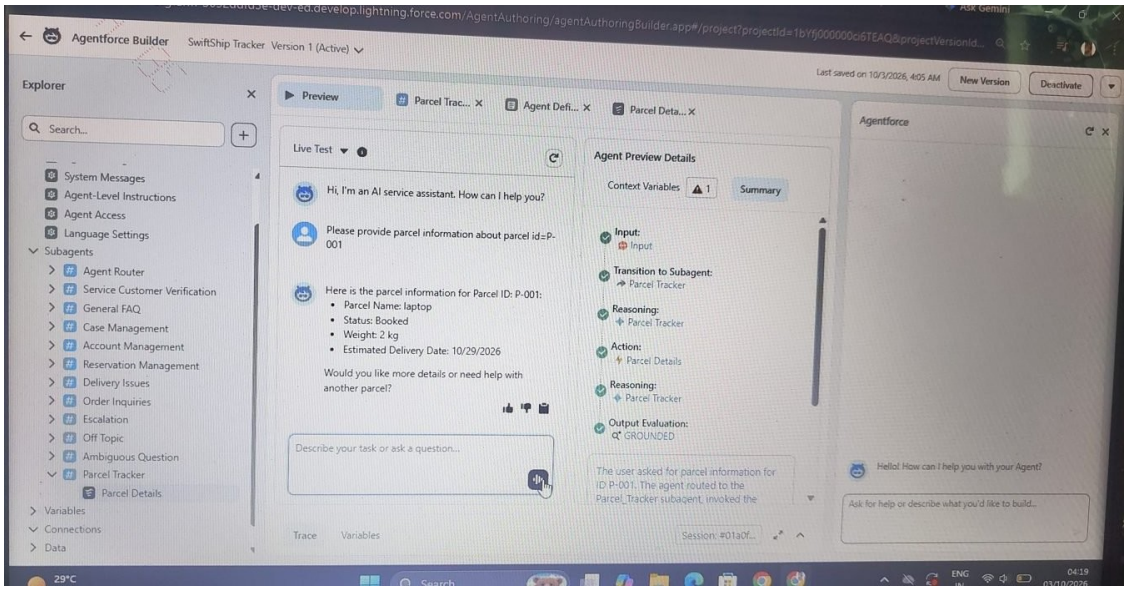


Figure 5: Agent response and reasoning trace in the Live Test

The agent returned the same values stored in the record (laptop, Booked, 2 kg, 10/29/2026). The trace shows the request was routed to the Parcel Tracker subagent, the Parcel Details action ran, and the output was evaluated as GROUNDED.

3. Scope of Testing

Real-world production testing (UAT or sandbox deployment) was not performed in this project because:

- Developer Edition orgs are standalone environments and do not support production-linked testing environments.
- Enterprise-level Agentforce deployment involves sandbox testing, staging environments, and CI/CD automation, which are outside the scope of this academic/prototype implementation.

4. Documentation

The project is documented in this report, which covers:

- **Planning and design:** problem statement, empathy map, requirements, data flow, and solution architecture.
- **Project management:** sprint backlog and schedule.
- **Build guide:** step-by-step instructions for creating objects, tabs, fields, the prompt template, the Flow, the permission set, and the agent.
- **Testing evidence:** screenshots of verification and live testing.

5. Maintenance, Monitoring & Troubleshooting

Ongoing maintenance ensures that Parcel, Sender, Receiver, and Delivery data remain accurate, synchronized, and up to date. Regular monitoring helps detect:

- misrouted agent queries
- failed Flow executions
- performance delays in Agentforce responses

Troubleshooting activities include:

- resolving configuration and permission errors
- updating Flow logic
- refining subagent descriptions and scope
- improving prompts to enhance user interactions

Although this project is implemented in a Developer Org for learning purposes, these practices are essential in real-world enterprise deployments. They keep AI-powered service agents reliable, scalable, and aligned with evolving customer service operations and business goals.

7. Functional & Performance Testing

During testing, screenshots were captured from Salesforce to validate agent setup, subagent configuration, Flow execution, permission assignment, data retrieval, and the agent's live responses. All features were verified for correct functionality.

No.	Feature tested	Result observed	Status
1	Agent listed and active	SwiftShip Tracker agent appears in Agentforce Agents (Version 1 active)	Pass

2	Subagent configuration	Parcel Details Inquiry has the correct description, scope, and linked Parcel Details action	Pass
3	Flow	Parcel Details flow is Active: Get Parcel Records → prompt action → output	Pass
4	Permissions	Swift Ship permission set assigned to EinsteinServiceAgent User (Success)	Pass
5	Agent activation	Version committed; activation confirmation shown	Pass
6	Data retrieval	Prompt for P-001 returned laptop, Booked, 2 kg, 10/29/2026, matching the record	Pass
7	Topic routing and grounding	Request routed to Parcel Tracker subagent; output evaluated as GROUNDED	Pass

Performance: the agent returned the parcel details in the Live Test without errors or retries, and the response was grounded in the Salesforce record. Detailed response-time measurements were not recorded in this prototype.

8. Advantages & Disadvantages

Advantages

- Centralized parcel and customer information in Salesforce.
- Better visibility into parcel booking and delivery tracking.
- Automates key logistics processes using Salesforce Flow.
- Agentforce enables conversational customer support and parcel-status assistance.
- Prompt Builder supports AI-powered responses and self-service.
- Reduces manual effort for customers, support teams, and administrators.
- Provides timely notifications and updates during the delivery process.
- Experience Cloud can give customers convenient access to tracking information.
- Supports integration with external logistics systems.
- Can scale to support multiple locations and growing parcel operations.

Disadvantages

- Requires accurate and up-to-date parcel and customer data.
- Depends on Salesforce configuration and automation.
- Agentforce responses depend on properly configured prompts and available data.
- External logistics integrations may increase implementation complexity.
- Flow and automation logic require regular maintenance when business processes change.
- Users need appropriate Salesforce or Experience Cloud access to use the system.
- Initial Salesforce implementation and configuration can require significant effort.

- Expanding to additional locations or integrations may require further customization.

9. Conclusion

The SwiftShip Tracker demonstrates how Salesforce can be used to build a centralized parcel management and tracking solution. The project integrates custom objects, relationships, Flows, Apex, Email Alerts, Experience Cloud, Agentforce AI, Prompt Builder, security controls, and reporting to support parcel booking, delivery tracking, status updates, and customer communication.

The integration of Agentforce AI with Prompt Builder and Salesforce Flow provides a conversational approach to parcel tracking, enabling customers to obtain shipment information without depending entirely on manual support. The project therefore provides a practical foundation for a scalable and user-friendly parcel management solution.

10. Future Scope

- Extend Agentforce capabilities for more parcel-management queries.
- Enhance real-time delivery tracking.
- Expand customer self-service through Experience Cloud.
- Add more advanced delivery-performance analytics.
- Integrate additional external logistics systems.
- Improve AI prompts and conversational experiences.
- Introduce enterprise-level sandbox and CI/CD deployment.
- Expand automation for additional parcel operations.

11. Appendix

- **A. Custom Objects:** Parcel, Delivery, Sender, Receiver
- **B. Standard Objects:** Account, Contact, Case, Task, EmailMessage, User
- **C. Automation:** Flows, Approval Processes, Email Alerts
- **D. Apex:** Apex Classes, Batch Apex
- **E. AI:** Agentforce AI, Prompt Builder
- **F. UI:** Lightning App, Lightning Record Pages, Experience Cloud
- **G. Security:** Profiles, Roles, Permission Sets, Sharing Rules, Field-Level Security
- **H. Analytics:** Reports & Dashboards
- **I. Testing:** Flow, Agentforce, data retrieval, security and permission validation

Thank You